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Integration of single-cell multi-omics and spatially resolved transcriptomics in neuroblastoma datasets

Master Thesis

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List of abbreviations

ALT: Alternative Lengthening of Telomeres

LFC: Log Fold Change

LR: Low Risk

NB: Neuroblastoma

PCA: Principal Component Analysis

scRNA-seq: single-cell RNA sequencing

scATAC-seq: single-cell Assay for Transposase-Accessible Chromatin sequencing

SRT: Spatially Resolved Transcriptomics

TF: Transcription Factor

UMAP: Uniform Manifold Approximation and Projection

EMT: Epithelial Mesenchymal Transition

Zusammenfassung

Mit räumlich aufgelöster Transkriptomik können Molekularbiologen die Morphologie der Genexpression von Hunderten oder Tausenden von Genen in Gewebeproben visualisieren. Diese neue Technologie hat den Weg für neue Integrationsmethoden geebnet, um räumlich aufgelöste Transkriptomik Daten mit anderen Einzelzelldaten zu integrieren um räumliche Auflösung von Epigenomik oder Proteomikdaten bzw. Transkriptionswerte für mehr Gene zu erreichen. Hier teste ich die neuesten Integrierungsmethoden, um Einzelzellsequenzierungsdaten auf 10x Xenium räumlich aufgelöster Transkriptomik Neuroblastom Proben zu kartieren und ihre Leistung zu vergleichen. Ich verwende die trainierten Modelle, um Einzelzellwerte Genaktivität und Transkriptionsfaktor-Aktivitätswerte zu kartieren, um einen multimodalen, räumlich aufgelösten Datensatz zu erhalten. Ich finde, dass Tangram und MaxFuse hervorragend für die Kartierung von scRNA-seq abschneiden, wobei MaxFuse Tangram bei der Kartierung von Transkriptionsfaktor- und Genaktivität übertrifft. Mit den neuen integrierten Daten kann ich die Zellgruppen mit einer räumlich eindeutigen Anordnung charakterisieren, um zu versuchen, ihre Rolle in der Tumormikroumgebung zu bestimmen und den Wert der räumlichen Integrierung von verschiedenen Omics Datenarten.

Abstract

With spatially resolved transcriptomics, molecular biologists can visualise the morphology of gene expression of hundreds or thousands of genes in the sample tissue. This new technology has paved the way to new integration methods that combine spatially resolved transcriptomics data with single-cell data allowing for spatial resolution of other modalities as well as superior gene coverage. Here, I test cutting-edge methods to map single-cell RNA-sequencing data onto 10x Xenium spatial transcriptomics Neuroblastoma samples and compare their performance. I used the trained models to further map single-cell gene activity and transcription factor activity scores to obtain a multimodal spatially resolved dataset. I find Tangram and MaxFuse to perform outstandingly for mapping single-cell RNA sequencing with MaxFuse surpassing Tangram for the purposes of mapping transcription factor and gene activity. With the new integrated data, I characterise the clusters of cells with a spatially distinct arrangement to attempt to infer their role in the tumour microenvironment and show the benefits of multiple omics data types.

1. Introduction

1.1 Spatial Transcriptomics

Through the past decade, single-cell omics-sequencing technologies have pushed the boundaries on the information that can be gained from biological samples, and the field of method development has contributed by developing a variety of methods to analyse single cell data. Single-cell data allows to identify and characterise cell types and can be leveraged to infer gene regulatory networks and transcription factor (TF) activity based on their transcriptional footprint (Tang *et al.*, 2009; Patel *et al.*, 2014; Zeisel *et al.*, 2015; Kameneva *et al.*, 2021; Farrell, 2023). Similarly, in the past 6 years since its inception in 2018, spatially resolved transcriptomics data has revolutionised the field once again.

Spatial resolved transcriptomics (SRT) offers the insight of spatial morphologically resolved information (Ståhl *et al.*, 2016; Liu *et al.*, 2021; Cheng *et al.*, 2023) and can help find patterns in cell type arrangement and gene expression alike. With spatial transcriptomics we can leverage the benefits of single-cell transcriptomics with the additional benefit of spatial resolution. Cell type decomposition in the context of tissue architecture can illuminate principles of self-organisation and zonation as well as study tumour microenvironment and cancer evolution (Xavier da Silveira dos Santos *et al.*, 2019; Cable *et al.*, 2022; Tavares-Ferreira *et al.*, 2022). In prior studies, spatial transcriptomics has been leveraged to characterise the tumour microenvironment and revealed the spatial arrangements of tumour cell states (Arora *et al.*, 2023; Denisenko *et al.*, 2024; Liu *et al.*, 2024) and for the identification of spatial gene regulatory patterns that can serve as a prognostic marker for carcinoma (Arora *et al.*, 2023).

1.2 Spatial Transcriptomics Technologies

1.2.1 Sequencing based technologies

SRT technologies can be divided into sequencing and imaging-based methods. Sequencing based methods rely on placing the probes on spatially barcoded arrays and sequencing the mRNA captured by each array. With sequencing based methods, it is possible to retrieve gene transcript counts for the whole transcriptome, albeit with a lower resolution dictated by

technical limits imposed by the size of the array grid that the tissue is divided in (Williams *et al.*, 2022; Cheng *et al.*, 2023; Li *et al.*, 2023). Popular sequencing-based methods include 10x Visium, Stereo-seq, Slide-seq or Light-seq.

1.2.2 Imaging based technologies

Conversely, imaging-based methods achieve single-cell resolution by using in situ hybridization of transcript-specific fluorescent probes combined with high resolution fluorescence microscopy. Imaging based methods furthermore allow for the localization of transcripts at a sub-cellular resolution, an achievement previously impossible in single-cell data (Williams *et al.*, 2022). The main drawback of in-situ imaging methods lies in the maximum size of the library of genes they can detect, which varies by method but always lies below the whole transcriptome size (Cheng *et al.*, 2023). Additionally, the assignment of transcripts to cells requires either the imputation of the cell boundaries in the sample or segmenting the cells only around their nuclei, thus discarding all transcripts that are not overlapping with a nucleus. Cross-cell contamination of transcripts due to poor cell segmentation can proved to be disruptive for the analysis (Fu *et al.*, 2024). There are many established spatial transcriptomics assays, in this study we use 10x Xenium, which only allows for a panel size of ~300 genes at subcellular resolution (see **Methods**). Similarly to Xenium, STARmap allows for a panel size of ~1000 genes but only at a single-cell resolution. Other technologies, like MERFISH and seqFISH have recently achieved resolutions of ~ 10.000 genes due to breakthroughs in multiplex probe design but are practically more limited (Williams *et al.*, 2022; T.-Y. Chen *et al.*, 2023). The platform MERSCOPE by VizGEN which relies on MERFISH technology only supports a maximal panel size of 1000 genes (Vizgen, 2020) and studies that employ seqFISH/seqFISH+ use smaller panel sizes ranging from a hundred to 3360 genes (Shah *et al.*, 2016; Takei *et al.*, 2021).

1.3 Integration of SRT with scRNA-seq

A common solution to overcome the panel size limit is to perform integration of SRT and scRNA-seq. This integration can be performed in a variety of computational strategies but broadly involves computing similarity between cells of the two modalities based on the common genes between the two datasets. (Biancalani *et al.*, 2021; Cheng *et al.*, 2023; Li *et al.*, 2023). The methods can be broadly categorised in methods that map multiple cells from the scRNA-seq onto the SRT cells and those that only map one cell each (**Figure 0**). Additionally,

the methods compute the similarity between cells in different ways. Some measure similarity between cells individually computing similarity scores based on the transcription profile (e.g. Tangram, cytoSPACE) while others co-embed cells of each modality into one latent space and compute their distances in the latent space to find the most similar matches (e.g. MaxFuse, spaGE). For more details on the *modus operandi* of the methods used in this project, see section **2.10** and **Tab.4**.

A review published in 2022 presented a benchmark of the available tools to perform integration of scRNA-seq and SRT data. The study identifies Tangram and spaGE as the top performing methods to accurately map transcription counts from single-cell to spatial transcriptomics. Furthermore, the study found the mapping quality to be directly proportional to the sparsity of the spatial count matrix and that the log normalisation of spatial counts resulted in consistent decreased performance in each tool. Since 2022, more tools have been developed and published to perform this integration, such as MaxFuse and CytoSPACE (S. Chen *et al.*, 2023b; Vahid *et al.*, 2023). These methods, unlike Tangram and spaGE do not infer transcript counts for each gene in the scRNA-seq panel and cell in the SRT data, but provide a one-to-one matching index that assigns each SRT cell its most similar counterpart in the scRNA-seq panel. They leverage different strategies to compute the similarities between the cells in two modalities without inferring any missing values (**Figure 1, Tab.4**). Thus, their mapped transcript counts are directly sampled from the original scRNA-seq data and retain the sparsity of the original matrix.

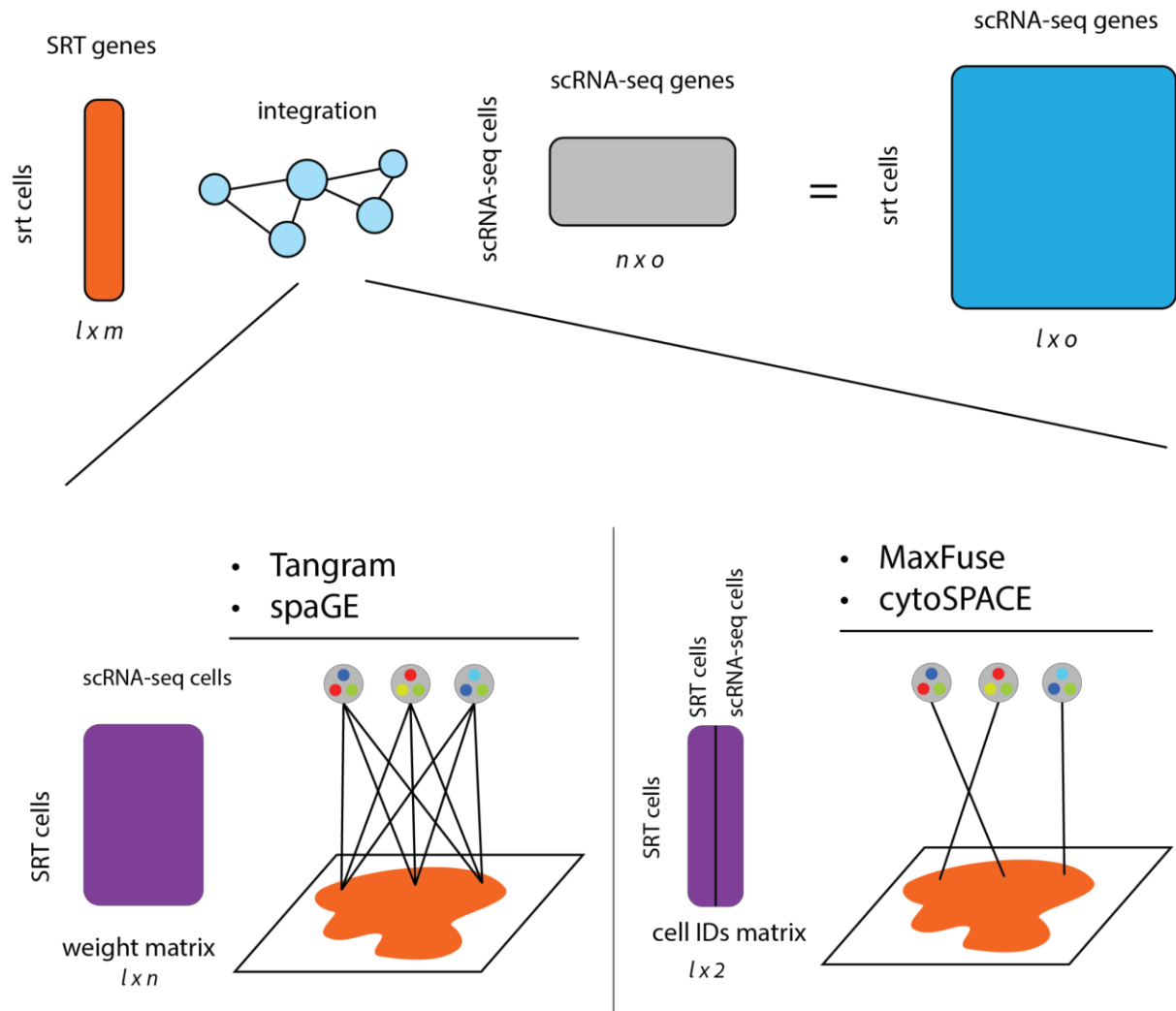


Figure 1. Benefits of scRNA-seq mapping onto SRT data and mapping strategies. 10x Xenium sample data contains tens to hundreds of thousands of cells but only transcript counts for 295 genes. scRNA-seq data contains thousands of cells per subtype and ca. transcript counts for 12,273 genes. By integrating the two datasets, we can achieve spatially resolved transcriptomics data for 12,273 cells. Some spaGE and Tangram map the scRNA-seq data by computing an optimised weight matrix to transfer information from multiple single cells to each Xenium cell, MaxFuse and CytoSPACE compute cell index pairs to assign each SRT cell the best possible match from the scRNA-seq (see **Tab.4, Methods**).

1.4 The goal of this project

In this study I seek to test these methods to build and validate a multimodal integration pipeline to project single-cell transcriptomics and epigenomics data onto SRT data. I used Neuroblastoma SRT data obtained by the 10x Xenium assay, which has already been

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successfully used in cancer research to reveal the composition of the tumour microenvironment in breast cancer carcinoma in an accurate and reproducible way (Janesick *et al.*, 2023).

This project was conducted in the following steps (**Figure 2**):

1. Select the best integration method(s) to map scRNA-seq as well as TF activity and gene activity onto the data
2. Validate the mapping of scRNA-seq, TF activity and gene activity data onto the SRT data
3. Leverage the spatially resolved multi-omics biological knowledge to characterise NB cells in select samples.

As previously discussed, scRNA-integration is performed to increase the gene coverage, in this case from 295 to ~12.273 genes. TF activity instead can offer more interpretable features than transcriptomics due to the high dimensionality of scRNA-seq data. It is a measure of a transcription factor's influence over the expression of its target genes in a cell and is a measure of both its transcriptional activity as well as the post-transcriptional regulation affecting its target genes' mRNA (Badia-i-Mompel *et al.*, 2022). Prior studies have leveraged the power of TF activity data computed from scRNA-seq and ATAC-seq data to investigate the role of TFs' in the evolution of cancer as well as a diagnostic tool for cancer subtype identification (Ulz *et al.*, 2019; Shiah *et al.*, 2023). The TF activity in this project was computed by integrating scRNA-seq data and scATAC-seq data, which measures the accessibility of chromatin regions (see **Methods**), but there is no consensus method and metric to quantify the activity of a transcription factor (Ma *et al.*, 2020).

Gene activity is exclusively derived from scATAC-seq data as well as prior knowledge of regulatory elements in the genome of interest (see section **2.2**) and is a measure of the accessibility of the gene to transcription, regardless of its expression or repression (Wendt *et al.*, 2014). Gene activity can be used on its own to e.g. find characteristic epigenetic signatures that match cell types or stratify tumour samples by prognosis, or can be integrated with other single-cell modalities to formulate mechanistic hypotheses for new gene regulatory mechanisms (Pliner *et al.*, 2018).

Mapping these data types spatially onto the SRT samples will help cancer researchers by opening up new doors to spatially characterise regulatory mechanisms in the cells' nuclei, e.g. by identifying new spatially resolved gene transcription and chromatin accessibility patterns.

, 2022).

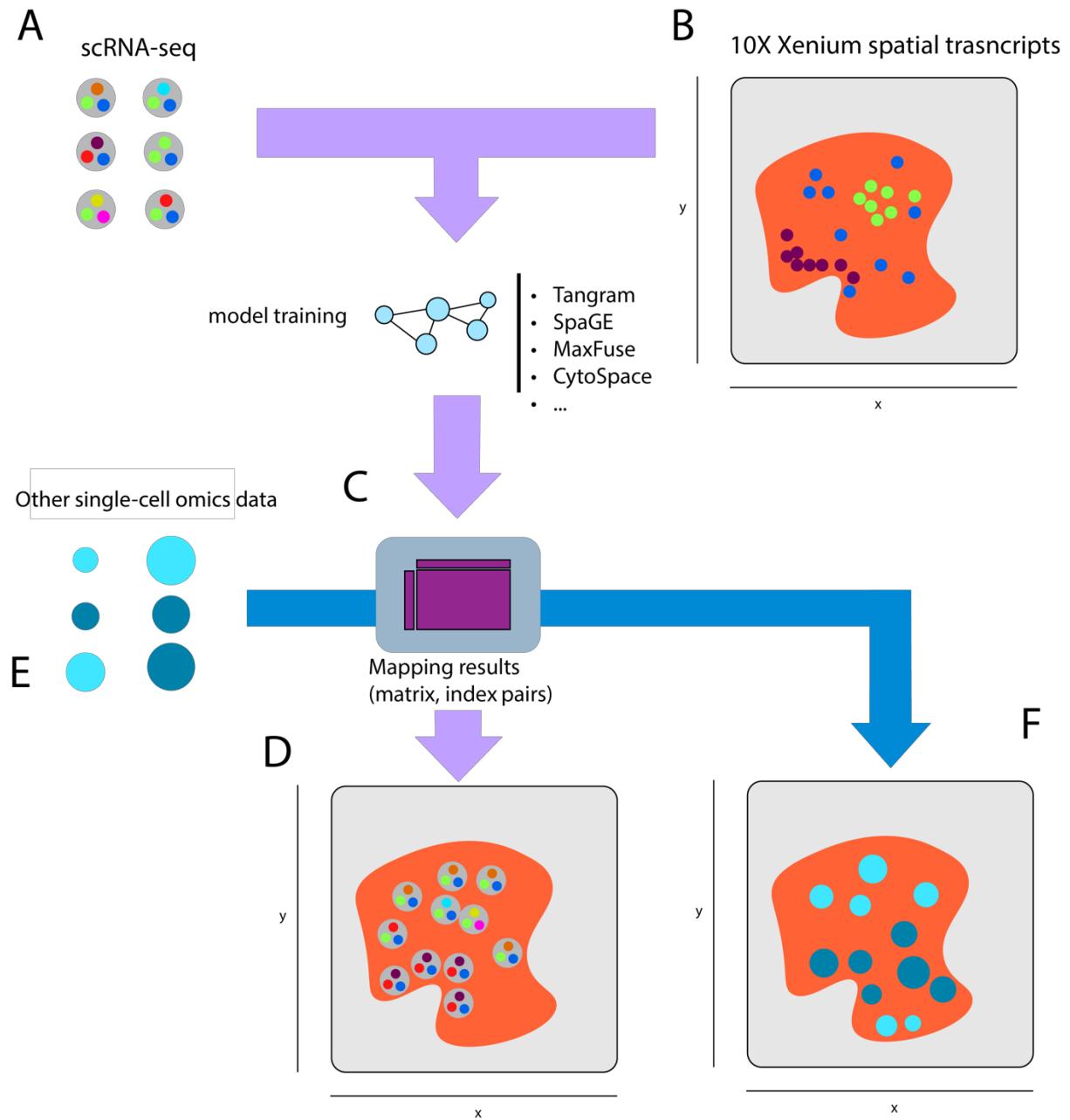


Figure 2. Overview of spatial integration of single-cell data. Single-cell RNA-seq data (A) and spatial transcriptomics data (B) can be integrated with several state-of-the-art tools (C) to map the whole transcriptome of single-cells onto the spatially resolved data, achieving spatially resolved whole transcriptome (D). Methods like Tangram, CytoSPACE and MaxFuse additionally return the mapping model results that can be used to map further single-cell modalities (E) such as single-cell gene activity and single-cell TF activity onto the spatially resolved data (F).

1.5 Neuroblastoma

Neuroblastoma (NB) is a paediatric cancer originating from the sympathoadrenal lineage of neural crest cells, and the most prevalent extracranial solid tumour among children and predominantly arises from the adrenal gland. It is characterised by a high degree of both inter- and intra-tumoral heterogeneity, which contributes to the varied prognosis and responses to therapy observed among patients. The prognosis for NB varies considerably: while children diagnosed before 18 months of age generally have a better chance of survival, many others face a median survival expectancy of about five years (Huang *et al.*, 2013; Boeva *et al.*, 2017).

1.5.1 Development and cell type composition in neuroblastoma

Neuroblastoma does not occur in adults because it is an “arrested development” type of cancer which can only occur in developmental stages of the embryo because it originates from the neural crest. The neural crest is a structure specific to the embryogenesis involved in the development of the peripheral nervous system including neurons, glial cells, melanocytes, Schwann cells (Cheung *et al.*, 2013).

The precursor cell type for NB is sympathoadrenal cells, which migrate from the ectoderm of the embryo’s ectoderm undergoing epithelial-mesenchymal transition to move to the dorsal side of the aorta. Healthy sympathoadrenal cells differentiate based on the activity of different transcription factors PHOX2A, PHOX2B, GATA2 and develop to form the adrenal glands, important endocrine structures located on top of the kidneys. (Cheung *et al.*, 2013).

Schwann cell precursors derived from this lineage can go on to further differentiate to Schwann cells as well as a bridge state from which they can either become chromaffin cells or neuroblasts. Neuroblasts are neuron precursors in a state of arrested development and are the primary malignant cell type of NB. (Zeineldin *et al.*)

1.5.2 Neuroblastoma heterogeneity

The most relevant oncogene is *MYCN*, a member of the MYC family of oncogenes and is amplified in 20-25 % of NB cases and associated with a high-risk disease profile and poor outcomes (Huang *et al.*, 2013). It influences several biological processes critical for tumorigenesis, including cell proliferation and apoptosis. Amplification *MYCN* usually results in relapse and resistance to chemotherapy, thus *MYCN* NB is generally considered to be the most lethal subtype (Norris *et al.*, 2012; Huang *et al.*, 2013). TERT and ALT NB are subtypes

characterised by the telomere repair enzyme TERT and the Alternative Lengthening of Telomeres (ALT). While patients diagnosed with these subtypes of NB have better outcomes, they are still lethal in most cases unlike the low-risk (LR) subtype of NB. LR NB has an average 90 % recovery rate as it is characterised by little to no MYCN expression (Monclair *et al.*, 2009; Huang *et al.*, 2013; Martínez-Pacheco *et al.*, 2023). The preprint “A spatial cell atlas of neuroblastoma reveals developmental, epigenetic and spatial axis of tumour heterogeneity” (Patel *et al.*, 2024) sought to build a NB atlas with 55 tumour samples from 51 patients across different ages, stages, gene marker expression and treatment stage. They present a classification of malignant cells in 3 subtypes characterised by expression profile and spatial heterogeneity. Malignant mesenchymal cells are the least immunogenic of the three, while adrenergic neuroblasts and proliferating sympathoblasts are more immunogenic and closer expression profile to the neural crest cells. The markers for these malignant subtypes are listed in **Tab.1**.

Table 1. NB malignant cell heterogeneity marker genes/TFs.

Malignant cell subtype	Gene markers	Marker TFs
mesenchymal (MES)	FN1, ACTA2, COL1A1, EGFR	ELK4, EGR3, MAFF, IRF1, IRF2, FLI1, RUNX2, FOSL1, FOSL2
adrenergic (ADRN)	NRG3, NTRK3, DBH, ERBB4, CHGB	HAND2, PHOX2A, PHOX2B, ISL2, GATA2
proliferating sympathoblasts (SYMP)	MKI67, TOP2A	

We can leverage the spatial mapping of these genes and transcription factors to confirm whether they are expressed/active in separate regions of the NB samples to verify if the internal neuroblast heterogeneity is preserved after the data integration.

2 Datasets and Methods

2.1 Spatially Resolved Transcriptomics (SRT) data

The spatial transcriptomics data was generated with the 10X Xenium assay by 10X Genomics. The assay detects transcripts through in-situ imaging and rolling circle amplification of padlock probes and can detect up to 312 select genes with an XY localization precision < 30 nm with a resolution of $0,2 \mu\text{m}/\text{pixel}$ (10x Genomics Xenium, 2024). Xenium allows measuring the gene expression of up to 312 genes. In this study, the Xenium panel was composed of 295 selected marker genes for NB, including neural differentiation markers (e.g. Neuropeptide Y), immune cell differentiation markers (e.g. *CD3*) and typical Hallmark genes for NB (e.g. *MYC*). The data was delivered in two batches of multiple samples each (**Tab.2**, **Tab.S4**).

Table 2. 10x Xenium NB samples statistics after preprocessing and nuclei segmentation (see following sections). Some biological samples were extracted from the sample patient (e.g. sample 217) and thus share the same sample Id while having different cell Ids.

Plate	sample Id	subtype	Number Cells	of Number Genes	of Sparsity	Median Counts per cell
0010562	576	LR	217023	294	0,8412	89
0010562	639	TERT	36988	245	0,8846	40
0010562	808	MYCN	191903	210	0,7607	121
0010685	217	LR	233044	257	0,7987	125
0010685	325	MYCN	84695	247	0,7953	116
0010685	827	ALT	210792	212	0,7667	128
0018281	833	ALT	66132	295	0,815	128
0018319	526	ALT	65489	295	0,8276	121
0018252	476	ALT	140507	295	0,824	116
0018319	644	MYCN	58491	295	0,8217	150
0018319	596	TERT	24581	295	0,8146	120
0018281	398	LR	134830	295	0,8034	160

0018249	61	ALT	42208	295	0,8128	134
0018281	369	TERT	138242	295	0,8344	107
0018249	395	LR	194631	295	0,811	129
0018249	720	TERT	54248	295	0,8362	99
0018252	614	MYCN	21749	295	0,8329	102
0018252	827	ALT	249665	295	0,8017	179
0018319	217	LR	123464	295	0,7973	169
0018249	325	MYCN	73384	295	0,7977	165

Unlike single-cell transcriptomics data, the SRT samples must be kept separate as their spatial organisation cannot be summarised in a tabular format. Instead, the spatial distribution of feature values remains distinct for each sample. Thus, the thesis focuses on a small number of these samples to maintain the results simple to review.

Descriptive statistics about each sample are noted in Tab S3. Cell type composition of each sample can be found in **Figure S1**.

2.2 Single-cell RNA-seq, ATAC-seq and TF activity data

The single-cell datasets to map onto the spatial data was provided by Dr. Robin Droit. There is partial overlap between the patients of origin of the spatial samples and the single-cell samples, but the single-cell observations were merged by subtype to have larger single-cell datasets, regardless of the patient of origin. This was done for two reasons; first, there are two orders of magnitude between the number of single-cell observations and spatial observations per sample, merging datasets of the same subtype can help bridge this gap and reduce the number of repeated observations in the case of one-to-one mapping tools like MaxFuse or CytoSPACE. Second, not every project operates with paired samples across modalities and a universally applicable validation was sought in this project.

scRNA-seq. The scRNA-seq dataset was pre-processed with the R packages Seurat and Signac (Stuart *et al.*, 2021; Ahlmann-Eltze *et al.*, 2023) and annotated with CellAssign using the cell type marker genes in **Tab.1**. The scRNA-seq data contains 12.273 genes and there is an overlap of 164 genes between the scRNA-seq data and the SRT data.

TF activity. The single-cell Transcription activity scores were computed with DecoupleR (Badia-i-Mompel *et al.*, 2022) for 122 TFs using a GRN generated by CellOracle (Farrell, 2023).

Gene activity. The single-cell gene activities were derived from scATAC-seq data using the method Cicero (Pliner *et al.*, 2018). The data contains activity scores for 4191 genes.

DecoupleR and Cicero return a binned distribution of transcription factor activity and gene activity scores respectively, many similar single-cells are assigned the same scores. When these data types are mapped by MaxFuse with its one-to-one mapping, it is impossible to compute a fully connected cell neighbourhood network. To overcome this issue, additive noise generated from a random zero-centred normal was added to this data to allow the computation of a fully connected graph between cells to visualise their separation with UMAP. The noise was added to each feature (TF/gene) separately, the standard deviation was selected to be proportional to the feature's own. This noise addition was used exclusively for the generation of plots for MaxFuse results and was not included in any testing or other analysis involving said data.

2.3 Spatial Data preprocessing

The spatial transcriptomics samples were processed by removing cells with non-zero counts for less than 5 % of the gene panel (rounded down) or 14 genes. Additionally, genes were excluded if their transcripts were found in fewer than 1% of cells. This stringent filtering of cells, as opposed to genes, was decided due to the limited number of genes and the substantially large number of cells per sample. Additionally, this order was chosen to avoid removing marker genes of underrepresented cell types. This was done to only keep easy cells as well to reduce the sparsity of the spatial transcript counts matrix, defined as the fraction of entries in the matrix equal to 0.

2.4 Preprocessing of single-cell transcription counts

The scRNA-seq and SRT datasets were normalised to account for variable sampling efficiency and log-transformed (with pseudo counts) to make subsequent statistical methods such as UMAP applicable. Although more sophisticated preprocessing methods exist, they have not outperformed log-normalisation in a benchmark study on single-cell RNA-seq preprocessing (Ahlmann-Eltze *et al.*, 2023). According to a study attempting to benchmark all available methods for scRNA-seq integration with SRT data, using raw counts for the SRT data

invariably yielded better results than normalised counts. As such, the raw matrix counts for each sample were saved and used for the integration step instead of the log normalised values (Li *et al.*, 2023).

2.5 UMAP

Uniform Manifold Approximation and Projection (UMAP) was used to reduce the dimensionality of our gene expression data. UMAP is a method to project high dimensional data to 2 dimensions that preserves the local and global structure of the data, allowing for effective visualisation and clustering. UMAP was run with the `scanpy.pp.neighbors()` and `scanpy.tl.umap()` functions using 10 neighbours per cell and a minimal distance of 0,25 (McInnes *et al.*, 2020).

2.6 Leiden clustering

Cells were clustered using the Leiden algorithm, which is an advanced method for detecting communities in large networks. This algorithm iteratively refines clusters by optimising the modularity, ensuring high-quality, stable partitions. Leiden clustering has proven to guarantee fully connected communities of observations unlike its predecessor, the Louvain Clustering (Traag *et al.*, 2019). Leiden clustering was run with the `scanpy.tl.leiden()` function with a resolution parameter of 1.

2.7 Shannon Entropy

The Shannon Entropy was used to quantify the separation of cell types in the data by computing the entropy of Leiden clusters in relation to their cell type composition. The Shannon Entropy is defined as:

$$H(X) = - \sum_{x \in X}^x p(x) * \log(p(x)) \quad (1)$$

Where x is a cell type out of all cell types X and $p(x)$ are the probability of observing said cell. The term $p(x)$ drives the term within the sum towards 0 if p is close to 0, while the term $\log(p(x))$ drives it towards 0 when $p(x)$ is close to 1. Because of this, the value H is 0 when one value $p(x)$ for x in X is equal to 1 and all other $p(x)$ values in X are zero. H is highest when $p(x)$ is

equal to $1/n$ where n is the number of cell types, thus all cell types are equally represented in said cluster. As such, the better a method preserves the cell type homogeneity within clusters, the lower the entropy value of each cluster is.

To compute the total entropy for a sample, the entropies of each cluster were summed up and weighted by the size of the Leiden cluster (Feutrill *et al.*, 2021) according to the following equation 2.

$$H_{tot} = \sum_{l \in L} H(s) * \frac{n(l)}{N} \quad (2)$$

Where l is a Leiden cluster from the total set of Leiden clusters L , $n(l)$ is the size of said cluster and N is the total number of cells in the sample. This method is a simplified adaptation of a proposed method for assessing homogeneity of single-cell data groups (Liu *et al.*, 2020).

To test the difference in entropy distributions between different cell segmentation types in section 3.3, a Wilcoxon signed-rank test was used as it allows to compare paired distribution of values (Rey *et al.*, 2011). The Wilcoxon signed-rank test tests the probability that two distributions of paired observations are extracted from the same distribution. Like the Wilcoxon rank sum test, it does not assume that the underlying distribution of the data is normal.

2.8 Differential Gene Expression Analysis

To identify the most discriminative genes for a given cell group, differential gene expression (DGE) analysis was conducted using the scanpy function `scanpy.tools.rank_genes_groups()`, employing Wilcoxon rank-sum testing. The same method was used to rank genes and TFs by their activities. The Wilcoxon rank sum test was chosen because it does not require the assumption of normality in the data (Rey *et al.*, 2011). Because TF and gene activities are not normally distributed, the Wilcoxon rank sum test can be equally applied to each data type in a comparable manner.

Benjamini-Hochberg correction was chosen to adjust the p values (Haynes, 2013). The significance threshold for the p values was set at 5×10^{-5} , as Wilcoxon rank sum testing produces lower p values than t -tests. Because occasionally p values get low enough to be rounded down to 0 by scanpy, these p values were set equal to the lowest adjusted p value

returned in the multiple tests. The significance threshold for log fold change (lfc) was set at 1 for positive lfc and -1 for negative lfc.

2.9 CellAssign

The scRNA-seq observations' cell type was determined by my colleague Dr. Robin Droit with CellAssign, a cell annotation method from the scvi-tools that uses marker gene expression to annotate cells without a reference population. CellAssign was initially used on the SRT data as well, but when the segmentation method was decided to be nuclei segmentation, the number of counts was too low for CellAssign to run and the cells were annotated using label transfer from the scRNA-seq data using CytoSPACE. The marker genes per cell type used by Dr. Droit to annotate the scRNA-seq data are listed in **Tab.S2**, to improve the interpretability of the data for this project, the cell types were condensed into fewer cell type labels, which were used in this analysis as well.

2.10 SRT and scRNA-seq integration methods

2.10.1 CytoSPACE

CytoSPACE maps observations between the single-cell and spatial transcriptomics data with a one-to-one matching algorithm that seeks to minimise the distance between the gene expression profiles of each cell pair (Vahid *et al.*, 2023). CytoSPACE was ultimately only used to transfer cell type annotations from the scRNA-seq data obtained with CellAssign onto the spatial transcriptomics data.

2.10.2 Tangram

Tangram is a tool designed to map scRNA-seq data onto spatial transcriptomics data. Tangram computes a similarity score between each pair of observations in the two sets of data based on shared features and returns a mapping matrix object of size $m*n$ where m is the number of spatial observations and n is the number of single-cell observations.

The matrix is trained with a loss function that maximises the similarities between single-cell and spatial observations as well as the similarity of gene expression levels across the entire sample. Each single-cell observation is assigned a similarity score for each spatial observation, weights are then normalised with a softmax function to add up to one to represent a probability

vector. Tangram projects gene expression by matrix multiplication with multiple observations with the learned mapping matrix (Biancalani *et al.*, 2021).

Tangram was run in ‘single-cell’ mode using transcript count data in both the SRT and scRNA-seq data and normalisation and log-transformation was performed after the inference. To map TF activity and gene activity with Tangram, the mapping matrix computed on the scRNA-seq integration was normalised so that the sum of each weight would add up to 1 for each spatial observation instead of single-cell observation. This step was performed to better represent the original single-cell distribution of values and because the TF and gene activity scores obtained were already normalised and scaled, a second normalisation on them was deemed questionable.

2.10.3 MaxFuse

MaxFuse is a tool designed to accurately integrate weakly linked modalities. It operates by embedding observations in a latent space using shared features through a technique known as manifold alignment. This technique projects the data from different modalities into a common space where similar features are aligned. The process iteratively smooths the embedding while matching cells, refining the alignment and discarding poor matches in each cycle. This iterative refinement enhances both robustness and accuracy. MaxFuse is modality-agnostic, relying on the principle that the chosen modalities have some degree of linkage (Biancalani *et al.*, 2021).

MaxFuse can be run with separate hyperparameters for the two modalities. The main hyperparameters are `svd_components` (single variable components), the number of principal components used to represent the observations in the latent space, and `n_neighbors`, the number of neighbours per cell when building the neighbourhood graph. Because we have a high number of cells in the SRT data and want to minimise the noise produced by cell-cell contamination, the `svd_components` hyperparameter was set lower for SRT than scRNA-seq data while `n_neighbors` was set higher (**Tab.4**).

Table 3. MaxFuse hyperparameters tuning

	SRT data	scRNA-seq
<code>svd_components</code>	15	20
<code>n_neighbors</code>	25	15

Mapping TF and gene activity with MaxFuse did not require any further steps, the same matching of indices was used to transfer each modality. As the TF and gene activity data was already processed, no further normalisation or transformation was performed.

2.10.4 spaGE

spaGE (Spatial Gene Enhancement) combines single-cell RNA sequencing (scRNA-seq) data with spatial transcriptomics data by projecting both datasets into a common spaGE of $m \times n$ where m is the total number of cells and n are all the shared genes between the two panels. spaGE embeds the cells of both modalities using principal component analysis and single variance decomposition to align and cluster the observation of the two modalities. The gene expression of the scRNA-seq observations is mapped onto the spatial observations as a weighted average based on the distance of the observations in the latent space (Abdelaal *et al.*, 2020). spaGE was run according to its provided tutorial on GitHub and with a number of principal components equal to half of the number of genes as recommended in the documentation.

Table 4. Machine learning based methods for integrating scRNA-seq methods.

Method	Mapping strategy	Features	Reference
Tangram	- computes softmax weights for each single-cell observation for each spatial observation - returns weight matrix, gene expression inferred via linear combination	- can construct pseudo cells to represent each cell type, expedite the mapping process at the cost of accuracy	(Biancalani <i>et al.</i> , 2021)
spaGE	- co-embedding and clustering of cells from SRT and scRNA-seq data - infers gene expression of SRT cells by weighted average of scRNA-seq cells in the same cluster	- does not provide the user with latent space clustering results, hard to use results for different modalities	(Abdelaal <i>et al.</i> , 2020)
CytoSPACE	- single observation assignment based on similar profile	- can be provided with estimation of spatial cell type proportions	(Vahid <i>et al.</i> , 2023)

MaxFuse	- iterative co-embedding and clustering of cells from both datasets - returns one-to-one matches between cells of the two datasets	- any two linked -omics data types (e.g. transcriptomics and proteomics) - independent setting of hyperparameters for each modality	(S. Chen <i>et al.</i> , 2023)
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2.11 Leave-one-out cross validation

Leave-one-out cross-validation (LOOCV) was implemented to evaluate the quality of gene expression inference. One gene was excluded from the training dataset, and its mapping quality was measured using the Pearson correlation between the observed values and the predicted values. This process was repeated for 15 genes from the list of overlapping genes ranked by their sparsity in the spatial dataset, sampling 5 from the top, middle, and bottom of the ranking respectively. The results were averaged for each group of 5 genes for better visualisation.

2.12 Hypergeometric testing

A hypergeometric test is a statistical test that utilises the hypergeometric distribution to determine the significance of observed successes in a sample drawn from a finite population (Rivals *et al.*, 2007). To evaluate the validity of the usage of TF targets from the GRN, hypergeometric tests were conducted to test if targets of upregulated TFs were differentially overexpressed and/or if downregulated TFs were differentially underexpressed. The hypergeometric test was conducted with the `scipy.stats.hypergeom.sf()` function.

2.13 enrichR

enrichR is a statistical tool to compute the enrichment of biological processes. The cells were characterised by running enrichR on differentially expressed genes and TFs based on differential gene expression analysis and TF activity analysis (see **Differential gene expression analysis** previously in this section). The p value threshold for each was set at $10e-5$. For gene expression, the log fold change was used as the metric with a threshold for the absolute $lfc = 1$ and genes with an $lfc > 6$ were discarded as artefactual product of the sparsity of the matrix. For TF activity the difference in gene expression was computed by subtraction and the threshold for the absolute difference was also set as 1. Log fold change could not be used as a metric for TF activity due to the scaling of the TF activity scores.

Materials and Methods

Because there are ca. 10^4 measured genes but only 122 TFs, to balance the importance of TF activity and gene expression, the information of the differential Transcription Factors was augmented by using the inferred GRN (see section 2.2) to find downstream targets of the TFs. (Chen *et al.*, 2013; Kuleshov *et al.*, 2016). The genes were computed from the GRN instead of using enrichR's transcription factor targets library because the GRN was inferred from the single-cell data by Dr. Robin Droit and is more specific to this dataset. The differentially active TFs and their targets were combined with the differentially expressed genes to obtain a more comprehensive profile of the cells' state.

3 Results

3.1 Preliminary analysis of 10x Xenium SRT Neuroblastoma samples

3.1.1 Spatially resolved localization of transcripts

I began by visualising the gene expression in SRT samples, focusing on important marker genes of NB samples, for example, *NPY* for neuroblasts, *PTPRC* for immune cells and *PDGFRB* for mesenchymal cells (**Figure 2**). These genes were selected for their low sparsity within their respective cell type. I selected sample 576, a low-risk stage 1 NB sample, and 808, a MYCN-amplified stage 4 NB sample, to demonstrate the results of scRNA-seq integration methods. The gene expression patterns are spatially heterogeneous: in sample 576, the three genes were expressed in distinct regions consistent with the sample's cell type composition. In contrast, sample 808 was predominantly composed of neuroblasts, *PTPRC* and *PDGFRB* are expressed in very small groups of cells that were not randomly scattered but located in spatially distinct regions.

These differences in gene marker expression are corroborated by cell type annotation with CytoSPACE which shows notable compositional differences between the samples: sample 576 was highly heterogeneous as indicated by the marker gene expression, featuring distinct regions characterised by neuroblasts, Mesenchymal cells, and immune cells (**Figure 3A, S1**). Conversely, sample 808 is almost entirely composed of neuroblasts with occasional interspersed immune cells and a small mesenchymal region at the bottom of the spatial plot.

Results

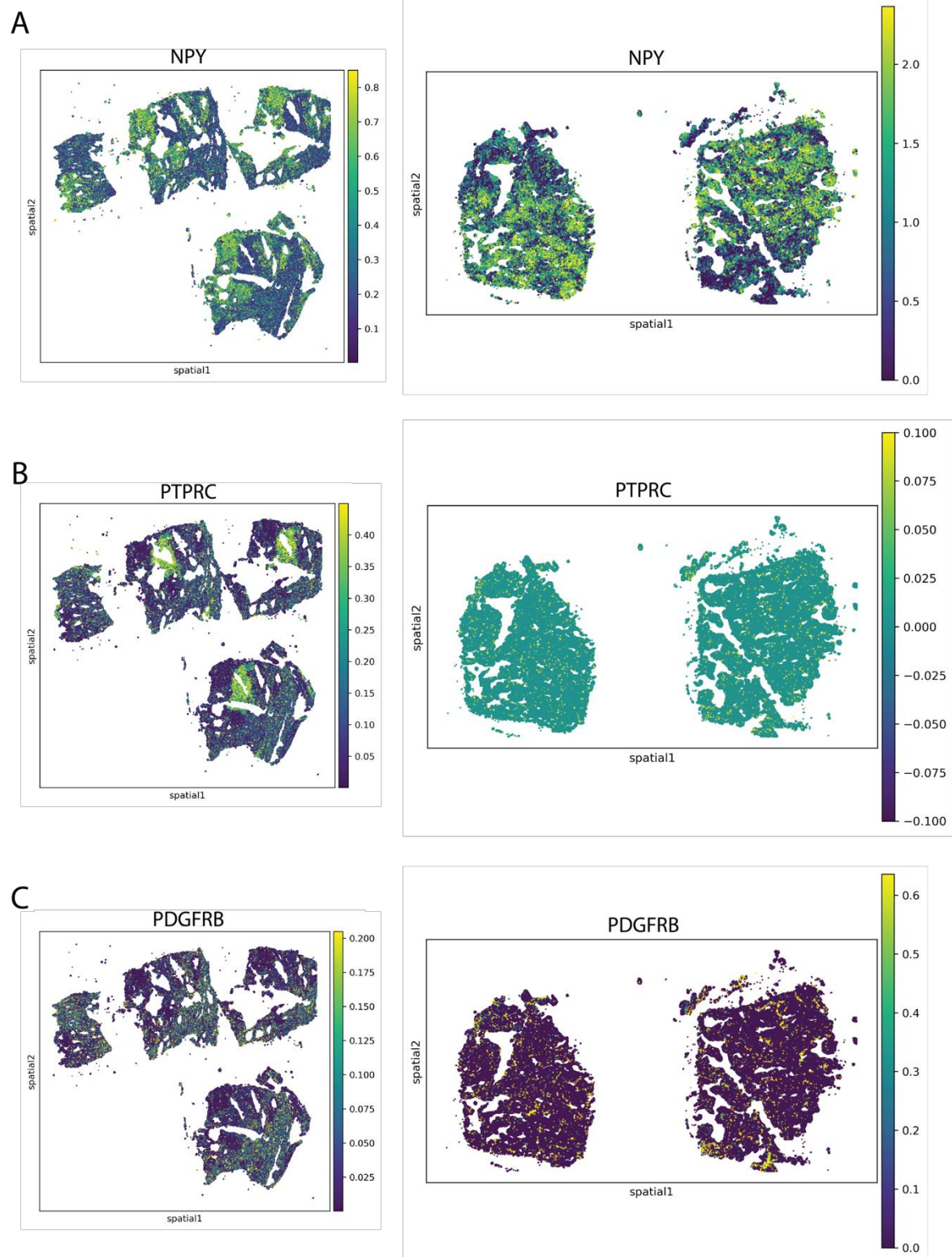


Figure 3. Spatial expression pattern of marker genes in sample 576 (left) and 808 (right). Spatial scatterplots of log-normalised spatial transcript count distribution across the samples 576 and 808. NPY (A), or neuropeptide Y is a marker for neuroblasts, PTPRC (B) is an immune cell marker and PDGFRB (C) is a mesenchymal marker. The different distribution of transcripts between the samples mirrors their different cell type composition (**Figure S1**).

3.1.2 Separation of cell types in low dimensional representation and spatial identity of Leiden clusters

The UMAP visualisation of the spatial data shows that cell clusters appear connected (**Figure 4B**) by transitional cells. These intermediate “transition” cells are normal occurrence in SRT data and indicate the presence of cells with genes corresponding to multiple cell types. Since Leiden clustering groups cells in the same connecting lines of cells together (**Figure 4C**), we can verify whether these mixed groups exhibit internal heterogeneity. Spatial visualisation of the Leiden clustering of cells in samples 576 (LR) and 808 (MYCN) (**Figure 4D**) indicates that these intermediate clusters can have specific spatial localizations (e.g., cluster 9 of sample 576). This suggests that the mixed groups do not arise independently from the spatial localization of their constituent cells. Instead, because of the propensity for cross-contamination of RNA transcripts in imaging-based SRT data, it is likely that their cells have been wrongfully assigned transcripts from a nearby cell of a different type (see **Methods**).

Results

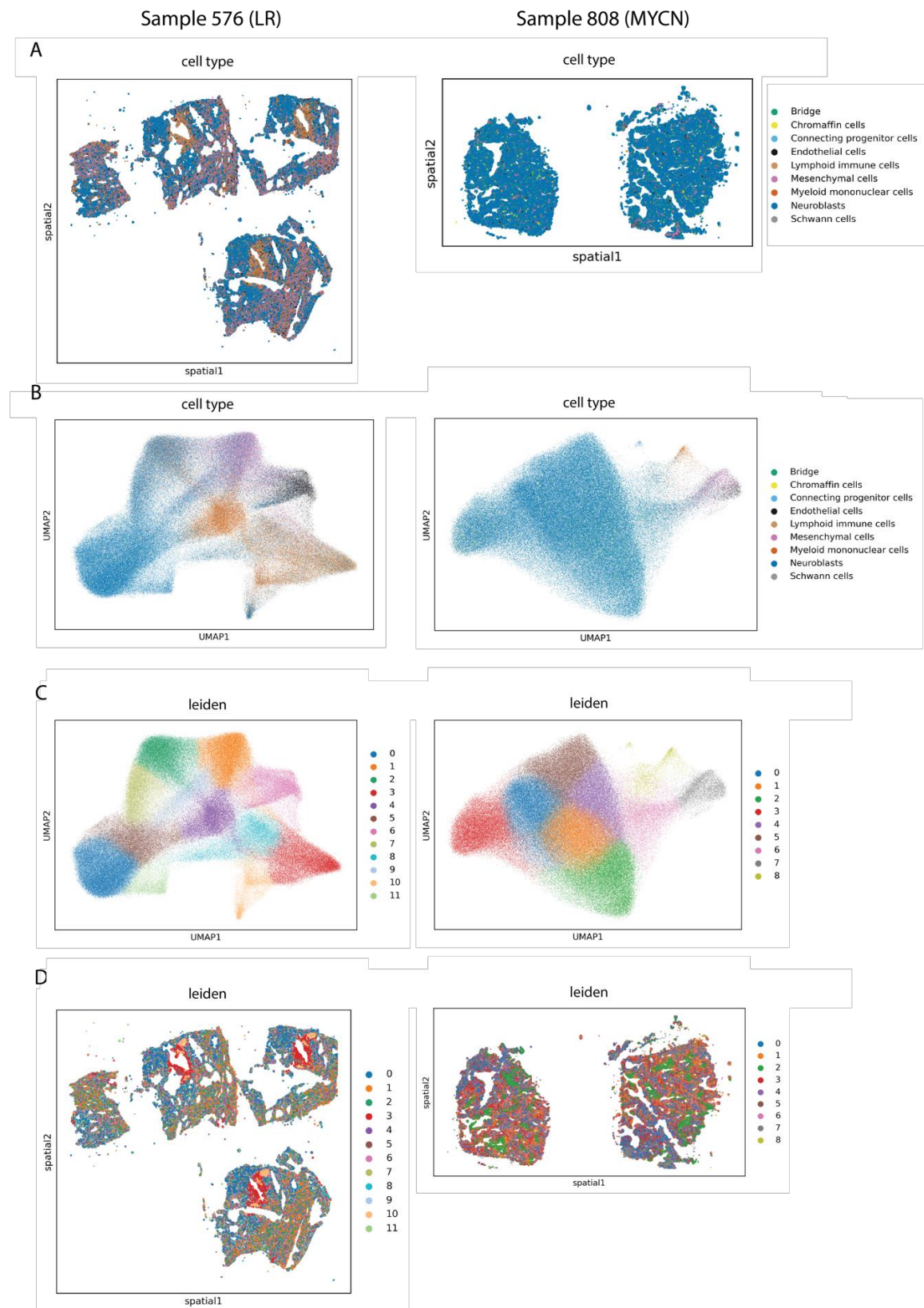


Figure 4. Sample 576 and 808 composition and clusterization. Spatial scatter plot of CytoSPACE cell type annotation (A) and UMAP projection of SRT data annotated by cell type (B) and Leiden clustering (C). Leiden cluster mapping onto spatial scatterplot (D). UMAP projection and Leiden clusterization are computed based on log-normalised counts spatial from the spatial transcript matrix.

3.2 Contrast of cell segmentation and nuclei segmentation

Because the Xenium assay method for cell segmentation infers cell boundaries without using cell membrane staining (see **Materials and Methods**), I attempted to mitigate contamination effects by considering only transcripts from the original Xenium data that overlap with the nucleus of cells. This information is provided by the Xenium assay results. This segmentation reduces the number of transcripts per cell but will decrease the risk of cross-cell contamination. The chosen criteria to evaluate the benefits of this procedure are to count the number of cells from the same cell-type in the neighbourhood graph of each sample as well as comparing the Shannon entropy change before and after nuclei segmentation of transcripts. The number of neighbours of the same cell type are averaged and normalised by the size of each Leiden cluster that they are in.

Nuclei segmentation improves the average cell neighbourhood homogeneity in samples with low counts and leads to a decrease in Leiden cluster entropy in all samples (**Figure 5**). For each sample, I computed their cells' average number of neighbours of the same type in their neighbourhood graph (**Figure 5A**). Sample 639 benefits the most from the nuclei segmentation, followed by samples 576 and 325. Notably, these are the samples with the lowest median transcripts per cell. Only in sample 808 does the nuclei segmentation led to a comparatively small decrease in cell neighbourhood homogeneity. This result suggests that nuclei filtering may disproportionately benefit samples with low counts and return similar results in the others.

Furthermore, weighted Shannon entropy sum instead measures the cell type homogeneity within each cluster significantly lower across the NB sample panel (**Figure 5B**). The loss of entropy also benefits sample disproportionately more than the other 5 shown here.

These results suggest that nuclei segmentation has a positive effect on the quality of the cell type clustering in the data as it reduces cross-cell contamination of transcripts. These results were corroborated by comparing the UMAP projections of sample 576 before (**Figure 4B**) and after applying nuclei segmentation. The projection after nuclei segmentation leads to greater separation between cell types, it removes a central group of cells and leads to thinner connecting cells between groups (**Figure 5C**). Similarly to the previous metric, the UMAP projection of sample 808 does show any visible improvement.

Results

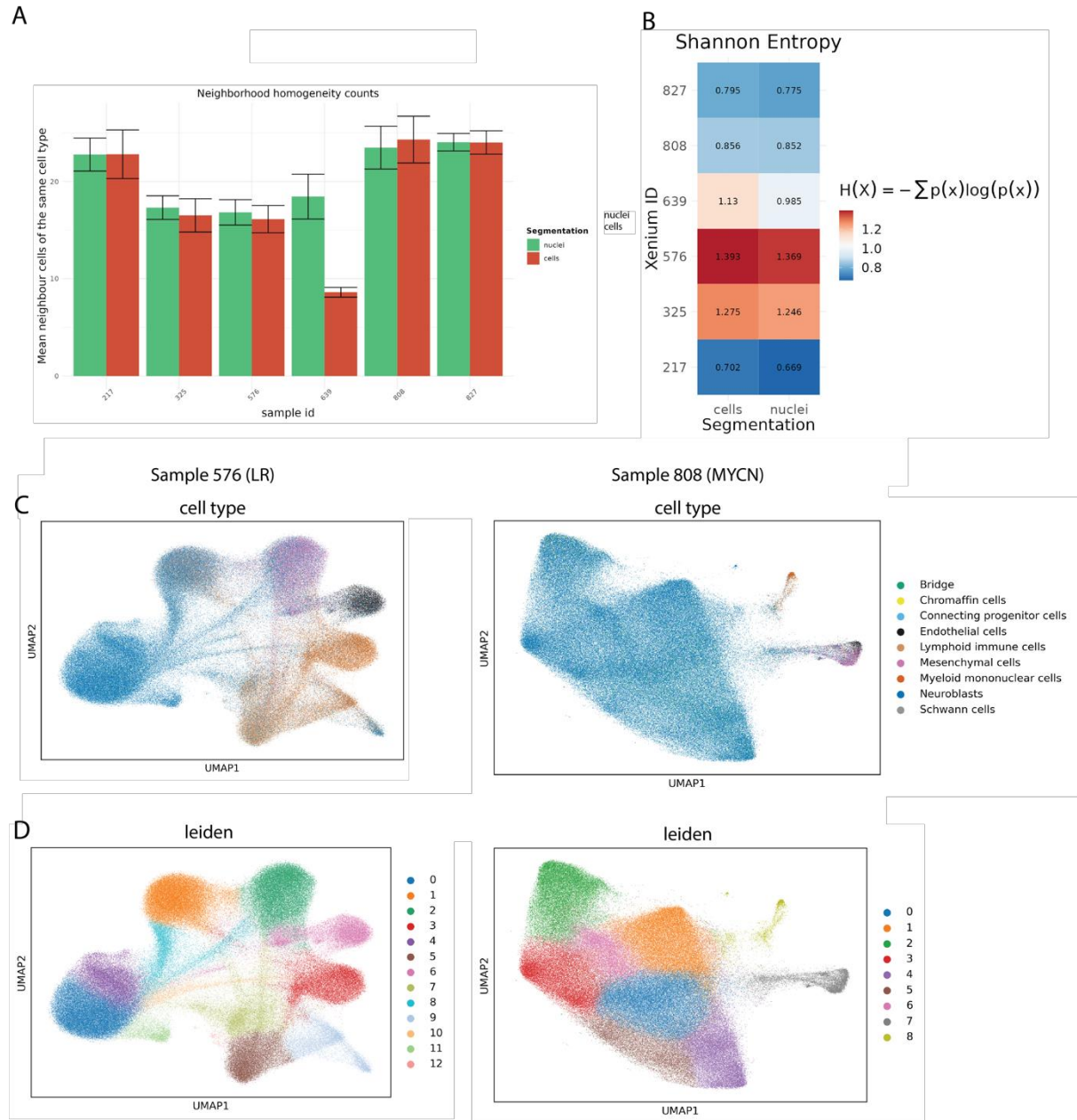


Figure 5. Nuclei segmentation effect on cell separation. (A) Bar plots illustrate the average number of cells of neighbours of the same cell type per cell before using cell (green) and nuclei (red) segmentation. (B) Shannon entropy of Leiden clusters regarding cell type composition is compared between nuclei and cell segmentation. Nuclei segmentation shows reduced entropy across all samples (here only 6 are presented for simplicity). Wilcoxon signed rank test: $n = 20$, $Z\text{-value} = -3.0594$, $p\text{-value} = 0.0022$. The UMAP projection of SRT data after applying nuclei segmentation to the transcript count coloured by (C) cell type and (D) Leiden cluster assignment.

3.3 Investigation of scRNA-seq data's suitability for mapping

3.3.1 Separation of scRNA-seq data by neuroblastoma subtype and cell type

In the next step, I compared different methods to map external scRNA-seq datasets into SRT data. I first visualised the clustering of cells in the scRNA-seq data I mapped onto the SRT data to visualise how the cells are clustered on a UMAP. The cells cluster both by subtype and by cell type. (**Figure 6A, B**). To verify that cells of the same subtype don't cluster by subtype independently of cell type, I compute the composition of Leiden clusters in terms of subtype and cell type next. The clusters are largely homogenous (**Figure 6C, D**) both in terms of cell and NB type. Only cluster 21, is composed evenly of neuroblasts, mesenchymal cells and lymphoid immune cells. TERT and ALT tumours are grouped in the same clusters, which is consistent as they are both defined by constant activation of telomere repair mechanisms.

3.3.2 Count matrices' similarity and sparsity

Based on the findings of the benchmarking review on SRT and scRNA-seq integration methods (Bin Li *et al.*, 2022), I compared the counts per cell distributions between the two data types, because the spatial and single-cell counts must have similar expression levels and it is suggested that single-cell spatial integration tools work better on counts rather than log-normalised values. The count distributions, normalised by the total number of genes in each dataset, show gene expression levels similar to the scRNA-seq data with similar modes of counts per cell per gene (**Figure 7A, B**). The nuclei segmentation values are predictably lower than cell segmentation counts. Because we are dealing with sparse arrays, I compare the sparsity of the data with the benchmark's estimation (Bin Li *et al.*, 2022) of method performance based on sparsity (**Tab.5**). Our samples score poorly, with fractions of 0,15 to 0,25 of non-zero entries in the spatial count matrix and according to the benchmark, we can expect a Pearson correlation of 0,2 to 0,4 between measured and inferred transcript counts.

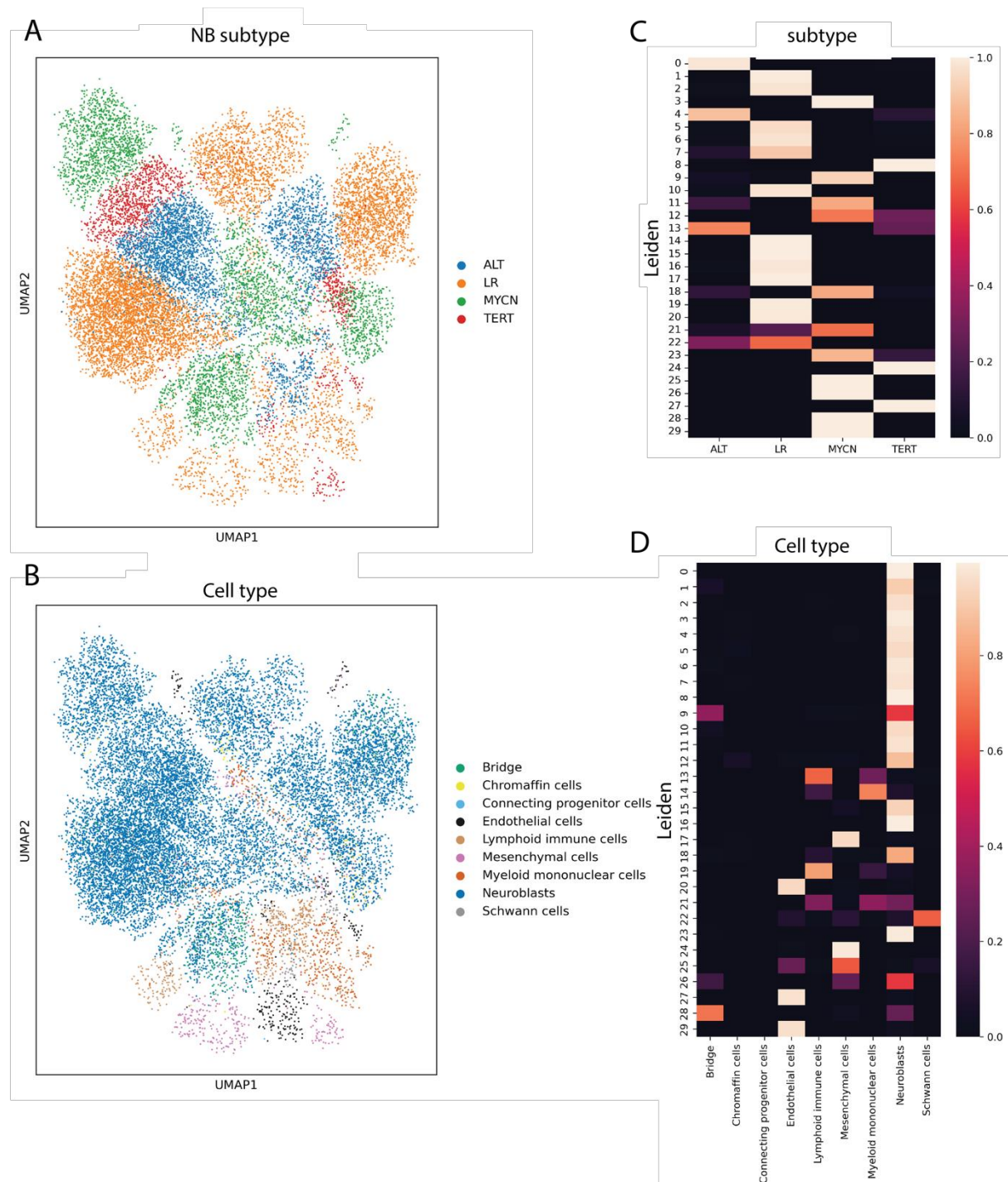
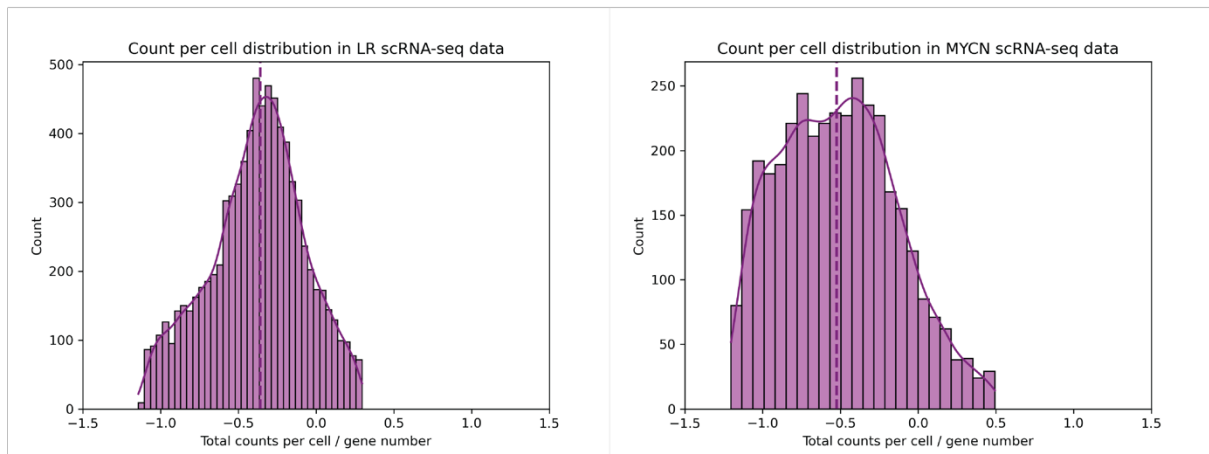


Figure 6. Separation and composition of scRNA-seq data. (A) UMAP projection of the entire scRNA-seq dataset computed on log-normalised counts and coloured by subtype and (B) cell type. The composition of Leiden clusters by subtype (C) and cell type (D) computed as a fraction of the entire cluster size.

Results

A



B

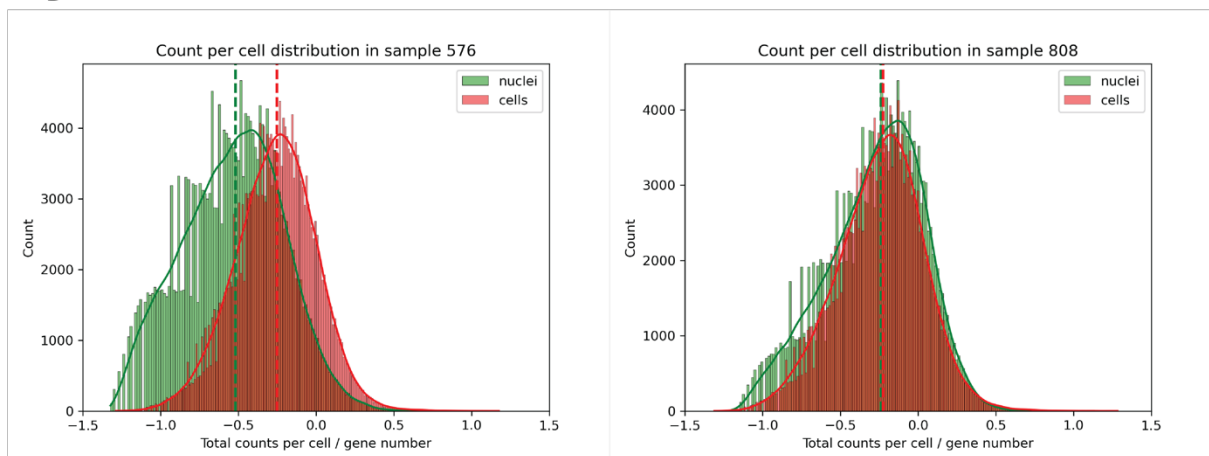


Figure 7. Total counts per cell SRT vs scRNA-seq data. Histograms of total counts per cell normalised by the total number of genes of the scRNA-seq data (A) for LR and MYCN cells. Histograms of the SRT count before (red) and after (green) nuclei segmentation (B) for sample 576 (left) and 808 (right). Median values are indicated by the dashed line. The x-axis is log transformed.

Table 5. SRT count matrix sparsity per sample.

Xenium Id	1 - sparsity
217	0,201
325	0,205
576	0,159
639	0,115
808	0,239
827	0,233

3.4 Evaluation of scRNA-seq mapping performance

3.4.1 Leave-one-out cross-validation (LOOCV)

I selected methods to map the scRNA-seq data onto the spatial transcriptomics data based on the mapping strategy (see **Tab.2**), their features beyond gene expression mapping, and their performance in leave-one-out cross-validation (LOOCV) (**Figure 8, Figure S2**). In terms of features, CytoSPACE, MaxFuse, and Tangram possess an essential feature for this project that spaGE lacks: they return a mapping object that can transfer other modalities onto the spatial data. CytoSPACE and MaxFuse provide a one-to-one matching of cell indices of the two modalities, while Tangram returns its mapping matrix with cell probabilities. This does not prevent spaGE from being selected to map scRNA-seq, but it is convenient to limit the number of methods employed by the pipeline.

I ran LOOCV on 15 genes per sample, as described in the section **2.10**. As previously explained, we have an expected average Pearson correlation between measured and imputed gene expression of 0,2 - 0,4 (Bin Li *et al.*, 2022).

CytoSPACE and MaxFuse performed as expected (**Figure 8A, B**) based on benchmarking metrics. Furthermore, their one-to-one mapping strategy returns a sparse matrix effectively doubling the number of 0's in the gene expression vectors used to compute the Pearson correlation compared to the former two methods. As such, they might be expected to score below the expected correlation coefficients of methods like Tangram and spaGE.

spaGE and Tangram instead exceeded expectations (**Figure 8C, D**). In most samples, the top 5 and middle 5 genes reach Pearson correlation coefficients above 0,5, while only the bottom 5 genes score close to the expected values. The results may exceed expectations due to the fact that some of the scRNA-seq data correspond to the same samples as the ones profiles in the SRT.

Results

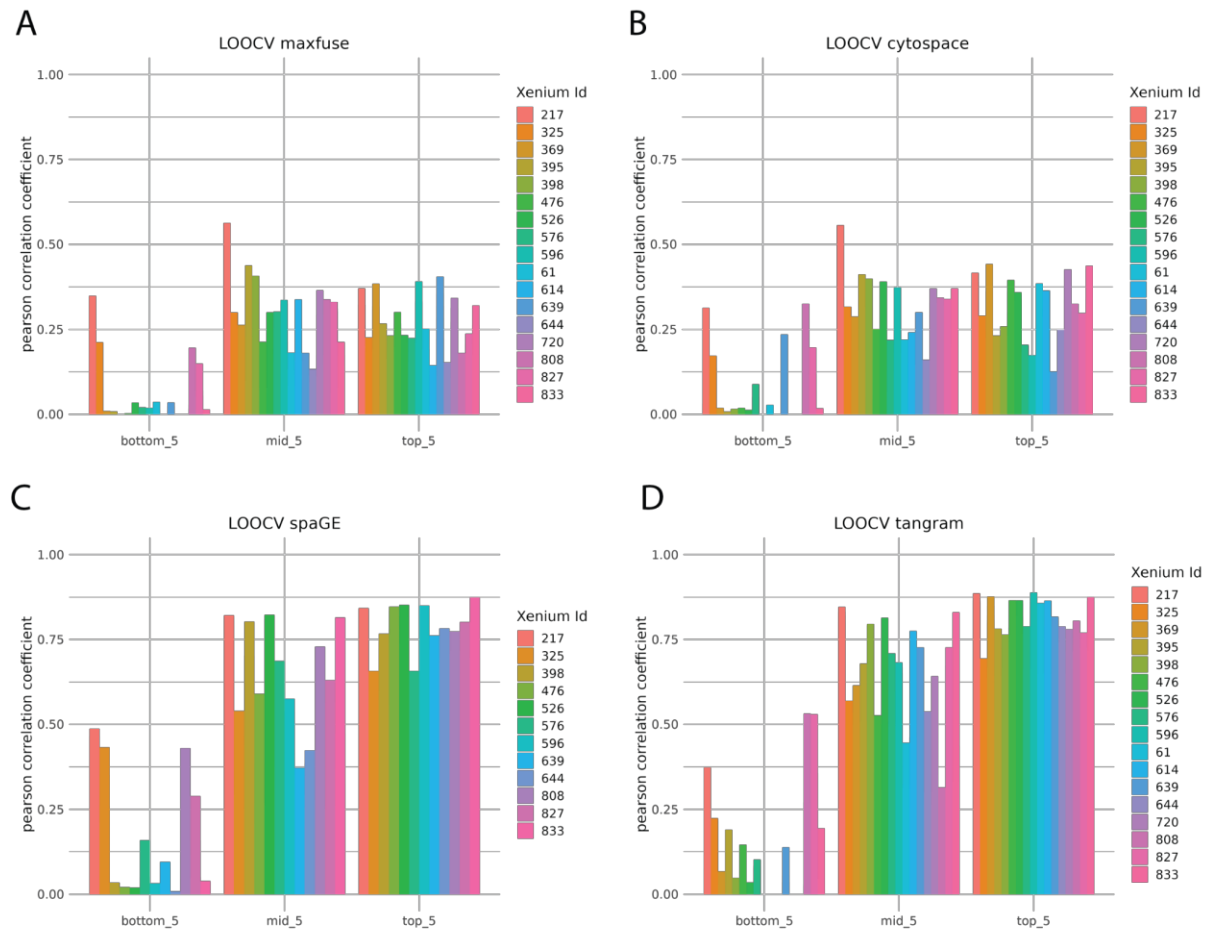


Figure 8. Leave-one-out cross validation results. Pearson correlation coefficients computed between the computed Xenium spatial transcript counts and the inferred gene expression values by each method. The correlation values are averaged for each sample for the three groups of each 5 genes from the bottom, middle, and top 5 genes ranked in ascending order by the number of non-zero entries in the spatial transcript counts matrix. Results are shown for MaxFuse (A), CytoSPACE (B), spaGE (C) and Tangram (D).

When I compared the UMAPs of the resulting gene expression matrices, the results of spaGE mapping result in a very poor low dimensional representation and clusterization of cells. This is corroborated by the spatial visualisation that the Leiden clusters are clearly influenced by an artefactual error (**Figure S3, S4**). Due to the lack of similar effects in the mapping by other methods and the availability of the similarly performing Tangram, spaGE was discarded.

I decided to keep only one of the two methods which maps cells pairwise between modalities. MaxFuse and CytoSPACE have comparable scores in the LOOCV analysis, I decide to keep MaxFuse, as it supports any pair of linked modalities (e.g., transcriptomics and proteomics) if we wish to add a modality from a different single-cell dataset in the future. In the following section, I will continue to show results on samples 576 and 808.

Results

In the following sections, I repeated the same tests on single-cell RNA-seq, TF activity and gene activity mapped onto spatial data each, to compare the performances of Tangram and MaxFuse. To reduce computational time, the mapped RNA-seq data were filtered to only consider the 1000 most variable genes in accordance to scanpy tutorials, while the TF activity and gene activity data was not filtered due to the lower number of features.

3.4.2 Leiden cluster distribution in UMAP space of the inferred transcript values

Tangram mapping results lead to a UMAP with fewer separate groups of cells compared to the original Xenium count matrix (**Figure 9C**) while the UMAP computed on MaxFuse's mapped transcripts results in a plot with separate and distinct groups of cells. I plotted the Leiden clusters computed on the original Xenium counts onto the new UMAPs to visualise their arrangement in the new data. (**Figure 9C and D**). In sample 576, the majority of the prior clusters were preserved, with the exception of cluster 0 and cluster 4 which were both composed of neuroblasts and merged in the Tangram UMAP and cluster 7, which was dispersed between clusters 3 and 6 (**Figure 9C**). Notably, despite MaxFuse enhancing the separation between cells in the UMAP, cluster 0 and 4 were still merged. Unlike in the Tangram results, cluster 7 was preserved (**Figure 9D**).

3.4.3 Validation of mapped results overview

The validation of each mapping steps is divided in the following tests:

Sample 576: Low risk, higher cell type heterogeneity (**Figure S1**)

1. I computed the weighted sum of the entropies of Leiden clusters computed on the imputed values. A compendium of all entropy results can be found in **Figure S5**.
2. I applied the scanpy function `sc.tl.rank_genes_group()` to find differentially expressed genes, testing each cell type against the rest. The top 5 genes for each cell type were then z-scaled and their average expression rates are compared across all cell types. I don't comment on highly ranked genes if they were included in the original gene panel to focus on the mapping quality of new genes. I also don't consider long noncoding RNA genes because of the lack of literature on non-protein coding genes.
3. I visualised the spatial expression of marker genes/TFs of the three neuroblast subtypes described in section 1.4 in neuroblasts in neuroblasts (**Tab.1**). This sample features

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spatially distinct neuroblast subpopulations and it is important that the integration methods maintain this heterogeneity.

Sample 808: MYCN, largely composed of neuroblasts (**Figure S1**)

1. I removed visualised the separation the neuroblast population from the remaining cells.
2. I showed, when possible, when the Leiden clustering computed from the inferred data preserved spatially distinct isolated cells in cluster 2 in this sample visible in **Figure 4D**.
3. I visualised the expression of marker genes/TFs of the three neuroblast subtypes described in section 1.4 in neuroblasts (**Tab.1**). I opted to show the expression/activity both with spatial resolution as well as in the UMAP's latent as this sample is less spatially heterogenous but still carries neuroblast heterogeneity.

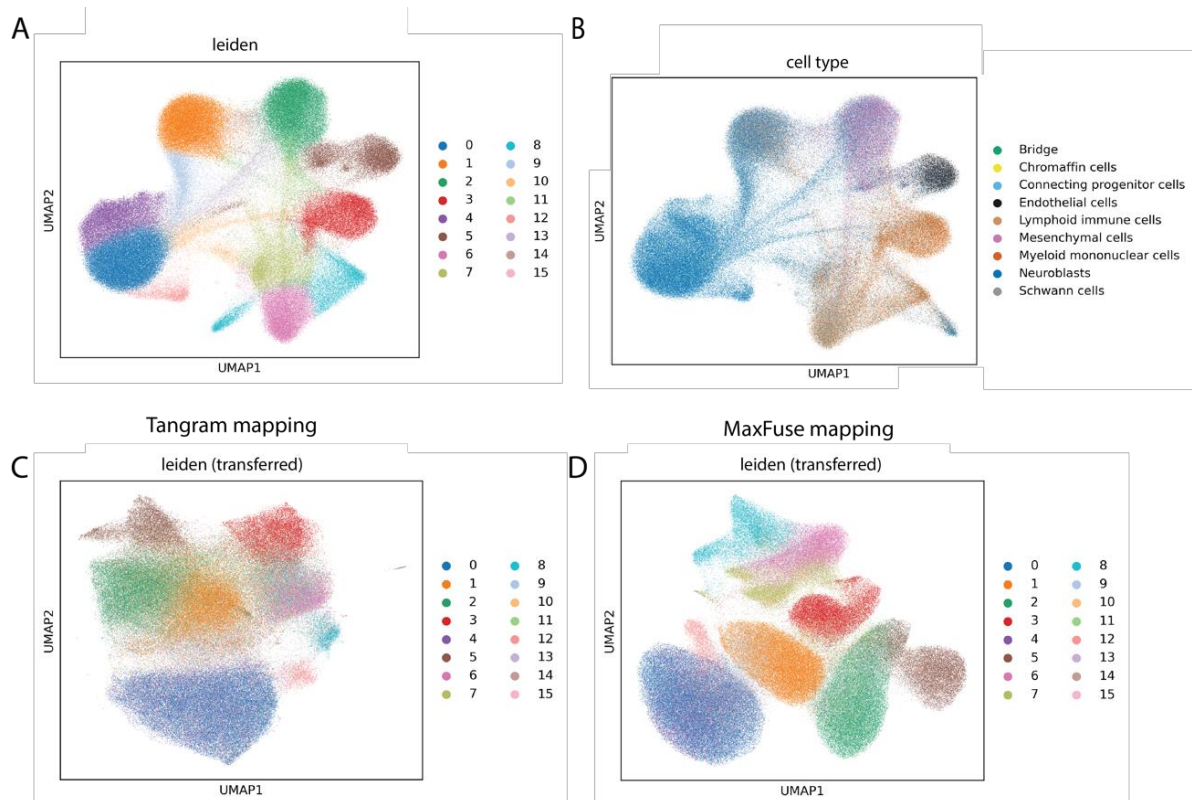


Figure 9. Separation of the original Leiden clusters onto mapped expression UMAP. UMAP computed from log-normalised counts of the original 295 genes panel coloured by Leiden cluster (A) and cell type (B). UMAPs computed on the top 1000 most variable genes from the 10,000 genes mapped by Tangram (C) and MaxFuse (D) coloured by their Leiden cluster assignment in plot A.

3.4.4 Cell type homogeneity of Leiden clusters

UMAP projection of the Tangram scRNA-seq inference, different cell types were still separated across the plot (**Figure 10A.1**) with contamination by neuroblasts in areas with mesenchymal, immune, and endothelial cells. Due to these contaminations, the Leiden clustering could only partially capture differences in cell type. Only clusters 0, 5, 7 and 9 were more homogeneously composed of neuroblasts (**Figure 10C**). Eight out of eleven clusters had a predominant cell type constituting the absolute majority of the cluster, such as mesenchymal, endothelial, and myeloid mononuclear cells. However, no clusters of any cell type other than neuroblasts were represented by more than 60% in any given cluster.

In contrast, the inferred gene expression with MaxFuse led to a very clear separation of cell types when visualised in a UMAP (**Figures 9A.2 and B.2**). The cell groups were close to each other but had clearly defined boundaries, unlike the Tangram mapping, with less overlap between cell types within Leiden clusters. Nevertheless, contamination of clusters with mesenchymal cells and immune cells by neuroblasts still occurred. The Leiden cluster composition in this dataset showed higher homogeneity (**Figure 9D**); 9 out of 13 clusters were composed of cells of a specific cell type by a fraction of 60%, and the differences in distributions of myeloid and lymphoid cells in immune cell clusters, such as clusters 3, 4, and 9, were more pronounced than in the Tangram clustering.

The original data had a weighted entropy score of 1.369. Tangram's weighted entropy score was 1.527, indicating an increase in Leiden cluster entropy, while MaxFuse mapping resulted in a weighted entropy of 1.306, suggesting a clearer separation than Tangram and an improvement over the original data.

I observed a significant ($n = 20$, $p\text{-value} = 6.4 \times 10^{-5}$) difference in cell type clustering across all samples by mapping scRNA-seq data with MaxFuse over Tangram; MaxFuse returns better results in almost every sample (**Figure S5**).

Results

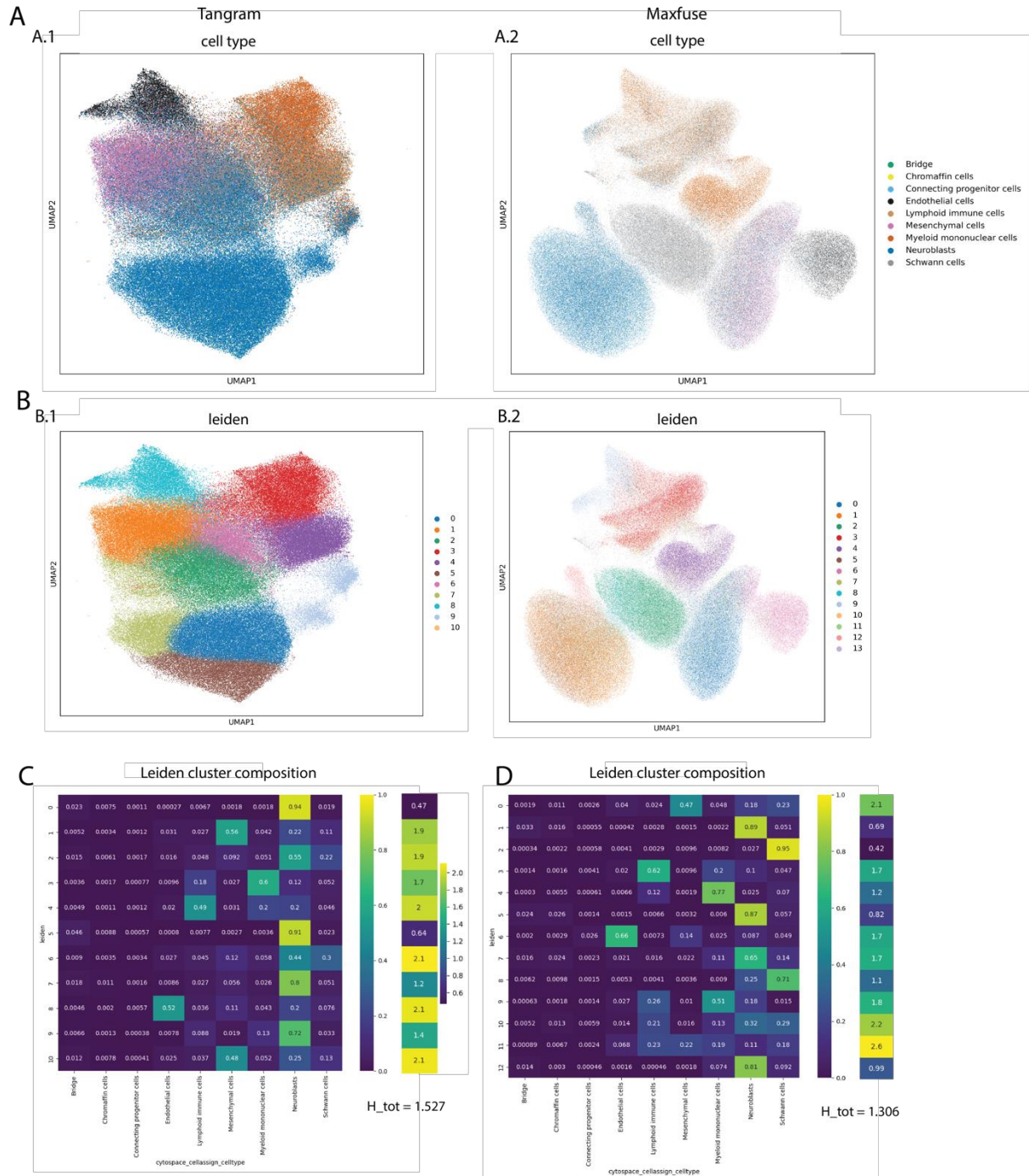


Figure 10. Separation of cell types by imputed gene expression in sample 576. UMAP of transcript counts inferred by Tangram and MaxFuse in sample 576 coloured by cell type (A) and Leiden clusters (B). UMAP values were calculated on log-normalised inferred transcript counts. Heatmap plot of cell type (columns) composition per Leiden cluster (rows) computed on transcript counts inferred by Tangram (C) and MaxFuse (D). To the right of each matrix, each cluster's Shannon entropy is returned with the weighted total sum at the bottom.

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3.4.4 Gene ranking analysis

Marker genes separate the cell types in distinct cluster: neuroblasts, bridge cells, and chromaffin cells had similar expression levels of each other's markers and are separated from the remaining pairs of cell types: myeloid and lymphoid immune cells, endothelial cells and connecting progenitor cells, and Schwann and mesenchymal cells (**Figure 11**).

The differentially expressed genes (DEGs) per cell type, not already included in the original panel, were validated by prior literature. For neuroblasts, the top marker gene *EPHA5* is implicated in axon neuronal development (Apellániz-Ruiz *et al.*, 2017), while *TUBB3* is a neuron-specific variant of Tubulin (Maris *et al.*, 2008).

For mesenchymal cells, the top genes were three collagen variant encoding genes: *COL6A3*, *COL3A1*, and *COL1A2*. Collagen is a fundamental component of connective tissue and is primarily produced by mesenchymal cells (Muckian *et al.*, 2002; B. Li *et al.*, 2022) *UACA* is a pro-apoptotic protein expressed ubiquitously across the body (Ohkura *et al.*, 2004).

The top DEGs for myeloid mononuclear cells included *CD74*, a gene involved in the stabilisation of MHC class II complexes, and *DOCK2*, which is implicated in the migration of leukocytes in the tumour microenvironment (Wang *et al.*, 2016; Ji *et al.*, 2021).

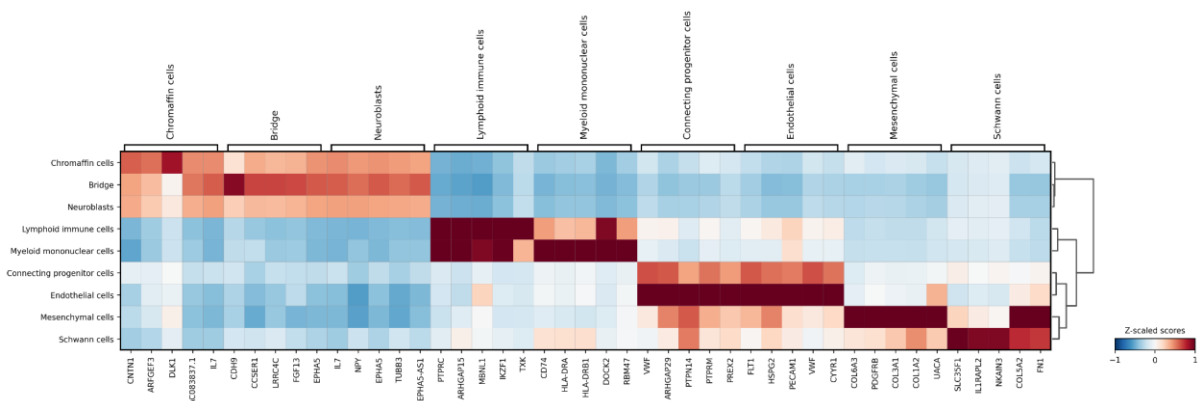


Figure 11. Gene expression of high-ranking genes by cell type computed on scRNA-seq Tangram results. The mean expression of top 5 ranking genes per cell type is z-scaled across cells for better visualisation. The dendrogram is constructed from the 1000 most variable genes in the data with `sc.tl.dendrogram()`.

The gene expression values mapped with MaxFuse lead to a similar gene expression-based clustering between cell types as with Tangram, albeit the differences of immune cells from the remaining cell types were greater compared to the Tangram mapping and mesenchymal cells

3.4.5 Spatially inferred expression of neuroblast subtype marker genes in sample 576

The imputed expression of markers in malignant cells is shown in **Figure 12** and I observed similar spatial expression patterns between Tangram and MaxFuse mapping. *NTRK* was mapped in regions further away from the healthy mesenchymal tissue, while *EGFR* expression was localised in the border area between neuroblasts and Mesenchymal cells. Finally, the *TOP2A*, the marker for highly proliferative neuroblasts, was expressed in small group of cells in according to both methods (see section 3.7 for a biological characterisation of these neuroblasts). Compared to the mapped Tangram values (**Figure 12A**), MaxFuse had a sparser distribution of counts due to its one-to-one mapping of sparse scRNA-seq data (**Figure 12B**).

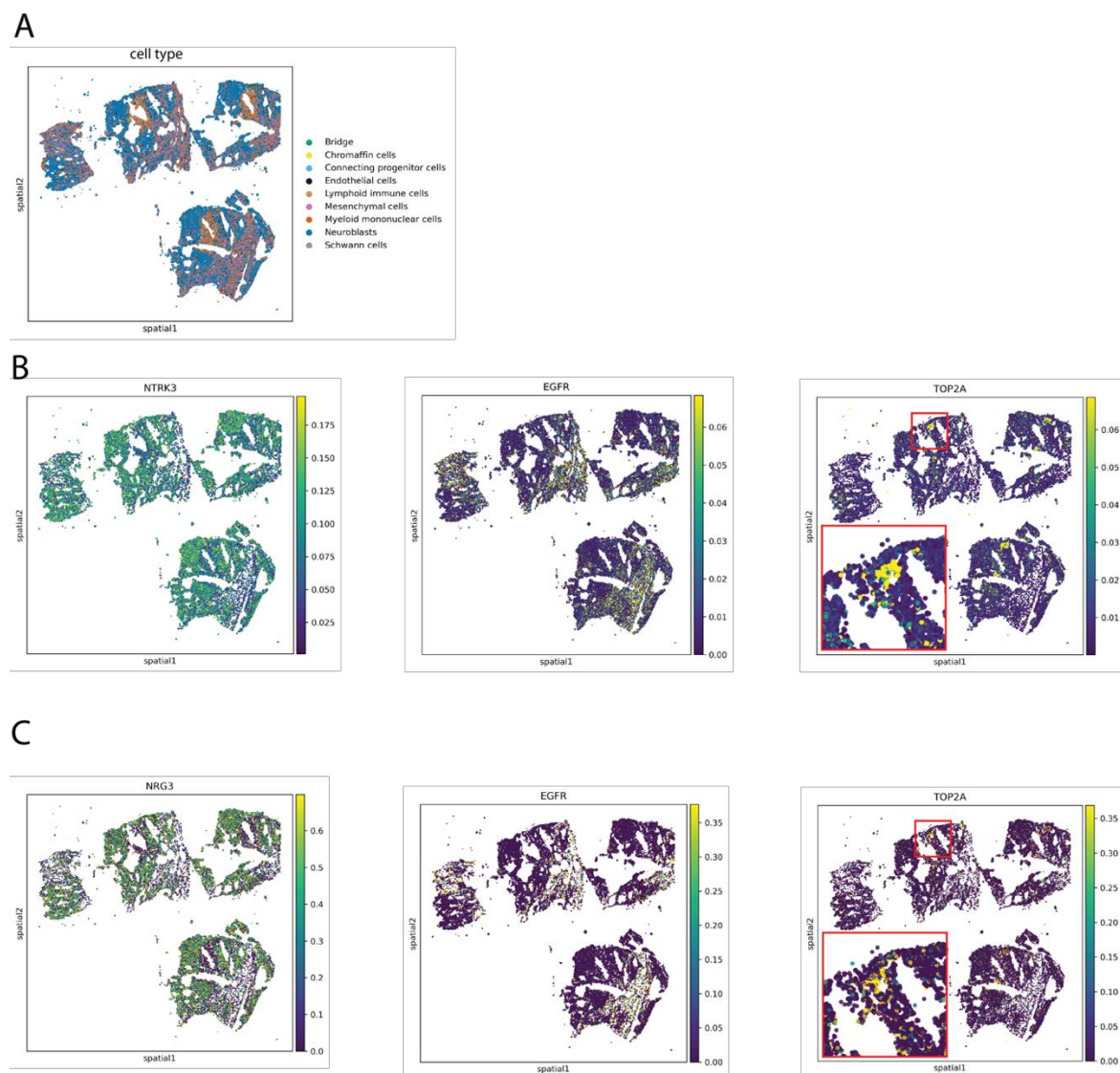


Figure 13. Spatial expression of neuroblast subtype markers in LR neuroblasts. Log-normalised counts for markers of adrenergic neuroblasts, mesenchymal neuroblasts, and proliferating sympathoblasts with the highest

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variance in the imputed gene expression data inferred with Tangram (B) and MaxFuse (C). The figures only include cells labelled as neuroblasts.

3.4.6 Preservation of neuroblast spatial heterogeneity

In sample 808 the UMAP projection showed similar differences between values inferred with Tangram and MaxFuse. The Tangram mapping leads to a UMAP similar to the one based on sample 576, cell types were not separated in different groups but are clustered within one connected shape, while MaxFuse clustered the cell types in different groups only connected by filaments of cells (**Figure 14**), similarly as to the UMAP of the original spatial transcript data. Notably, in both UMAPs the Leiden clustering still grouped mesenchymal and endothelial cells in the same cluster.

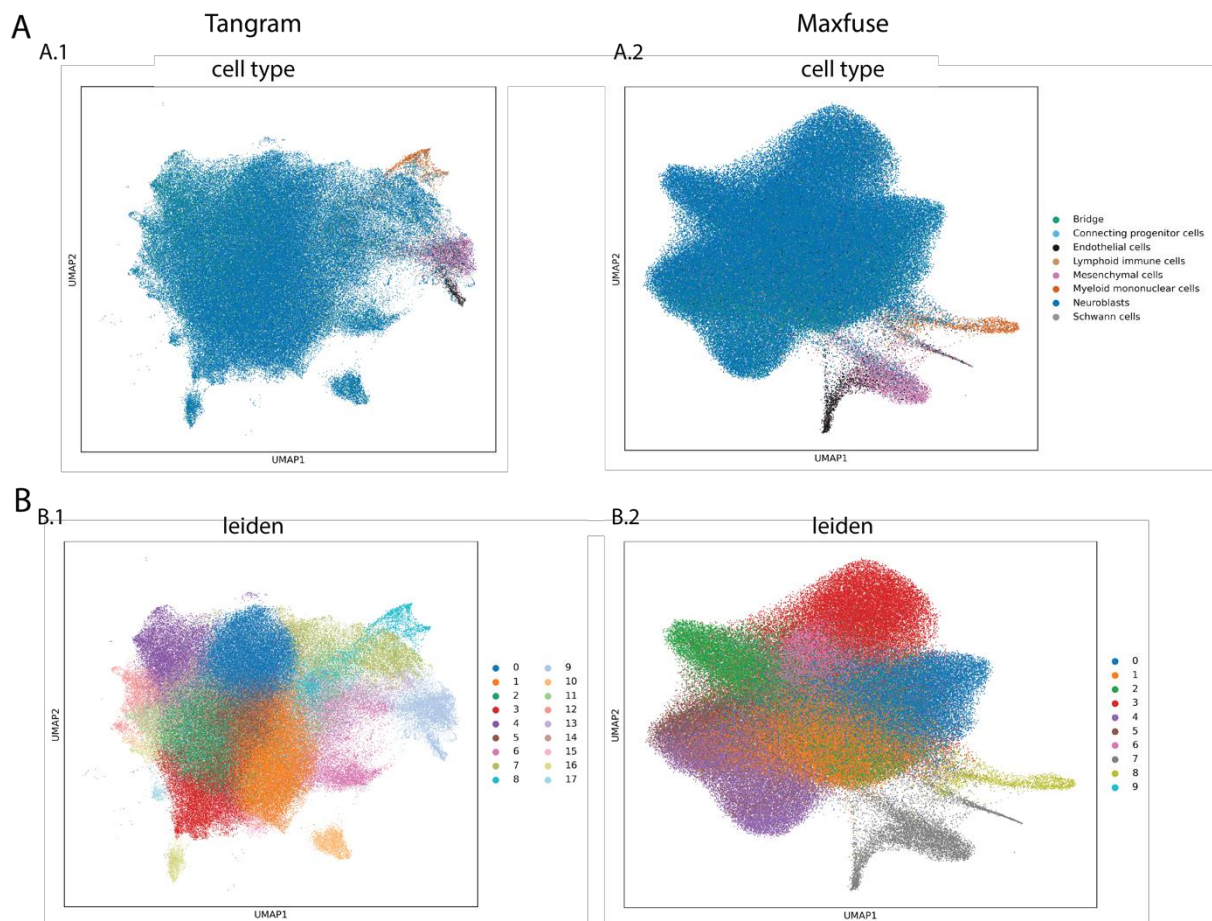


Figure 14. Clustering of cells by inferred gene expression in sample 808 (MYCN). (A) UMAP projection of transcript counts inferred by Tangram (A.1, B.1) and MaxFuse (A.2, B.2) sample 808 coloured by cell type (A) and Leiden clusters (B) and UMAP values were calculated on log-normalised inferred transcript counts.

Because the Leiden clustering of the original spatial count data showed a spatially distinct pattern in cluster 2 of separate small islands of cells, I sought to verify which of the mapping

tools capture that spatial pattern in its clustering. The comparison of the new computed Leiden clusters showed that the Tangram gene expression scores did not preserve the spatial pattern of cluster 2 observed in the clustering based spatial counts (**Figure 4D**), while with MaxFuse, cluster 3's spatial distribution resembles the spatial structure observed in the Xenium dataset (**Figure 15B**).

3.4.7 Inferred expression of neuroblast subtype marker genes of sample 808

Finally, the subtype's markers were visualised on the UMAP projection because this sample does not have the same level of spatial heterogeneity as 576 and finding the separation of the subtypes in the neuroblast population is a better indication of the separation between malignant cell subtypes. In Tangram, the adrenergic marker gene *DBH* and the proliferative marker gene *TOP2A* were expressed in different areas of the UMAP, while *EGFR* is expressed in few cells in the region close to the mesenchymal cells. With MaxFuse, the *ERBB4* and *TOP2A* were expressed in partially overlapping regions while *FNI* was still expressed in the filament connecting neuroblasts to mesenchymal cells (**Figure 16**).

Results

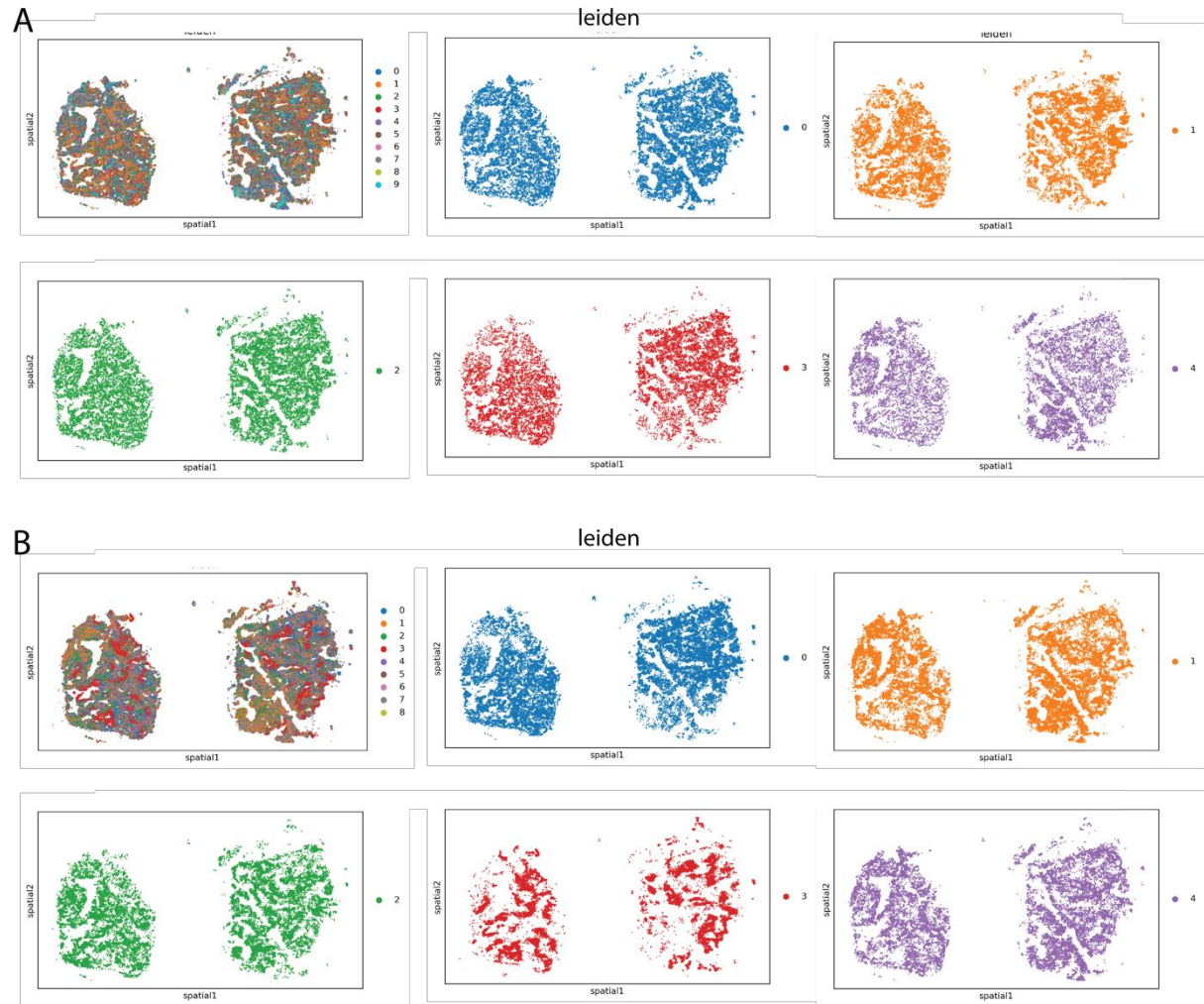


Figure 15. Top 5 Leiden cluster spatial distribution comparison. Sample 808 spatial scatter plot coloured by Leiden clusters computed on (A) gene expression values inferred with Tangram and (B) MaxFuse. Only the 5 largest clusters in each dataset are shown.

Results

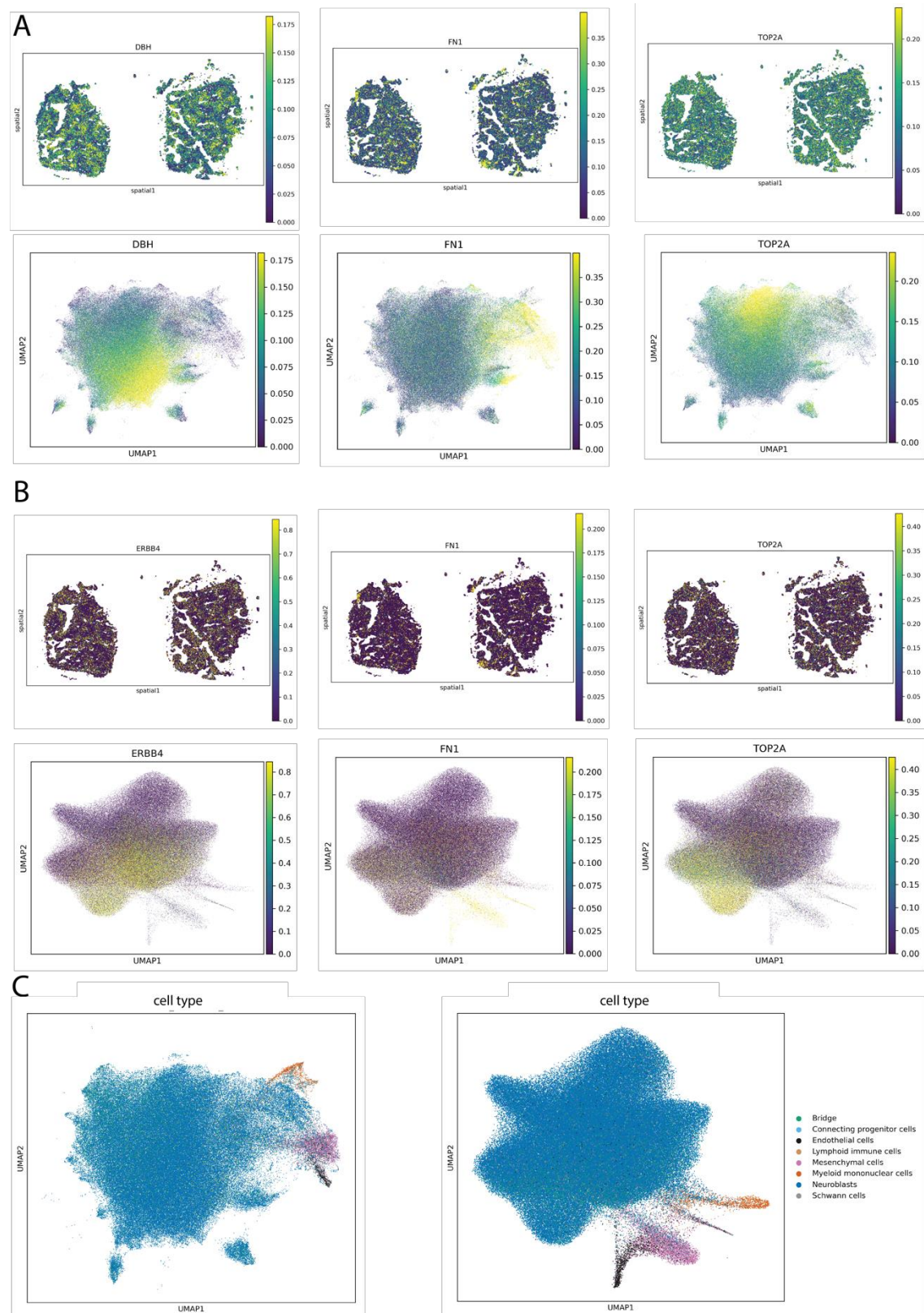


Figure 16. Expression of neuroblast subtype markers in MYCN neuroblasts. Log-normalised imputed counts for markers of Adrenergic neuroblasts (left), Mesenchymal neuroblasts (centre), and Proliferating (right) Sympathoblasts with the highest variance in the gene expression data imputed with Tangram (A) and MaxFuse (B). Cell type annotation of the same UMAPs for reference (C). The figures only include cells labelled as neuroblasts.

3.5 Validation of TF activity mapping

Next, I inferred transcription factor (TF) activity values across the SRT data cells. I followed the same procedure as with gene expression to validate the quality of the mapping.

3.5.1 Cell type homogeneity in Leiden clusters

First, I analysed the clustering of cell types within Leiden clusters using the mapped TF activities in sample 576. The UMAP separation of cell types between Tangram and MaxFuse followed a similar trend as in the gene expression data with exacerbated differences (**Figure 17**). The TF activity mapped with Tangram led to a single shape of cells with an internal separation between neuroblasts and non-neuroblasts, but there was no further separation between e.g. mesenchymal cells and immune cells. Endothelial cells line the right edge of the plot, and are largely clustered in cluster 30 (**Figure 17A**). The Leiden clustering on these results led to a much more fragmented clusterization, defining 32 clusters (**Figure 17B**) in contrast to the 11 clusters using the same clusterization parameters (**Figure 10B**). These clusters were mostly composed of neuroblasts, but in contrast to the gene expression data, clusters 11 and 21 were composed mostly of immune cells that distinguish between myeloid and lymphoid lineage. Mesenchymal cells remained clustered with other cell types, most prominently neuroblasts and endothelial cells (**Figure 17C**).

The MaxFuse UMAP (obtained by down sampling the dataset by 0,1 and adding random noise to the data (see “**2.2 Single-cell RNA-seq, ATAC-seq and TF activity data**” section), showed a sharp separation of the cells by cell type, with mild contamination between groups (**Figure 17A and B**). Neuroblasts, mesenchymal cells and immune cells were each represented as the majority in at least 1 cluster. Notably, clusters 7 and 8 were heavily inhomogeneous and appear to be the result of contamination between neuroblasts, and both lineages of immune cells. Unlike the gene expression results, TF activity mapped with the same indices did not separate Schwann cells enough to generate a Schwann cells-specific cluster and they were largely clustered with neuroblasts in cluster 1 (**Figure 17D**).

Despite these results, the separation of cell types was not consistently better with MaxFuse (**Figure S5**) and there was no significant effect across samples (Wilcoxon signed rank test, $n = 20$, $p\text{-value} = 0.1178$).

Results

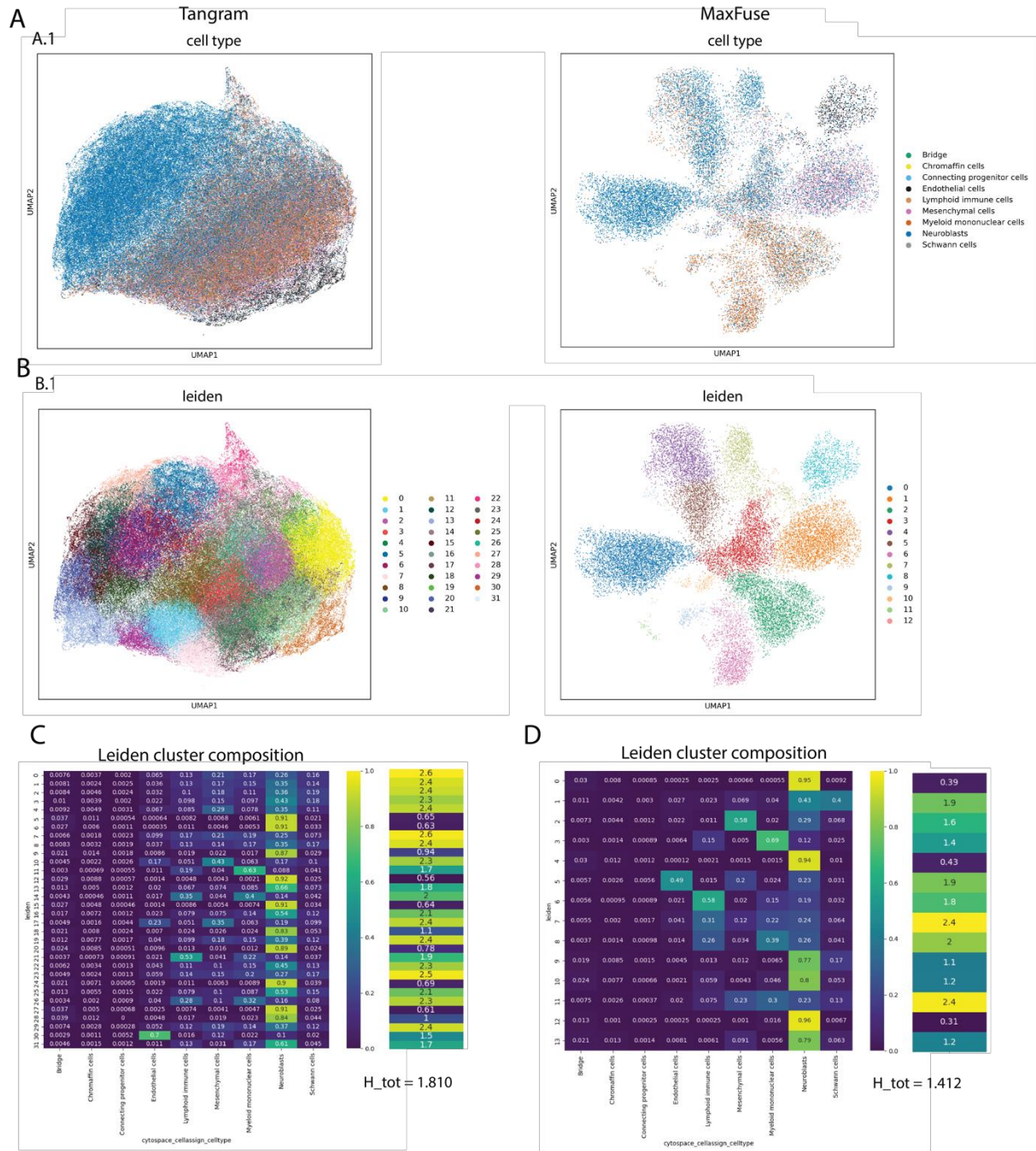


Figure 17. Clustering of cell types by inferred TF activity in sample 576. UMAP of transcription factor activity inferred by Tangram and MaxFuse in sample 576 coloured by cell type (A) and Leiden clusters (B). Heatmap plot of cell type (columns) composition per Leiden cluster (rows) computed on transcription activity inferred by Tangram (C) and MaxFuse (D). To the right of each matrix, each cluster's Shannon entropy is returned with the weighted total sum at the bottom. The MaxFuse UMAP was computed on 10% of the TF activity scores mapped by MaxFuse and augmented with noise (see section 2.2).

3.5.2 TF ranking analysis

TF activity separated the cell types in three groups, first chromaffin cells, bridge cells and neuroblast, then endothelial, mesenchymal and connecting progenitor cells, and finally myeloid,

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lymphoid immune cells and Schwann cells (**Figure 18**). The immune cells were clustered closer together compared to Schwann cells, but generally all further separation was small compared to that between the Chromaffin, Bridge and neuroblast cells cluster and the rest.

The top transcription factors for neuroblasts were DLX5, GATA2 and MYC. MYC is a well-known oncogene (Huang et al., 2013) and DLX5 has been implicated in various cancer types as it is also a regulator of differentiation (Sun et al., 2021). GATA2 is amongst the characteristic transcription factors of the Adrenergic/Proliferating regulon and is important for the upkeep of the endless proliferative potential of stem cells (Shu et al., 2015).

The Mesenchymal associated TF WT1 is an important transcription factor implicated in tissue development and maintenance and acts as an oncogene and a tumour suppressor gene depending on tumour type, though no study on its role in N currently exists (TOSKA et al., 2014). TEAD2 is active in the Hippo signalling pathway and is involved in embryonic organ development as well as mediating the epithelial-mesenchymal transition by regulating YAP1 expression (LANDIN-MALT *et al.*, 2016; Shim *et al.*, 2023). GATA4 is a TF active in the development and function of the mesenchymal tissue of the heart (Afouda, 2022).

Notable highly ranked TFs in Myeloid Mononuclear cells were CEBPA and CEBPD, two TFs involved in the inflammatory response of immune cells (Ullmann et al., 2021; Adamo et al., 2023).

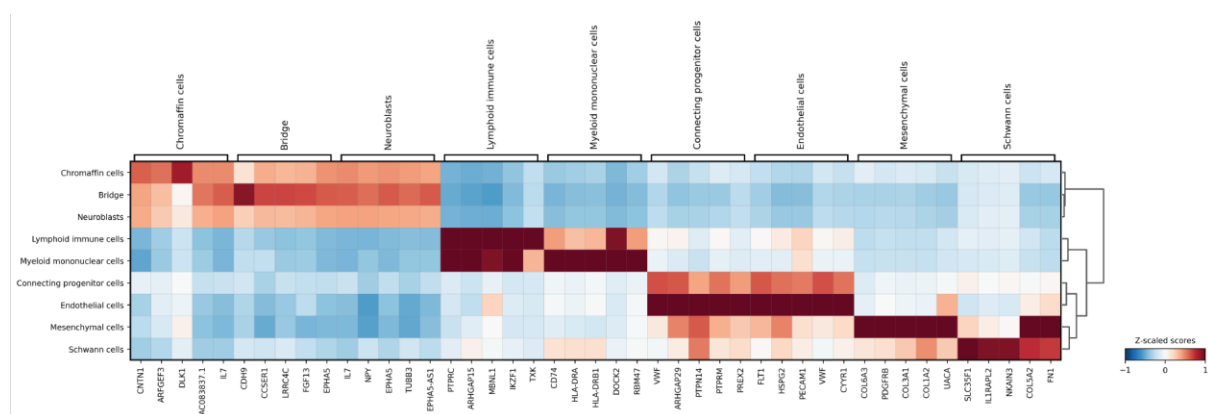


Figure 18. Activity of high-ranking TFs by cell type based on Tangram spatial mapping of single-cell TF activity. The mean activity of top 5 ranking genes per cell type is z-scaled across cells for better visualisation. The dendrogram is constructed from all TFs in the data with `sc.tl.dendrogram()`.

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In the MaxFuse mapping results we saw a clustering more similar to the scRNA-seq data, the Schwann cells were clustered with mesenchymal cells instead of immune cells (**Figure 19**).

There were re-occurring characteristic TFs within this clustering, such as GATA2, DLX and TFDP1 for neuroblasts and ELF1 and EHF for immune cells (**Figure 19**).. Other characteristic spatial TFs in the MaxFuse mapping for neuroblasts were DLX5, GATA2 like in the Tangram results, as well as SOX4, a TF implicated in the development of colon cancer (Lin *et al.*, 2013).

For the Mesenchymal cells, the aforementioned WT1, TEAD2 and GATA4 were the same discriminative TFs found in the Tangram results. Myeloid Mononuclear cells in this sample were distinguished by the tumorigenic TF EHF, and the pro-inflammatory TFs CEBPD and CEBPA, all of which were already detected in this sample in the Tangram mapping (**Figure 18**).

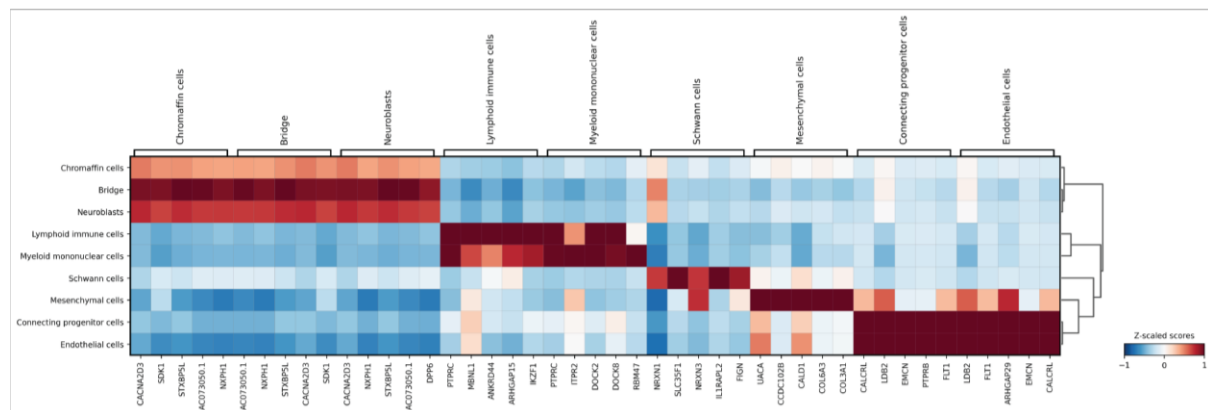


Figure 19. Activity of high-ranking TFs by cell type based on MaxFuse spatial mapping of single-cell TF activity. The mean activity of the top 5 ranking genes per cell type is z-scaled across cells for better visualisation. The dendrogram is constructed from all TFs in the data with `sc.tl.dendrogram()`.

These TFs are biologically relevant to their cell types or have been implicated in tumorigenesis or tumour progression in the case of neuroblasts. Potentially due to the smaller number of TFs compared to genes, most of the highest-ranking TFs were found by both Tangram and MaxFuse for the same cell types.

3.5.3 Spatially inferred expression of neuroblast subtype marker TFs in sample 576

There were only two transcription factors in the characteristic regulons of Adrenergic/Proliferating malignant cells and Mesenchymal malignant cells in our panel of TFs, EGR3 for the Mesenchymal cells and GATA2 for the Adrenergic/Proliferating malignant cells.

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Both tools mapped them to similar areas as the respective marker genes in the previous section with no notable changes (**Figure 20**).

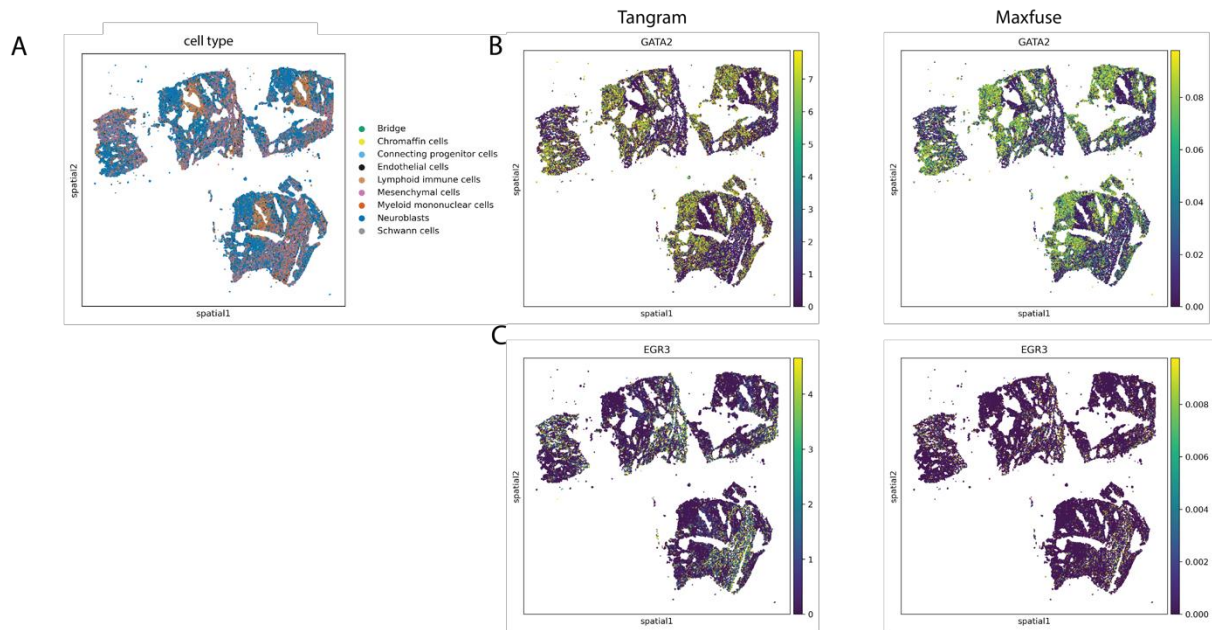


Figure 20. Mapped TF Activity of marker genes in LR neuroblasts. Spatial cell type distribution in sample 576 for reference (A). TF activity scores for adrenergic/proliferating marker GATA2 (B) and mesenchymal marker EGR3 (C) mapped by Tangram and MaxFuse. The figures only include cells labelled as neuroblasts.

3.5.4 Preservation of neuroblast spatial heterogeneity

In sample 808, the UMAP of the TF's mapped by Tangram led to an even smaller separation of neuroblasts from other cell types, as the small fraction of non-neuroblasts in this sample is inside the neuroblast area (**Figure 21A**). Similarly to sample 576, the Leiden clusterization with identical parameters as the spatially inferred gene expression results led to a much larger number of clusters. Meanwhile, inferred TF activity with MaxFuse in this sample separated the neuroblasts in multiple interconnected groups while the remaining cells were grouped together in a mixed group at the top of the UMAP projection (**Figure 21B**).

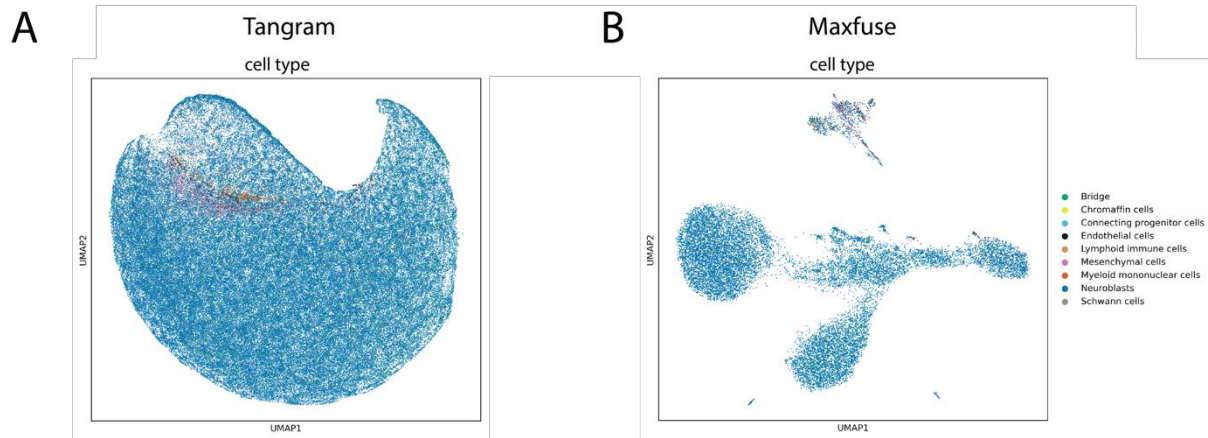


Figure 21. Clustering of cells by inferred TF activity in sample 808 (MYCN). UMAP of transcription factor activity inferred by Tangram (A) and MaxFuse (B) in sample 808 coloured by cell type. The MaxFuse UMAP was computed on 10% of the TF activity scores mapped by MaxFuse and augmented with noise (see section 2.2).

3.4.7 Inferred expression of neuroblast subtype marker TFs

The mapping of the characteristic TFs' activities with MaxFuse yielded both different but overlapping UMAP regions for the two marker TFs. The spatial patterns were not as distinct as in sample 576. In the UMAP projection computed from MaxFuse results' I observed overlap in the lower section of the plot, but the cells with high EGR3 values had lower GATA2 activity levels while GATA2's activity is high in the left portion of the UMAP (**Figure 22A**). Notably, EGR3 has higher TF activity values overall and is also active in neuroblasts in the mixed group of cells at the top of the plot where mesenchymal cells are found (**Figure 22B**). The Tangram UMAP projection was not considered on behalf of its poor separation of cell types.

These results suggested that Tangram and MaxFuse present the same strengths and weaknesses with TF activity mapping as they did with scRNA-seq mapping. The single-cell TF activity data mapping generally reduced the distinction between cell types compared to scRNA-seq in either method, demonstrated by the increased entropy scoring. Because the single-cell TF activity matrix, unlike the scRNA-seq count matrix, is not sparse, I consider MaxFuse to be the unequivocally better choice for the mapping of TF activities (see **Discussion**).

Results

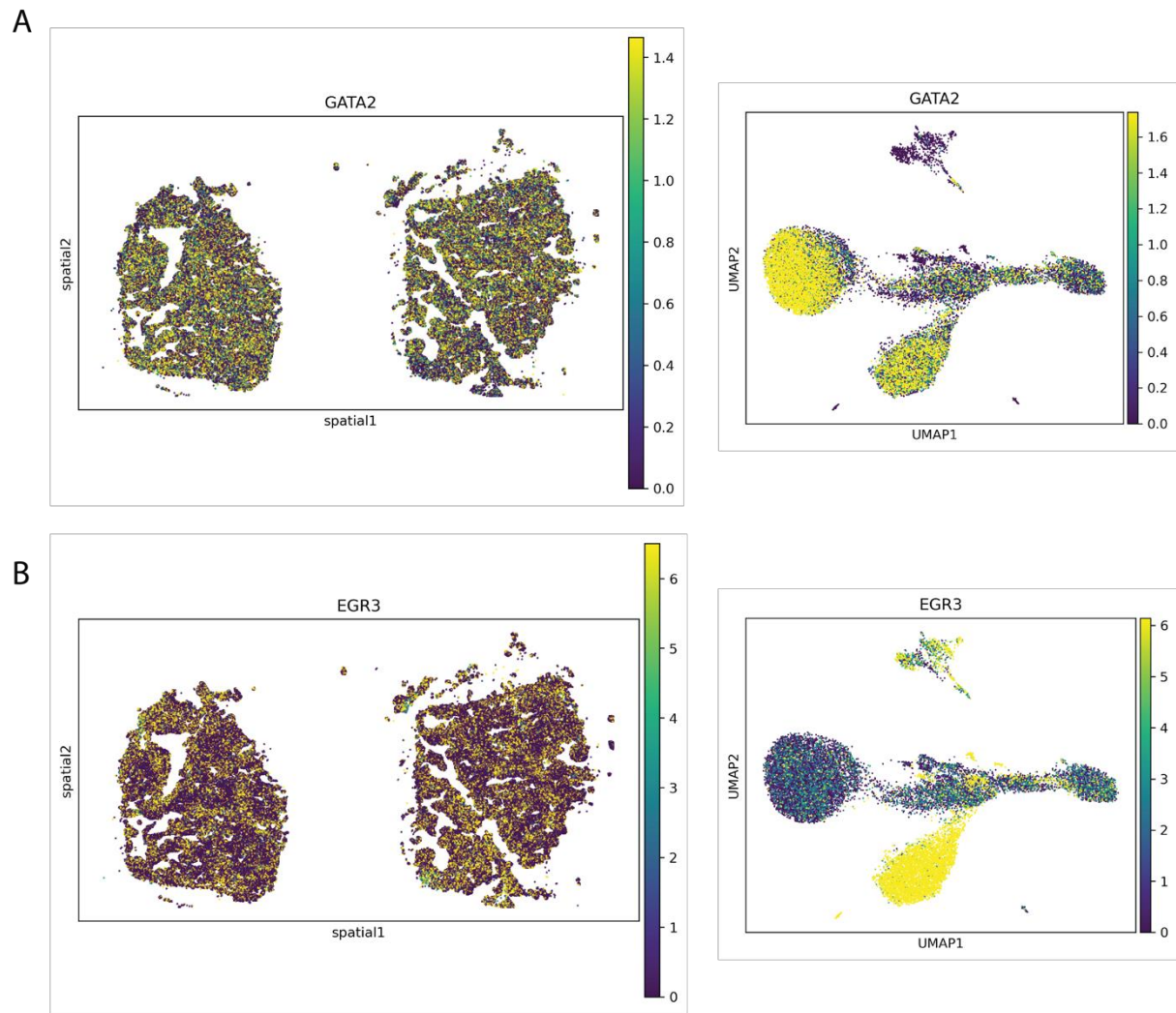


Figure 22. Mapped TF Activity of marker genes in MYCN neuroblasts TF activity values of GATA2 (A) and EGR3 (B) spatially (left) and in UMAP projection (right). The figures only include cells labelled as neuroblasts.

3.6 Mapping single-cell gene activity data

3.6.1 Cell type homogeneity in Leiden clusters

The gene activities derived from scATAC-seq mapped with Tangram led to a similar UMAP as the TF activities. The data separated the cells along the axis of malignancy but all non-neuroblast types were mixed together (**Figure 23A**) and the Leiden clustering found over 20 clusters, similarly to the transcription factor mapping results with Tangram (**Figure 23B**). Unlike in the TF activity mapping, all clusters were either majorly composed of neuroblasts or were mixed, such as the first and to a lesser degree the second cluster (**Figure 23B and C**).

MaxFuse mapping of gene activity led to a better separation of neuroblasts from the remaining cells. The Leiden clusters' composition of the gene activity levels is similar to the TF activity Leiden clusters' composition: with both methods, the Schwann cells and mesenchymal cells were both grouped together in hybrid clusters with neuroblasts (here: 1 and 2) and immune clusters could be divided in ones with a majority of myeloid cells and ones with a majority of lymphoid cells. The weighted entropy sum computed on Tangram clusters ranked above the spatial data total entropy again at 1,906 and MaxFuse exceeds it by a small margin at 1,403.

These results are consistently reproduced across all samples (Wilcoxon signed rank test, $n = 20$, $p\text{-value} = 9.89 \times 10^{-5}$) with even larger divergence of entropy scores between Tangram and MaxFuse (**Figure S5**).

Results

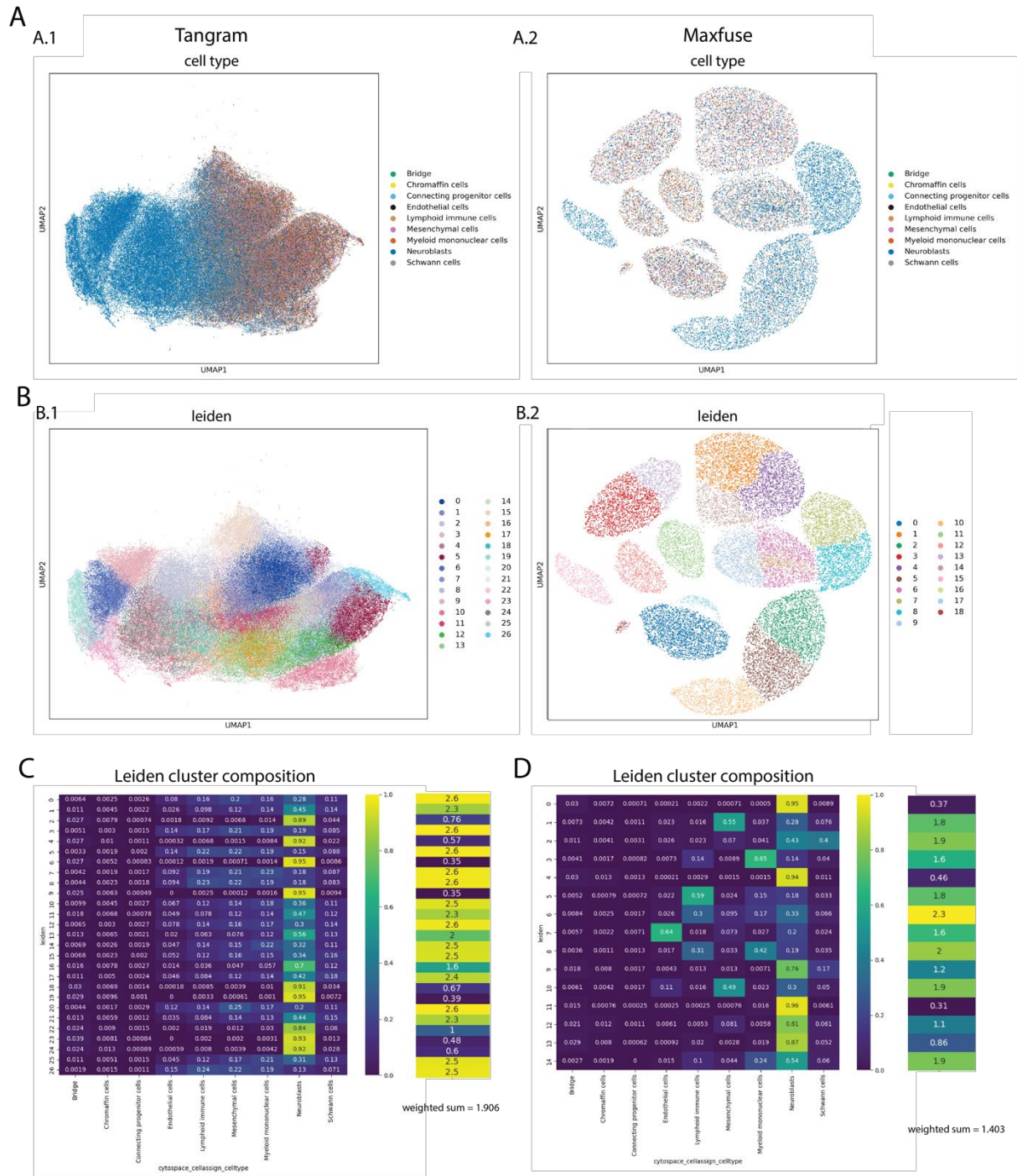


Figure 23. Separation of cell types by inferred gene activity in sample 576 (LR). UMAP of gene activity inferred by Tangram and MaxFuse in sample 576 coloured by cell type (A) and Leiden clusters (B). Heatmap plot of cell type (columns) composition per Leiden cluster (rows) computed on gene activity inferred by Tangram (C) and MaxFuse (D). To the right of each matrix, each cluster's Shannon entropy is returned with the weighted total sum at the bottom. The MaxFuse UMAP was computed on 10% of the gene activity scores mapped by MaxFuse and augmented with noise (see section 2.2).

3.6.2 Gene activity ranking analysis

Gene activity is a measure of a gene's accessibility computed from scATAC-seq data, and as such, it is not directly proportional to its expression. An active gene may be bound to a transcription factor (TF) and expressed, bound to a repressor and silenced, or neither. Nevertheless, I performed this measurement because the UMAP and clustering of the data still support a characteristic gene activity profile in malignant cells.

The high-ranking gene expression patterns exhibit a stronger separation between neuroblasts, bridge cells, and chromaffin cells compared to the previous two modalities. This separation is evident in recurring marker genes such as *LMTK3* and *IZUMO1* for the first group, and *ABR* and *LINC00511* for the second. These recurring markers cannot be explained by a small number of genes, as the dataset includes gene activities for over 5000 genes (**Figure 24**) and suggest that the separation between cell types is driven a by a small number of highly variable gene activities.

Notable neuroblast markers identified by gene activity included *LMTK3*, a tyrosine kinase with established oncogenic activity (Ditsiou et al., 2020), and *LRRC4B*, a synaptic adhesion protein that contributes to the formation of excitatory synapses (Zhang et al., 2021).

For mesenchymal cells, *CASC15* is a long non-coding RNA (lncRNA) and, despite its classification, it is worth mentioning for its documented tumour suppressor activity in NB (Russell et al., 2015). *DST* encodes a cell junction protein implicated in the epithelial-mesenchymal transition of head and neck cancer (Graves et al., 2014), and *ABR* encodes a regulator of cell signalling pathways in neurons and acts as a driver of myeloid cell differentiation (Namasu et al., 2017). In the myeloid immune cells, none of the highly ranked genes were pertinent to the immune response.

Results

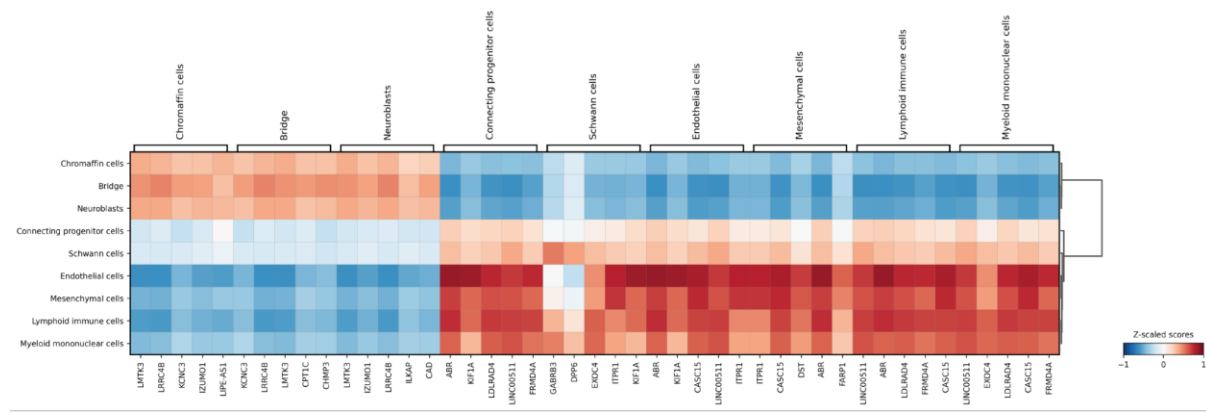


Figure 24. Gene activity of high-ranking genes by cell type computed on scRNA-seq Tangram results. The mean expression of top 5 ranking genes per cell type is z-scaled across cells for better visualisation. The dendrogram is constructed from all genes.

In the gene activity data inferred by MaxFuse, the distance between the neuroblast/chromaffin cells/bridge cells group and the remaining cells was less extreme but present (**Figure 25**). According to the MaxFuse mapping, the marker genes for NB were the hydratase *ABHD1*, expressed in mitochondria (Lan *et al.*, 2021) and with no specific links to cancer, neurons or cell proliferation; *IPO13* is responsible for the nuclear import in cells and is highly expressed in neurons (Wang *et al.*, 2013) and *DIAPH1* is associated with axon development as it is implicated in actin dynamics and cytoskeletal organisation (Ercan-Sencicek *et al.*, 2015). Like in the Tangram mapping, mesenchymal cells were marked by the tumour suppressor *CASC15* (Russell *et al.*, 2015).

These genes are less cell type specific, instead the test mainly found genes implicated in the nervous system regardless of cell type. This may be the result of representation bias in the original dataset, and can be partially explained by the ambiguity of gene activity data as it pertains to the actual function of the gene in a cell. It is noteworthy that the gene activities clearly separate neuroblasts, bridge cells and chromaffin cells from the remaining cells. For MaxFuse, we observe the further classification of Schwann cells and connect progenitor cells, myeloid and lymphoid cells and endothelial and mesenchymal cells together.

Results

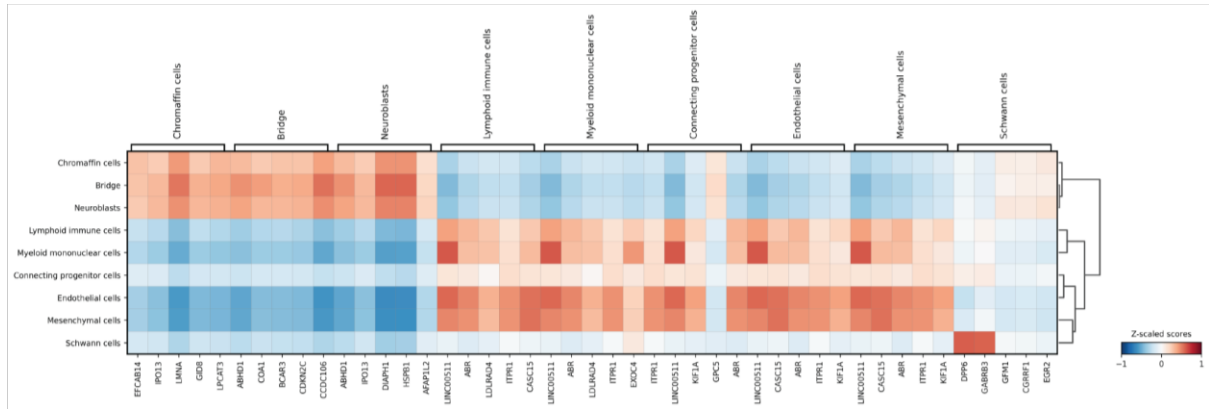


Figure 25. Gene activity of high-ranking genes by cell type computed on scRNA-seq MaxFuse results. The mean expression of top 5 ranking genes per cell type is z-scaled across cells for better visualization. The dendrogram is constructed from all genes.

3.6.3 Neuroblast separation and characterization of Leiden clusters in sample 808 with enrichR

In sample 808, the non-neuroblast cells were separated from the neuroblasts more clearly than TF activities mapped by Tangram. The Leiden clustering grouped them in one cluster with additional neuroblasts; similarly to the TF activities mapping, the sample was segmented in more Leiden clusters than the spatial counts and the inferred scRNA-seq data. MaxFuse based mapping led, similarly to sample 576, to a division of neuroblasts into multiple groups and the clustering of the non-neuroblasts in a mixed cluster. Unlike the sample 576 UMAP, these clusters had no connecting filaments between them (**Figure 26A and B**).

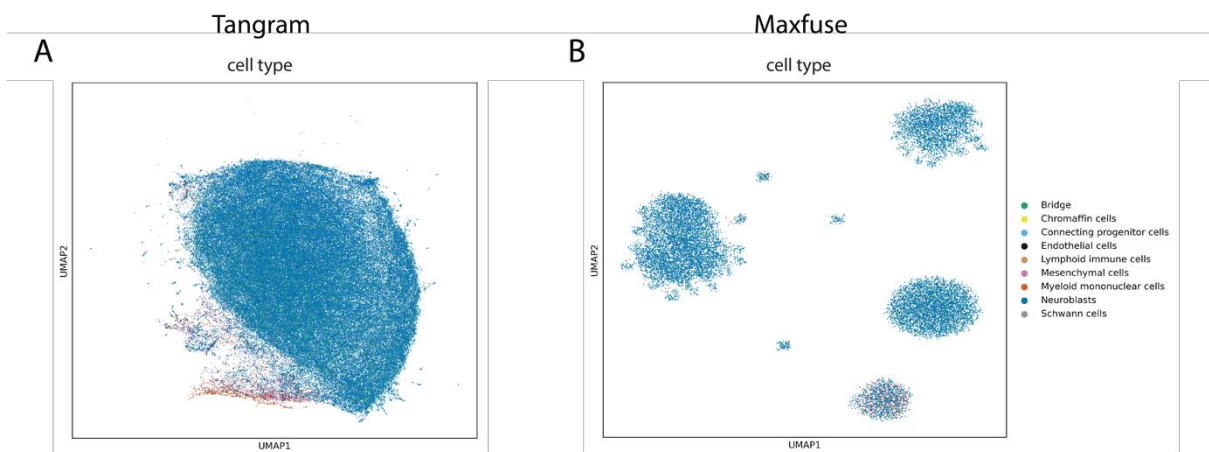


Figure 26. Clustering of cells by inferred gene activity in sample 808 (MYCN). (A) UMAP of gene activity data inferred by Tangram (A) and MaxFuse (B) sample 808 coloured by Leiden clusters (A.1, B.1) and cell type (A.2, B.2). The MaxFuse UMAP was computed on 10% of the gene activity scores mapped by MaxFuse and augmented with noise (see section 2.2).

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I compared the Leiden cluster spatial distribution in this sample next and in line with prior results, but the Leiden clustering neither mapping method produced spatially distinct Leiden clusters.

Evaluating the mapping of gene activities is more difficult and less straightforward than TF activity. To address this, I attempted to run enrichR on the three groups of neuroblasts generated by MaxFuse in **Figure 26A** which are clustered in the 4 largest Leiden clusters of the sample. With this method I can determine whether the differentially active genes have a shared functional identity or if the genes that separate them do not enrich any term significantly.

EnrichR's results identify several enriched terms in the 4 largest clusters (**Figure 27A, B**). Cells in cluster 0 appear to have reduced chromatin accessibility in genes involved in tissue invasion, proliferation and immune escape. Conversely, in cluster 1 and especially cluster 2 cells the same terms are positively enriched. If those genes are bound by TFs, these results would suggest that cluster 1 and 2 are more proliferative and invasive than cluster 0 cells. The enriched terms for cluster 3 seem contradictory, the mitotic spindle is enriched by accessible genes but inaccessible genes enrich the E2F targets. Generally, the enrichment indicates that biological information is preserved through the mapping of gene activities with MaxFuse.

After mapping three modalities and comparing the two tools' performance, MaxFuse surpassed Tangram in regards to preserving the separation between cells and mapping modalities beyond scRNA-seq. Tangram had an edge over MaxFuse in the leave-one-out cross validation scores and due to its ability to infer gene expression levels for each cell-gene combination so its implementation should still be considered only to map scRNA-seq data. For single-cell TF activity and gene activity values where the matrix is not sparse, MaxFuse is a safer choice, although it did not significantly improve the Leiden cluster homogeneity in TF activity mapping (**Figure S5**). I conclude that MaxFuse is unequivocally the best option out of the two for the purposes of this thesis (see more in the **4. Discussion**).

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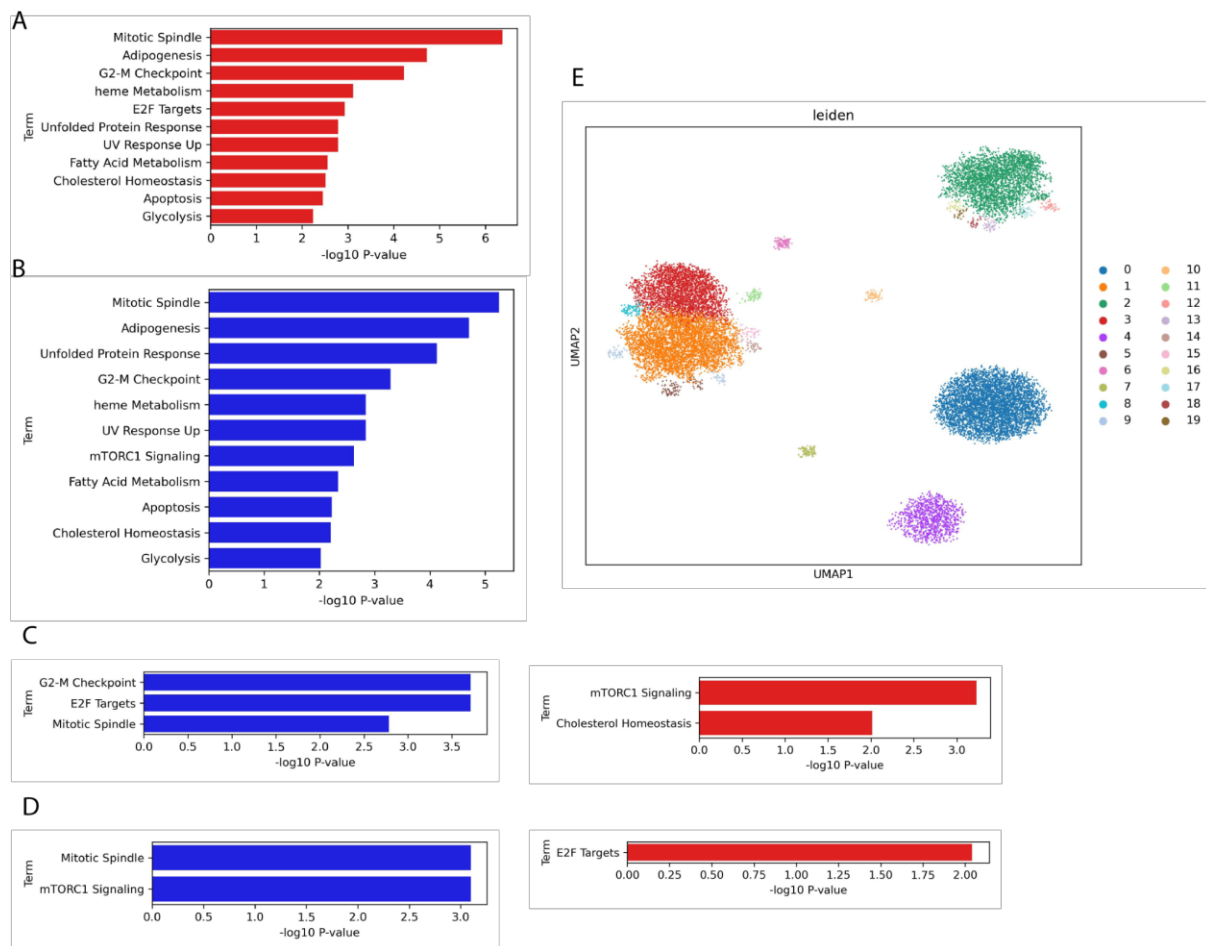


Figure 27. Enriched terms by differentially active genes. Enriched terms computed on differentially active genes in for cluster 0 (A), 1 (B), 2 (C) and 3 (D). Clusters 0 and 1 only have enriched terms for differentially accessible or inaccessible genes respectively. Terms enriched by accessible genes have blue bars and inaccessible genes have red bars. Leiden cluster distribution in the UMAP projection is shown for reference (E). Very small clusters arise due to the binning of the single-cell gene activity data (see section 2.2)

3.7 Characterization of neuroblasts expressing proliferating marker *TOP2A*.

I used my spatial integration results to characterise spatially distinct clusters of cells next and evaluated the benefits of multi-omics spatial mapping to gain biological knowledge. I selected spatially distinct populations of cells and used enrichR to derive their biological identity using the inferred scRNA-seq and TF activity scores. First, I characterised the small cluster of neuroblasts in sample 576 that express the proliferating marker *TOP2A* (Figure 28A).

3.7.1 Characterization of the proliferating neuroblasts' spatial neighbourhood.

This group of neuroblasts was located near the immune cell population of the sample and the edge of what could be a tear or a vascular channel in the tissue (**Figure 28B**). Visualising the spatial localization of the cluster and the distribution of endothelial cells along the gap, in addition to the presence of immune infiltration around this gap, suggested that it may be a vascular channel lined with endothelium (**Figure 28C**). The cells were largely grouped in Leiden cluster 8, which also included many immune cells around the aforementioned gap (**Figure 28D**).

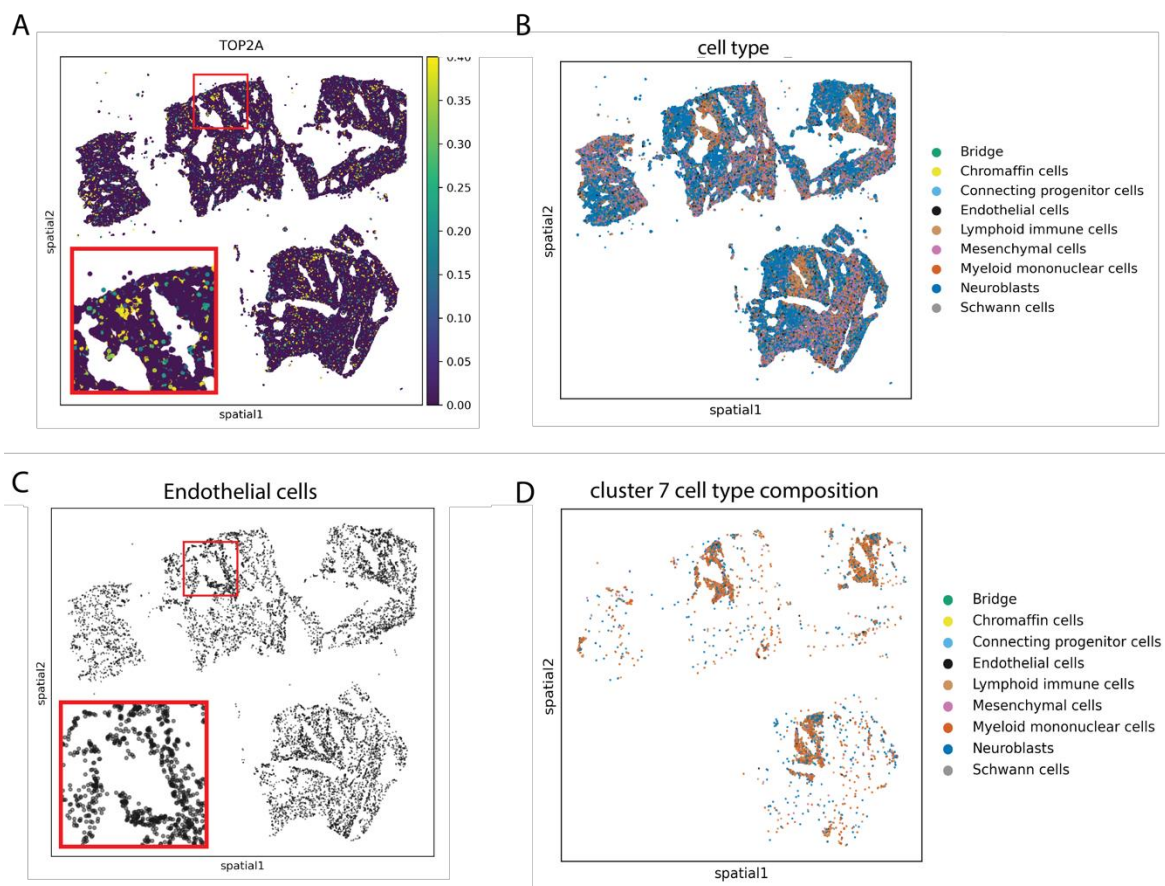


Figure 28. Spatial characterisation of neuroblast cluster expressing the proliferating marker TOP2A. (A) Expression of TOP2A across the spatially resolved sample. (B) Sample 576 coloured by cell type and (C) endothelial cells distribution in the tissue. (D) Cluster 8 distribution and cell type composition.

The cluster is composed of ca. 39,8 % myeloid mononuclear cells and 27 % neuroblasts (**Tab.6**). I filtered the cells in this cluster to only keep neuroblasts, bridge and connecting progenitor cells as I wanted to characterise the neuroblasts in this group and then computed the most discriminative TFs and genes values using `scanpy.tl.rank_genes_groups()`. I followed the

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steps described in **Figure 29** to find differentially active TFs and differentially expressed genes to run enrichR on.

Table 6. Sample 576, cluster 7 cell type composition.

Cell type	Fraction
Myeloid mononuclear cell	0,398
Neuroblasts	0,276
Lymphoid immune cells	0,219
Mesenchymal cells	0,043
Schwann cells	0,042
Endothelial cells	0,015
Bridge	0,004
Chromaffin cell	0,002
Connecting progenitor cells	0,002

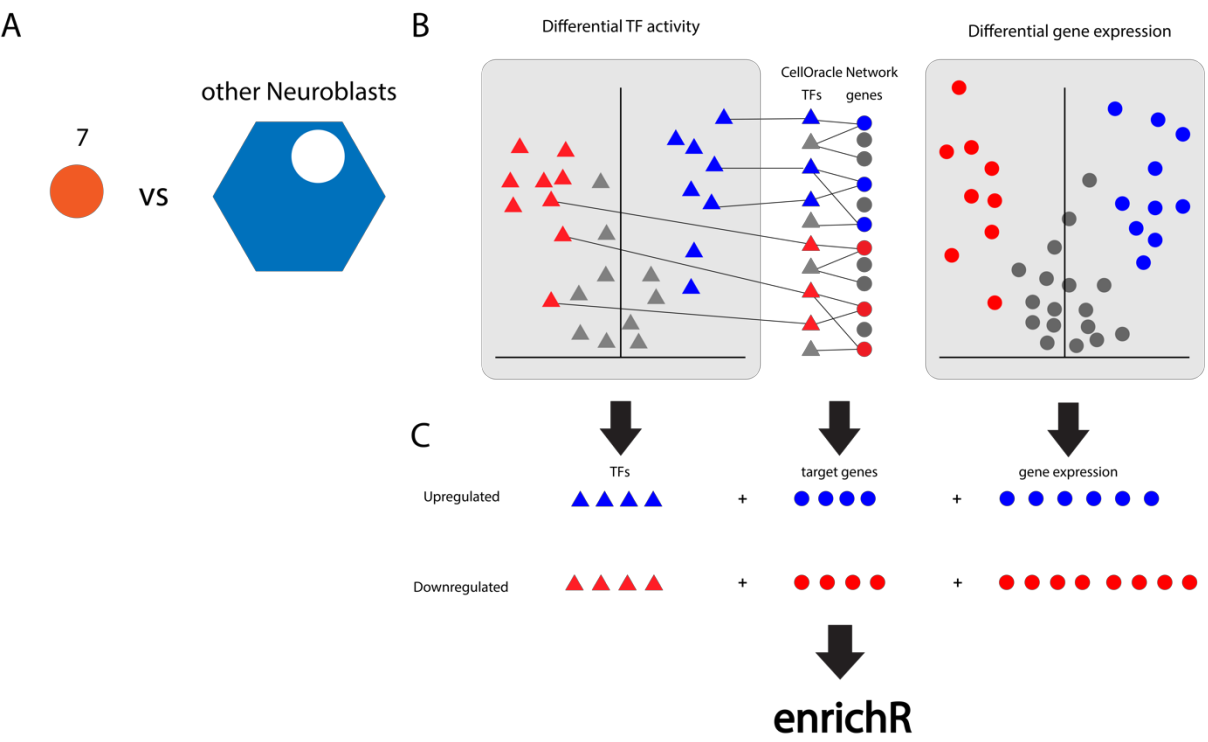


Figure 29. Multimodal differential analysis of neuroblasts in Leiden cluster 7 in sample 576. The sample is filtered to only consider the differences between neuroblasts (A). Differential analysis of transcription factor activity and gene expression are performed (B) and the up- and downregulated genes and TFs are considered for enrichment analysis with enrichR (C). Differentially active TFs are also used to select their significant target genes in the GRN computed by CellOracle.

3.7.2 Differentially expressed genes / differentially active TFs

In the differentially active TFs I found E2F1 and E2F7, two TFs members of the E2 family of transcription factors which uphold the cell cycle (Xie *et al.*, 2021), furthermore MYB plays a role in the regulation of the cell cycle as well (George *et al.*, 2014). Notably, FOXP3 appears to be less active in cluster 7 than in the rest of the neuroblasts and is generally considered a tumour suppressor albeit it was discovered to act as an oncogene in pancreatic cancer (W. Yu *et al.*, 2022) (**Figure 30 A**).

In the DGE analysis, notable overexpressed genes are *VAV3*, *LYN* and *ADAM28*. *VAV3* is an oncogene that has been shown to be a positive predictor of depth of invasion, nodal status, distant metastasis, stage and poor disease-free survival in colorectal cancer (Uen *et al.*, 2015). *LYN* encodes a kinase that promotes cancer migration and invasion (Choi *et al.*, 2010) and *ADAM28* is an oncogene associated with poor prognosis in pancreatic cancer (Wei *et al.*, 2019).

Amongst the underexpressed genes are *PDLIM4* which is associated with sensitisation of cancer cells to apoptosis (Boumber *et al.*, 2007) and *EGR1* which acts as a tumour suppressor gene by regulating cell proliferation (Zhang *et al.*, 2017). Notably, *YAP1* is an important gene that contributes to the Epithelial mesenchymal transition (Piccolo *et al.*, 2014a; Park *et al.*, 2019) similarly to the overexpressed *LYN* (**Figure 30B**).

These genes and TFs suggest a more proliferative and invasive profile for neuroblasts in cluster 7 than the remaining neuroblasts.

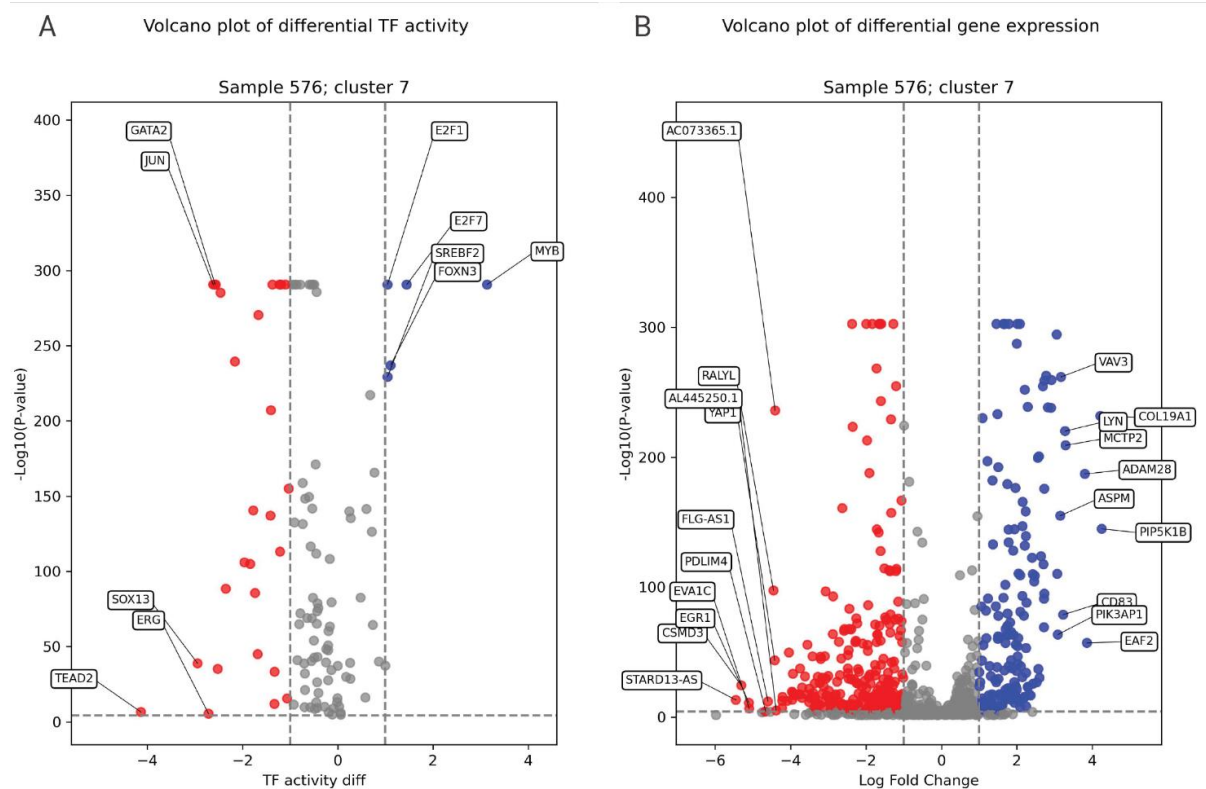


Figure 30. Differential TF activity and gene expression analysis. Neither analyses feature genes with a fraction of zero entries higher than the 0,95 to prevent fold change “explosions”. The difference of mean TF activity between neuroblasts of cluster 7 and the remaining malignant cells plotted against the p value for Wilcoxon rank sum test (A). Threshold for the activity difference was set at 1, p-value threshold was set at 10^{-5} . Log Fold Change (lfc) of gene expression plotted against p value for Wilcoxon rank sum test (B).

3.7.3 TF target inference through the GRN

To test whether the GRN targets of differentially active TFs may hold real biological information, I used a one-sided hypergeometric test. I computed whether the target genes of differentially active TF are more likely to be DEGs than other genes (**Tab.7**). The results show that target genes of upregulated TFs are disproportionately overexpressed, while there is no effect detected with underexpressed targets of the upregulated TFs.

Conversely, targets of downregulated TFs are still disproportionately overexpressed compared to the total genes and no effect is detected concerning the underexpressed targets of the downregulated TFs. The results indicate that the targets of the differentially active TFs are disproportionately overexpressed in cluster 7 neuroblasts compared to other neuroblasts regardless of the TFs activity. Based on these results, I did not include targets of downregulated TFs but did include the targets of upregulated TFs in the enrichR analysis.

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Table 7. Differential expression rate of target genes of up- and downregulated TF. The number of up- and downregulated targets of the TFs are weighted against the total number of up-and downregulated genes in the entire dataset with a hypergeometric test.

	DEGs (up)	DEGs (down)	All genes
All genes	2186	6509	11656
targets of upregulated TF	33	27	70
p-value	3.069 * 10e-8	0.997	
targets of downregulated TF	105	137	279
p-value	5.926 * 10e-14	0.987	

3.7.3 Characterization of proliferating cells by enrichR

I ran enrichR on the two TF/gene lists using the database MSigHallmark2021 built in enrichR (**Figure 31**). For upregulated genes and TFs, highly enriched terms were “G2-M Checkpoint”, “PI3K/AKT/mTOR” pathway and “E2F Targets” and indicate proliferative activity with pathways associated with the cell cycle (**Figure 31A**). The “TNF-alpha signalling” term is enriched in both up-and downregulated genes, this may be the result of the proximity of these neuroblasts to the immune cells and potentially immune escape from TNF-alpha repression (**Figure 31B**).

Terms enriched with downregulated genes and TFs instead are associated with the anti-tumoral DNA repair pathway meant to prevent skin cancer from UV light, (Lee *et al.*, 2020; van Loo *et al.*, 2023), apoptosis, the epithelial-mesenchymal transition, and angiogenesis (**Figure 31B**). The latter two suggested that unlike the other malignant cells, these neuroblasts are located already close to vascular tissue and do not require angiogenesis to access fresh oxygen and despite their growth, are not invading mesenchymal tissue or require blood vessel growth in their microenvironment. Running enrichR without the TF activity and GRN targets information yields similar enriched terms for the downregulated genes but important terms such as “G2-M checkpoint” and “E2F targets” are not significantly enriched without the information from TF activity scores (**Figure S5**).

Results

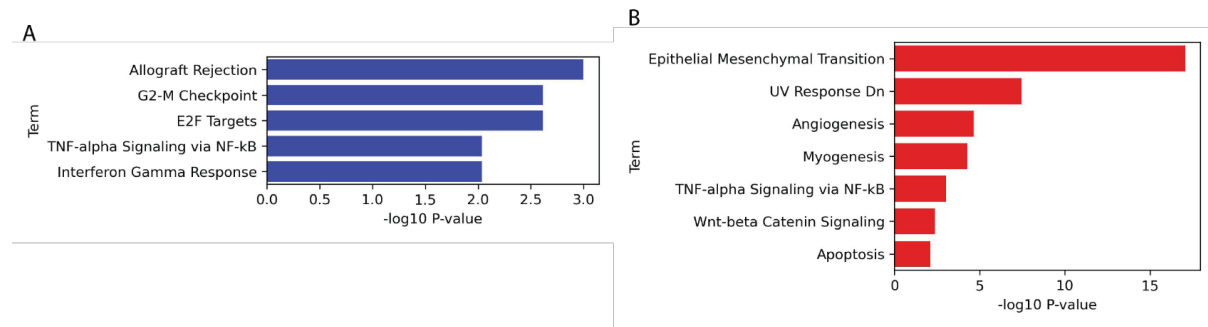


Figure 31. Significantly enriched enrichR pathways in sample 576 cluster 7. Enriched terms with an adjusted p-value < 0.01 for (A) upregulated and (B) downregulated TFs and genes.

3.8 Characterization of spatially concentric Leiden clusters.

Another spatially noticeable structure I identified is a ring-like structure in sample 398 (LR) (**Figure 32**). This low-risk tumour sample showed spatially heterogeneous clustering using spatial data. In this case I compared specifically cluster 0, 1, and 2 to identify a potential relation between these cells.

3.8.1 Cluster cell type composition

First, by comparing the composition of the clusters I observed a trend of decrease in neuroblasts per cluster towards from 0 to 2. Instead, cluster 1 and 2 appeared to have a higher percentage of neuroblast precursor cell types bridge and chromaffin cells (**Tab.8**). As such, I did not filter cells in these clusters by cell type but only ran comparisons in the inferred TF activity and gene expression between these clusters and disregarded the rest of the sample as clusters 0, 1 and 2 are largely homogeneous in terms of cell type (**Tab.8**).

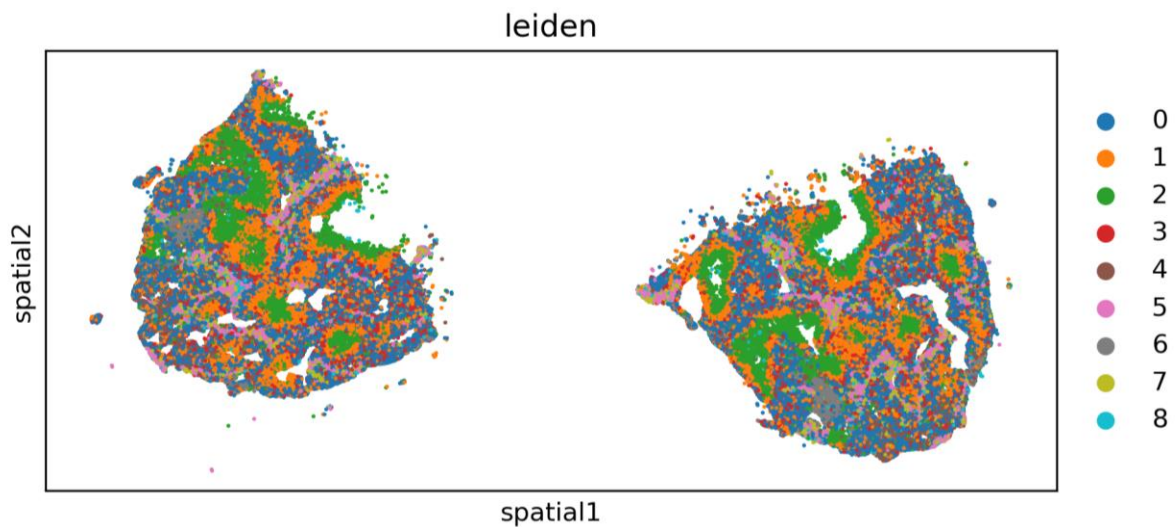


Figure 32. Spatial scatter plot of sample 398 coloured by Leiden cluster computed on gene expression values mapped by MaxFuse. The spatial distributions of cluster 1 and 2 suggest a stratified architecture with cells of cluster 2 forming spatially isolated islands separated by cells in cluster 1 from the surrounding neuroblasts in cluster 0.

Furthermore, because the spatial arrangement suggested that cluster 1 may be a transitional state between 0 and 2, I compared cluster 1 only to cluster 0 to have more interpretable results (**Figure 33C**).

Results

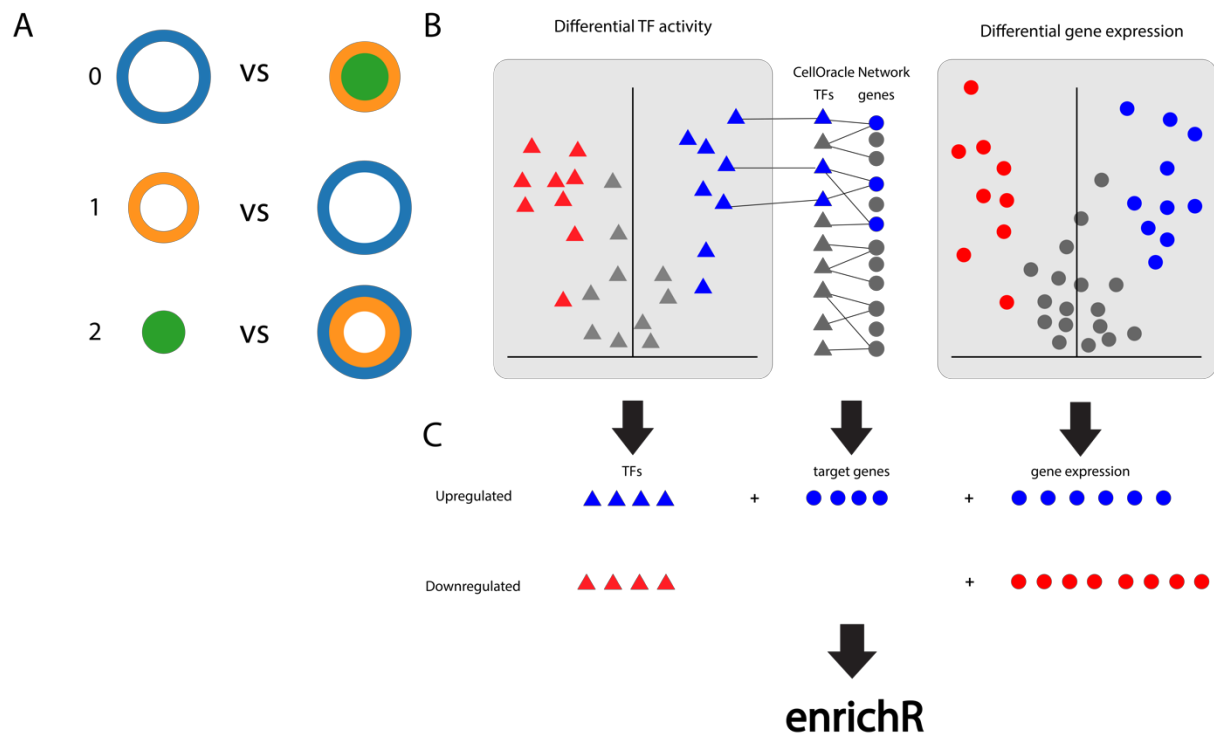


Figure 33. Overview of the multimodal differential analysis of clusters 0, 1, 2. Clusters 0 and 2 and compared to the remaining 2, while 1 is only compared to 0 to facilitate the interpretability of the results. Colours match the cluster colour is **Figure 32** (A). Differentially active TFs and differentially expressed genes are used to run the enrichR analysis, only targets of the differentially upregulated TFs are added to the list of genes due to the results of the hypergeometric test in the previous section (**Tab.8**).

Table 8. Cell type fraction composition of cluster 0, 1 and 2 in sample 398.

Cluster	0	1	2
Neuroblasts	0,957	0,942	0,847
Bridge	0,024	0,027	0,032
Schwann cells	0,007	0,007	0,012
Chromaffin cells	0,005	0,020	0,097
Mesenchymal cells	0,004	0,002	0,004
Endothelial cells	0,001	0,001	0,003
Lymphoid immune cells	0,001	0,001	0,002
Myeloid mononuclear cells	0,000	0	0,002
Connecting progenitor cells	0	0	0

3.8.2 Differentially expressed genes / differentially active TFs

An analysis of TF activity between the clusters located characteristic TFs for each cluster. NRF1, NR2F1, NR2F2 are transcription factors responsible for constitutive processes in the cell's metabolism such as cell respiration (**Figure 34B**). NR2F1 and NR2F2 in particular are hormone receptors that belong to the steroid/thyroid hormone receptor superfamily and are important for neural cell differentiation and development of the CNS. GATA3 is a TF associated with regulation of tumour growth and the downregulation of tumour growth and metastasis. SOX4 is an embryonic developmental TF responsible for cell fate determination. NR2F1, NR2F2 are comparatively less active in clusters 1 and 0.

In contrast to cluster 0, cluster 1 was characterised by MYC, GATA2 and CREB3 and DLX5 and JUN (**Figure 34C**). These TFs are all oncogenic TFs responsible for maintaining the cells pluripotency, protecting the cells from apoptosis and promote infiltration and metastasis (Huang *et al.*, 2013, p. 2; Shu *et al.*, 2015; Hu *et al.*, 2019; Tan *et al.*, 2021; Zeng *et al.*, 2024).

Finally, cluster 2 was characterised by ELK4, which promotes progression of the cell cycle as well the maintenance of pluripotency by maintaining stem-like characteristics (Peng *et al.*, 2016; Gao *et al.*, 2023). The TFs SOX11, MEIS1, TFAP2B and FOXP1 are the developmental TFs implicated in organogenesis and development of the CNS (**Figure 34D**). As mentioned before, cluster 2 and 1 both had lower activity scores for TFs NR2F1, NR2F2 compared to cluster 1; cluster 2 also shares a downregulated TF with cluster 0, TEAD2, which mediates cell proliferation, migration and epithelial mesenchymal transition (EMT) by controlling the expression of *YAP1* (Piccolo *et al.*, 2014b; LANDIN-MALT *et al.*, 2016; Shim *et al.*, 2023).

Results

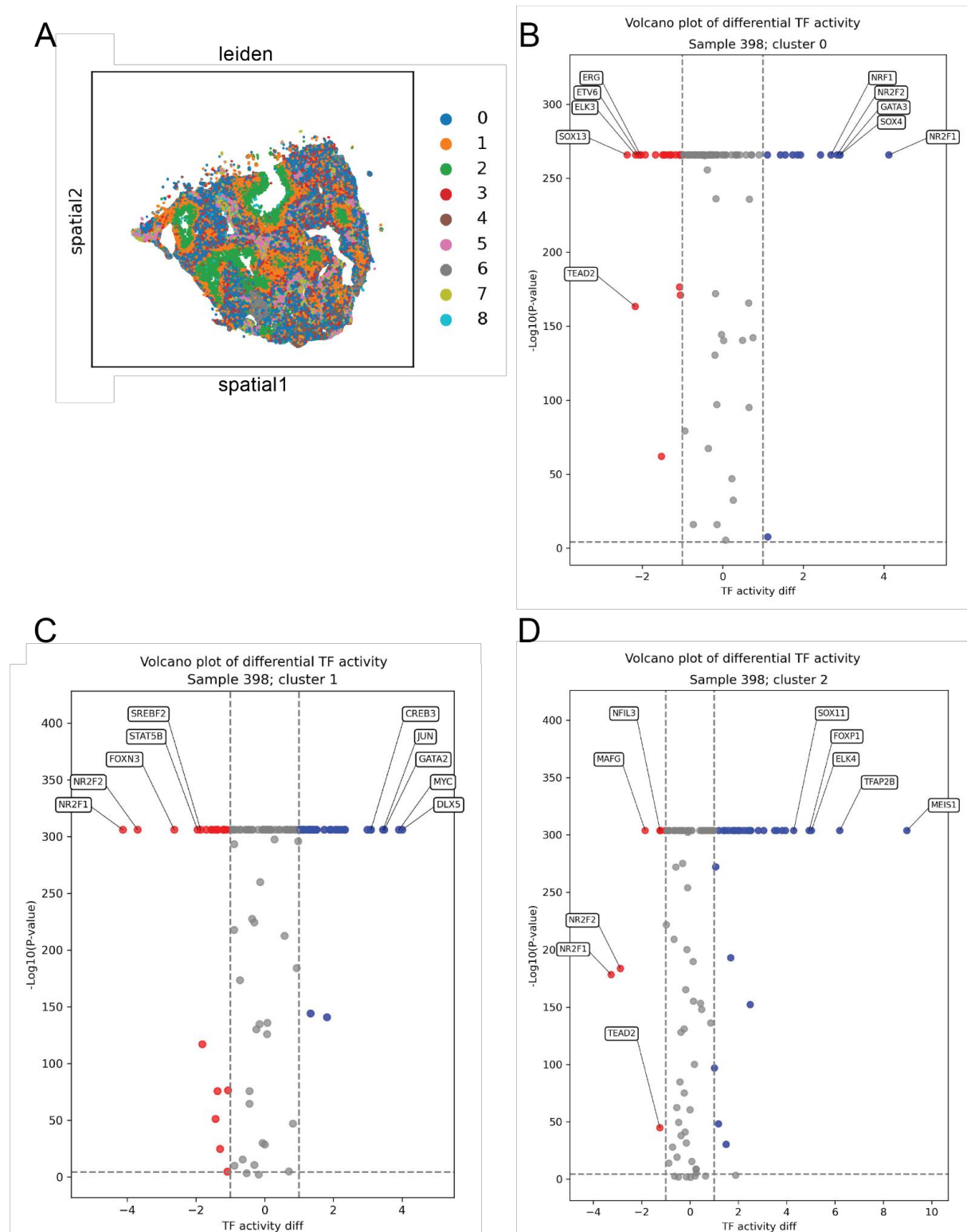


Figure 34. Differential TF activity analysis in sample 398 clusters. The Leiden cluster annotation of the spatial scatterplot (cropped) is shown for reference (A). The results of the differential TF activity analysis for cluster 0 (B), 1 (C) and 2 (D) are presented, the 5 TFs with the most extreme TF activity difference are highlighted with their name. The adjusted p value threshold is set at 5×10^{-5} . p values that were rounded to 0 were set to the lowest non-zero p value in the results.

Results

The DGE highlight similarities in the mapped gene expression values between clusters 1 and 2. These two clusters share multiple- and downregulated genes like *XIST*, *DACH2*, and the noncoding genes *CASC9*, *AC106895.2* and *LINC02055* (**Figure 35C and D**). *DACH2* is relevant to NB as it is characteristic to neural tissue and is a prognostic marker for metastasis in Carcinoma and *CASC9* is a promoter of proliferation and tissue invasion (Qiu *et al.*, 2022). *AC106895.2* has been found to be a prognostic marker for poor disease outcome in breast cancer (Shi *et al.*, 2020).

Cluster 0 and 2 also share underexpressed genes and *SLC5A7*, *ALI62493.1*; *SLC5A7* encodes a choline uptake transporter in cholinergic neurons and is expressed in mature, fully differentiated cells (Rodríguez Cruz *et al.*, 2021). *SLC5A7* is upregulated in cluster 0 (**Figure 35B**), along *LRBP1*, a gene disrupted in multiple types of cancer and is a potential tumour suppressor gene (G. Yu *et al.*, 2022).

3.8.3 Characterisation of clusters by EnrichR

The EnrichR analysis of cluster 0, which forms the majority of the sample, results in three terms that are significantly enriched by downregulated genes/TFs (p value threshold = 0,01): “G2-M Checkpoint”, “myogenesis” and “EMT” (**Figure 36B**). In their early stage, neural crest neuroblast precursors undergo EMT to migrate from the ectoderm to the neural crest, requiring high rates of myosin production and cytoskeleton modulation; thus, neuroblasts in cluster 0 may be stem-like and proliferative than the rest.

Cluster 1 presented upregulated oncogenic TFs such as GATA2 and CREB3, and consequently the enriched term “G2-M Checkpoint” associated with the cell cycle in contrast to cluster 0, (**Figure 36C**).

Finally, enrichR analysis of cluster 2 also associated this cluster with increased proliferation (E2F targets) as well as the transition of cells into a mesenchymal state (**Figure 36D**) and “apical junction proteins”. The “Epithelial-Mesenchymal Transition” terms and expression of endothelial cell proteins suggested a more undifferentiated state, as neuroblasts originated from the neural crest where they migrate from the ectoderm and undergo EMT as one of their first steps towards specialisation. Additionally, apical junction proteins can act as promoters of the epithelial-mesenchymal transition, tissue invasion and metastasis (González-Mariscal *et al.*, 2020).

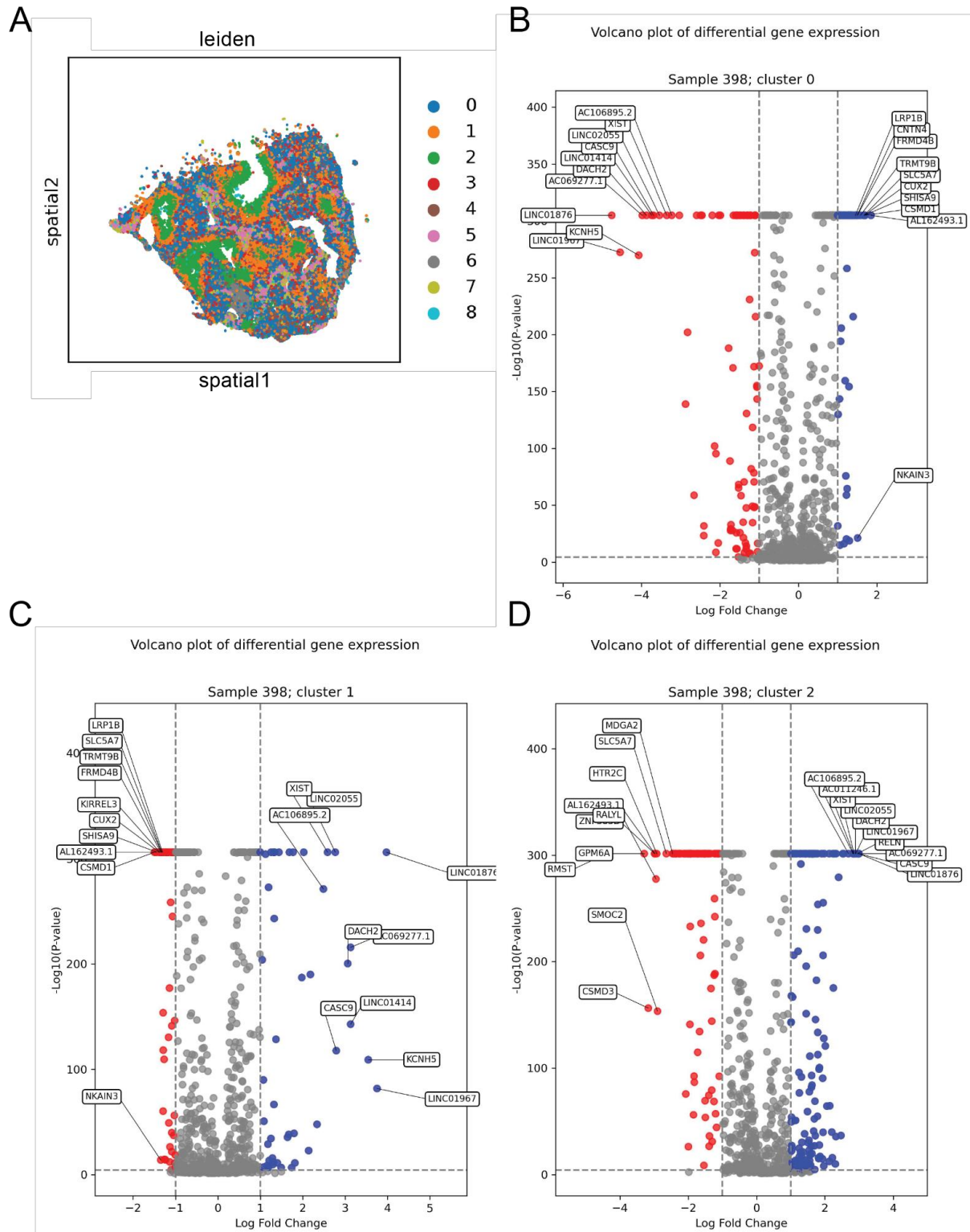


Figure 35. Differential gene expression analysis in sample 398 clusters. The Leiden cluster annotation of the spatial scatterplot (cropped) is shown for reference (A). The results of the differential TF activity analysis for cluster 0 (B), 1 (C) and 2 (D) are presented, the 5 TFs with the most extreme TF activity difference are highlighted with their name. The adjusted p value threshold is set at 5×10^{-5} . p values that were rounded to 0 were set to the lowest non-zero p value in the results.

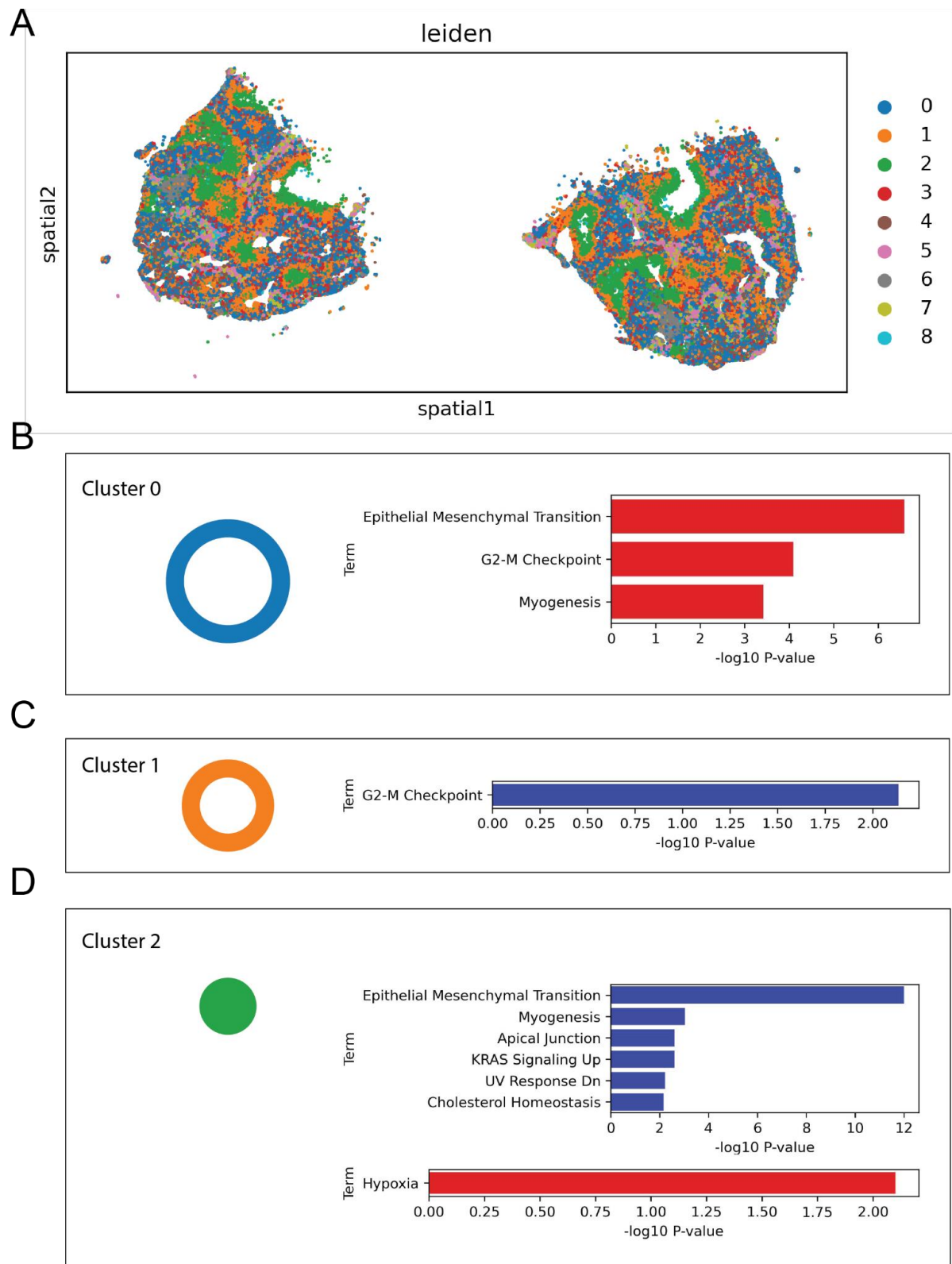


Figure 36. EnrichR analysis of clusters 0, 1, and 2. Spatial scatter plot coloured by Leiden cluster for reference (A). Enriched terms for cluster 0 (B), 1 (C) and 2 (D) are plotted with their adjusted p value. Only significantly enriched pathways (p value threshold = 0.01) are shown. Terms enriched with upregulated TFs/genes have blue bars, down regulated TFs/genes have red ones.

4. Discussion

4.1 Methodological validation of single-cell and spatial integration

Spatial transcriptomics data offers unique and unprecedented advantages to the field of molecular biology by allowing visualisation of the spatial expression of hundreds or thousands of genes, depending on the assay. However, it also presents challenges, such as the trade-off between gene panel size and the spatial resolution of the transcripts. In this project, I compared different methods to map single-cell data into spatially resolved transcriptomics samples to address this issue.

4.1.1 Nuclei segmentation improves the separation of cell types

During the preprocessing step of the analysis, I chose to segment the sample by counting only the transcripts overlapping with nuclei to limit cross-cell contamination (see section 3.3). The count of same-cell neighbours per cell yielded improved homogeneity of cell neighbourhoods in samples with lower counts and the weighted Shannon entropy per cluster showed that nuclei segmentation significantly improves cell type separation by clusterization in every sample, but by a larger margin in low count samples (**Figure 5B**). Furthermore, in sample 576, the cell segmentation-based data produced a UMAP with a cluster of immune cells in the centre (**Figure 4B**), which moved to the edge of the plot with fewer connecting cells (**Figure 5C**) after implementing nuclei segmentation. I believe that nuclei segmentation disproportionately benefits samples with lower sequencing depth due to the greater effect that contamination can have on the identity of a cell with fewer correctly assigned transcripts. Conversely, samples with many correctly transcripts may lose more identity information than they gain by removing transcripts with nuclei segmentation.

The negative effects of nuclei segmentation were noted in the step of cell type assignment. Before nuclei segmentation, we were able to run CellAssign on the samples with a small matrix of marker genes, but the results became inaccurate afterward, leading to a switch to CytoSPACE. Tumour cells grow in an uncontrolled and irregular manner, and infiltrating immune cells may have led to overlapping cells and transcripts, with the cytosol of one cell overlapping with the nucleus of another. If the same tools were applied to more organised tissue

with healthy, non-tumoral cells, I recommend re-running a comparative analysis between the cell and nuclei segmentation or trying alternative cell segmentation methods and selecting the best ones.

4.1.2 LOOCV and method selection

I used the benchmarking review of methods to map single-cell RNA-seq data onto spatial transcriptomics data as a guideline for the method comparison; but I could not use all of their metrics for evaluating the quality of tools because they were working with a large panel of annotated and analysed datasets with a reliable cell annotation, while my data was new and required to be annotated by the same tools I was testing. I chose the tools to test based on a combination of high performing tools such as Tangram and spaGE as well as trying new ones that had not been benchmarked against each other (**Tab.4**). The tool gimVI was part of the original selection, but was not included due to technical difficulties running it on our data, potentially due to the high sparsity of the SRT data.

spaGE and Tangram scored better than the MaxFuse and CytoSPACE in the leave-one-out cross validation step (see section 3.4, **Figure 8**). I believe that to be tied to their shared ability to infer gene expression values for every single cell and gene, MaxFuse and CytoSPACE suffer from the zero entries of the single-cell matrix. This leads to many poor matches when looking at one gene specifically, even though the matched cells' overall expression profiles are similar.

Despite the high Pearson correlation coefficients reached by spaGE, the inferred data did not separate cell types according to the UMAPs of both normalised and raw spaGE results (Figures **S3** and **S4**). This appears to be the product of a technical issue in the way the method runs, which is evident by the spatial distribution of the Leiden clusters in samples 576 compared to the mapping results of Tangram and MaxFuse on the same sample. Because the 10x Xenium assay device captures imaging data of the biological samples in square tiles, the square pattern of the Leiden clusters suggests that the SpaGE latent space is detecting a systemic difference between different tiles of the data and clustering the cells accordingly. This ultimately could result into cells being clustered based on the tile they were imaged if said effect determines the variation of the inferred gene expression more than real biological differences. While it would be interesting to delve deeper into how to fine tune SpaGE to overcome this issue, I preferred to proceed with Tangram since it produced comparable LOOCV scores and its results produced more coherent UMAP projections and Leiden clusters (**Figure 10**).

MaxFuse was chosen over CytoSPACE because CytoSPACE does not support the integration of weakly linked modalities but MaxFuse does. If in the future there was an desire to integrate the spatial data with single-cell proteomics data from a different cell population, one could not apply the same transfer matrix because of the different cell populations. Additionally, CytoSPACE uses a similar mapping strategy as Tangram as they both compute a similarity score between the gene expression profiles of cells, while MaxFuse's iterative network construction and refinement offer a different approach to the mapping compared to Tangram and CytoSPACE.

4.1.2 Mapping scRNA-seq with Tangram and MaxFuse

MaxFuse's LOOCV scores are not as impressive as Tangram's and spaGE's, they align with the benchmark's expectations. MaxFuse's better separation of cell types highlights the benefits of one-to-one mapping, which preserves the distribution statistics of the original scRNA-seq data and enhances differences between cells. This is a trend illustrated by the UMAP projections across multiple modalities (**Figures 10A.2, 17A.2, 23A.2**). However, the expression of marker genes for different subtypes of malignant cells appeared sparser and more inconsistent with MaxFuse in contrast to Tangram (**Figure 13**).

Tangram's continuous mapping matrix, in contrast to MaxFuse's one-to-one matching, results in diminished cell separation. Tangram's scRNA-seq results decrease cell type separation in every sample (**Table S4**). Despite this, neuroblast subtype marker genes are still mapped to distinct areas similar to MaxFuse (**Figures 13**) and the expression is present in most neuroblasts unlike in the results mapped by MaxFuse.

Attempts to improve mapping by spatially smoothing gene expression using diffusion along the spatial graph constructed with squidpy did not enhance the final gene expression mapping results. Including more genes, irrespective of their sparsity in the SRT dataset, also did not improve cell type separation. However, the scRNA-seq mapping from single-cell to Xenium data still leads to cells aggregating in cell type-specific regions in the UMAPs (**Figure 10**).

After computing the high-ranking genes for each method's result, I was pleased to see that the cell types were hierarchically clustered in line with the biological ground truth we expect. Chromaffin, bridge and neuroblast cells were always clustered together first, as were the two lineages of immune cells. Schwann and Mesenchymal cells were also clustered together as they both are part of the extracellular scaffolding for neurons. While this clusterization is not always

preserved exactly in the next sections, MaxFuse results generally maintain a biologically sound clusterization with even distances between each group.

4.1.3 TF and gene activity mapping

Beyond scRNA-seq, using Tangram and MaxFuse to map TF activities yielded mixed results. Normalising Tangram's mapping matrix for scRNA-seq values yielded good cell type marker TFs and the expected spatial distribution of neuroblast subtype marker TFs in sample 576. I observed a great increase in the number Leiden cluster per sample with no change of parameters, which might be a by-product of the additional Tangram matrix normalisation step I performed.

Table S4 and Figure S5 show the weighted Shannon entropy sums for each sample, method and -omics type that was mapped. I did not observe a significant difference in Shannon entropy between Tangram and MaxFuse TF activity results (**Figure S5**). This may be due to the low dimensionality of the TF dataset which compensates for the general smoothing effect of the Tangram Mapping matrix. Both tools infer spatially distinct neuroblast subtype marker TF activities in spatially distinct regions of sample 576 in line with the scRNA-seq mapping results. Based on other observations such as the UMAP projection of the data and the hierarchical clustering of cell types based of TF activity, I would still rely on MaxFuse to map TF activities.

When mapping gene activity, MaxFuse leads to a far better separation observable in the UMAP projection as well as in the signed Wilcoxon rank test (**Figure S5**). Gene activity data was more difficult to interpret, but the enrichR results conducted on the Leiden clusters suggest that the mapping results of gene activity carry meaningful information for the characterisation of the cells (**Figure 27**).

MaxFuse overall offers a better option than Tangram, with no significant downsides except the one-to-one mapping of binned scores resulting in too many similar cells, necessitating augmentation through additive noise to construct a fully connected neighbourhood graph (see section 2.2). This issue pertains to the input data and does not affect my assessment of MaxFuse's performance.

4.1.4 Conclusions on Tangram and MaxFuse

In summary, Tangram and MaxFuse offer distinct advantages and disadvantages, making them valid alternatives for mapping scRNA-seq data. The choice between the two should depend on the data characteristics and the specific aspects of the data that the researcher wants to

emphasise. Tangram provides a more nuanced interpretation of spatial data by inferring gene expression levels for every cell and gene, with its mapping matrix offering greater interpretability. In contrast, MaxFuse should be used to emphasise broader differences within the data, despite continuing to use sparse data post-mapping. For mapping TF and gene activity scores to spatial data, MaxFuse is superior to Tangram. Additionally, MaxFuse can map other well-linked modalities, such as proteomics and transcriptomics, offering further flexibility and utility in spatial transcriptomics studies.

I recognize that some evaluation criteria used in this section rely on qualitative results over quantitative. As the Xenium data was new and missing e.g. an official ground truth annotation of cell types, I used metrics such as cell neighbourhood composition and the weighted entropy to quantify the results I base my conclusions on. Nevertheless, the interpretation of the quality of UMAP projections, as well as the validation of marker genes/TFs through the literature were indispensable evaluation criteria to me as this project has a broad scope and many possible avenues to explore. I believe MaxFuse and Tangram are excellent tools to use for the integration of scRNA-seq data and SRT data generated by the 10x Xenium assay and that MaxFuse should be used to map other single-cell data types.

4.2 Biological characterization of spatially resolved clusters

4.2.1 Proliferating neuroblasts in sample 576

I characterised the spatially resolved Leiden clusters after validating the mapping methods to display the benefits of spatial transcriptomics and especially of mapping multimodal single-cell data. The enrichment analysis corroborated the proliferative profile suggested by the expression of the marker gene *TOP2A* in the neuroblasts of cluster 7. The cluster of highly proliferating cells in sample 576 is associated with high proliferation, low anti-tumoral DNA damage repair, low infiltration of mesenchymal tissue and low angiogenesis rates. (**Figure 31**). Due to its proximity to endothelial cells lining a gap in the tissue surrounded by immune cells, it is potentially a group of neuroblasts in the process of metastasizing into the bloodstream by maximising their proliferative rate while escaping immune attack (**Figure 28**).

The results of the hypergeometric test suggested that targets of upregulated TFs are more likely overexpressed, but the targets of downregulated TFs are not more likely to be underexpressed, on the contrary they are also overexpressed (**Tab.7**). Genes can be targeted by several TFs, as

such, a highly active transcription factor results in the expression of its targets, the opposite is not necessarily true. This can explain why targets of downregulated genes are not significantly more underexpressed than the rest, but does not explain why they are overexpressed. As the differential analysis is only run between neuroblasts, it is not explainable by a general overexpression in all neuroblasts either.

Due to the limited number of TFs, it is difficult to assess how many active TFs are targeting each gene. Additionally, TF activity changes results in a change in gene expression with a time delay, if the traits of neuroblasts in cluster 7 have evolved only recently in the tumour, there might be changes to the genome and the TF activity patterns that have not materialised in gene expression shifts yet. The inclusion of TF activity and their targets to the list of gene names passed to enrichR improved the results of enrichR, leading to the enrichment of the terms tied to the proliferation of neuroblasts that were seeking to validate the neuroblast subtype identification (**Figure S6**). This is evidence of the benefits of mapping multiple data types onto spatial transcriptomics data. Transcription factor activity leverages information contained in gene expression and chromatin accessibility data to infer more generalised information about the cellular processes active in the cells. The gene activity data was not included in this enrichment analysis due to its different biological significance. Further exploration of this multi-omics spatially resolved integration could attempt to use gene activity to filter target genes of active TFs by only keeping those that the TFs could have accessibility to.

4.2.2 Concentric neuroblast subpopulations in sample 398

In sample 398 instead we observe small circular groups of cells composed of neuroblasts grouped into one cluster (2) and surrounded by another group of neuroblasts clustered separately (1) that separate them from the largest “scaffold” cluster of neuroblasts (0) (**Figure 32**). The TFs and genes found by the differential analysis suggest that cluster 0 and 1 are less differentiated and more stem-like than cluster 2. The larger fraction of bridge cells and the enrichment analysis performed with enrichR suggests that cluster 2 is made of proliferating clusters of less differentiated cells that are in the process of invading mesenchymal tissue.

Many positively enriched terms of cluster 2 are present in cluster 0 as negatively enriched terms, while cluster 1 only features positively enriched G2-M Checkpoint, which is negatively enriched in cluster 0. Due to their spatial arrangement, and the down regulated proliferation of

0 compared to 2, it is possible that cells in cluster 2 may be radially expanding and transitioning into cluster 1 and finally cluster 0 cells (**Figure 36**).

These insights are possible due to the spatial resolution of the data, the high resolution of the Xenium assay combined with the whole transcriptome sequencing of cell-cell resolution and the epigenomic information in the TF activity data that we have access to.

4.3 Outlook

The additional data types increase our understanding of the tissue and can help us find new relationships between cell populations as well as drive the formulation of hypotheses through spatial organisation of variables in the tissue. This integration could be pushed further by applying methods to infer spatial cell-cell interactions based on the spatial co-expression of ligand and receptor protein pairs such as squidpy's ligrec, CellChat or CellphoneDB (Efremova *et al.*, 2020; Jin *et al.*, 2021; Palla *et al.*, 2022). For such tool I recommend to continue using MaxFuse results, as these tools are often designed to be run on sparse log-normalised count matrices and Tangram's output may not be suitable for them (Zhang *et al.*, 2021). Cell-cell communication is an aspect of the tumour microenvironment that holds much promise for therapeutic approaches but many of its aspects still require clarification (Dominiak *et al.*, 2020; AlMusawi *et al.*, 2021; Lee *et al.*, 2024). Exploring the cell-cell communication in the NB microenvironment could offer information about how tumour subpopulations interact and communicate as well as immune evasion strategies and as points of attack for new therapeutic approaches (Arora *et al.*, 2023; Patel *et al.*, 2024).

References

- Abdelaal, T. *et al.* (2020) ‘spaGE: Spatial Gene Enhancement using scRNA-seq’, *Nucleic Acids Research*, 48(18), p. e107. Available at: <https://doi.org/10.1093/nar/gkaa740>.
- Adamo, A. *et al.* (2023) ‘Identification and interrogation of the gene regulatory network of CEBPA-double mutant acute myeloid leukemia’, *Leukemia*, 37(1), pp. 102–112. Available at: <https://doi.org/10.1038/s41375-022-01744-5>.
- Afouda, B.A. (2022) ‘Towards Understanding the Gene-Specific Roles of GATA Factors in Heart Development: Does GATA4 Lead the Way?’, *International Journal of Molecular Sciences*, 23(9), p. 5255. Available at: <https://doi.org/10.3390/ijms23095255>.
- Ahlmann-Eltze, C. and Huber, W. (2023) ‘Comparison of transformations for single-cell RNA-seq data’, *Nature Methods*, 20(5), pp. 665–672. Available at: <https://doi.org/10.1038/s41592-023-01814-1>.
- Apellániz-Ruiz, M. *et al.* (2017) ‘Targeted Sequencing Reveals Low-Frequency Variants in EPHA Genes as Markers of Paclitaxel-Induced Peripheral Neuropathy’, *Clinical Cancer Research: An Official Journal of the American Association for Cancer Research*, 23(5), pp. 1227–1235. Available at: <https://doi.org/10.1158/1078-0432.CCR-16-0694>.
- Arora, R. *et al.* (2023) ‘Spatial transcriptomics reveals distinct and conserved tumor core and edge architectures that predict survival and targeted therapy response’, *Nature Communications*, 14(1), p. 5029. Available at: <https://doi.org/10.1038/s41467-023-40271-4>.
- Badia-i-Mompel, P. *et al.* (2022) ‘decoupleR: ensemble of computational methods to infer biological activities from omics data’, *Bioinformatics Advances*, 2(1), p. vbac016. Available at: <https://doi.org/10.1093/bioadv/vbac016>.
- Biancalani, T. *et al.* (2021) ‘Deep learning and alignment of spatially resolved single-cell transcriptomes with Tangram’, *Nature Methods*, 18(11), pp. 1352–1362. Available at: <https://doi.org/10.1038/s41592-021-01264-7>.
- Boeva, V. *et al.* (2017) ‘Heterogeneity of neuroblastoma cell identity defined by transcriptional circuitries’, *Nature Genetics*, 49(9), pp. 1408–1413. Available at: <https://doi.org/10.1038/ng.3921>.

References

- Boumber, Y.A. *et al.* (2007) ‘RIL, a LIM gene on 5q31, is silenced by methylation in cancer and sensitizes cancer cells to apoptosis’, *Cancer Research*, 67(5), pp. 1997–2005. Available at: <https://doi.org/10.1158/0008-5472.CAN-06-3093>.
- Cable, D.M. *et al.* (2022) ‘Robust decomposition of cell type mixtures in spatial transcriptomics’, *Nature Biotechnology*, 40(4), pp. 517–526. Available at: <https://doi.org/10.1038/s41587-021-00830-w>.
- Chen, E.Y. *et al.* (2013) ‘Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool’, *BMC Bioinformatics*, 14, p. 128. Available at: <https://doi.org/10.1186/1471-2105-14-128>.
- Chen, S. *et al.* (2023a) ‘Integration of spatial and single-cell data across modalities with weak linkage’, *bioRxiv: The Preprint Server for Biology*, p. 2023.01.12.523851. Available at: <https://doi.org/10.1101/2023.01.12.523851>.
- Chen, S. *et al.* (2023b) ‘Integration of spatial and single-cell data across modalities with weakly linked features’, *Nature Biotechnology*, pp. 1–11. Available at: <https://doi.org/10.1038/s41587-023-01935-0>.
- Chen, T.-Y. *et al.* (2023) ‘Spatial Transcriptomic Technologies’, *Cells*, 12(16), p. 2042. Available at: <https://doi.org/10.3390/cells12162042>.
- Cheng, M. *et al.* (2023) ‘Spatially resolved transcriptomics: a comprehensive review of their technological advances, applications, and challenges’, *Journal of Genetics and Genomics*, 50(9), pp. 625–640. Available at: <https://doi.org/10.1016/j.jgg.2023.03.011>.
- Cheung, N.-K.V. and Dyer, M.A. (2013) ‘Neuroblastoma: developmental biology, cancer genomics and immunotherapy’, *Nature Reviews Cancer*, 13(6), pp. 397–411. Available at: <https://doi.org/10.1038/nrc3526>.
- Choi, Y.-L. *et al.* (2010) ‘LYN is a mediator of epithelial-mesenchymal transition and target of dasatinib in breast cancer’, *Cancer research*, 70(6), pp. 2296–2306. Available at: <https://doi.org/10.1158/0008-5472.CAN-09-3141>.
- Denisenko, E. *et al.* (2024) ‘Spatial transcriptomics reveals discrete tumour microenvironments and autocrine loops within ovarian cancer subclones’, *Nature Communications*, 15(1), p. 2860. Available at: <https://doi.org/10.1038/s41467-024-47271-y>.

References

- Ercan-Sencicek, A.G. *et al.* (2015) ‘Homozygous loss of DIAPH1 is a novel cause of microcephaly in humans’, *European Journal of Human Genetics*, 23(2), pp. 165–172. Available at: <https://doi.org/10.1038/ejhg.2014.82>.
- Farrell, J.A. (2023) ‘An oracle predicts regulators of cell identity’, *Nature*, 614(7949), pp. 630–632. Available at: <https://doi.org/10.1038/d41586-023-00251-6>.
- Feutrill, A. and Roughan, M. (2021) ‘A Review of Shannon and Differential Entropy Rate Estimation’, *Entropy*, 23(8), p. 1046. Available at: <https://doi.org/10.3390/e23081046>.
- Fu, X. *et al.* (2024) ‘BIDCell: Biologically-informed self-supervised learning for segmentation of subcellular spatial transcriptomics data’, *Nature Communications*, 15(1), p. 509. Available at: <https://doi.org/10.1038/s41467-023-44560-w>.
- Gao, F. *et al.* (2023) ‘ELK4 Promotes Cell Cycle Progression and Stem Cell-like Characteristics in HPV-associated Cervical Cancer by Regulating the FBXO22/PTEN Axis’, *Balkan Medical Journal*, 40(6), pp. 409–414. Available at: <https://doi.org/10.4274/balkanmedj.galenos.2023.2023-4-66>.
- George, O.L. and Ness, S.A. (2014) ‘Situational Awareness: Regulation of the Myb Transcription Factor in Differentiation, the Cell Cycle and Oncogenesis’, *Cancers*, 6(4), pp. 2049–2071. Available at: <https://doi.org/10.3390/cancers6042049>.
- González-Mariscal, L. *et al.* (2020) ‘Relationship between apical junction proteins, gene expression and cancer’, *Biochimica Et Biophysica Acta. Biomembranes*, 1862(9), p. 183278. Available at: <https://doi.org/10.1016/j.bbamem.2020.183278>.
- Graves, C.A. *et al.* (2014) ‘The translational significance of epithelial-mesenchymal transition in head and neck cancer’, *Clinical and Translational Medicine*, 3, p. 39. Available at: <https://doi.org/10.1186/s40169-014-0039-9>.
- Hu, Y. *et al.* (2019) ‘Knockdown of CREB3 activates endoplasmic reticulum stress and induces apoptosis in glioblastoma’, *Aging*, 11(19), pp. 8156–8168. Available at: <https://doi.org/10.18632/aging.102310>.
- Huang, M. and Weiss, W.A. (2013) ‘Neuroblastoma and MYCN’, *Cold Spring Harbor Perspectives in Medicine*, 3(10), p. a014415. Available at: <https://doi.org/10.1101/cshperspect.a014415>.

References

- Janesick, A. *et al.* (2023) 'High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis', *Nature Communications*, 14(1), p. 8353. Available at: <https://doi.org/10.1038/s41467-023-43458-x>.
- Ji, L. *et al.* (2021) 'The role of Dock2 on macrophage migration and functions during *Citrobacter rodentium* infection', *Clinical and Experimental Immunology*, 204(3), pp. 361–372. Available at: <https://doi.org/10.1111/cei.13590>.
- Jin, Y. *et al.* (2017) 'CACNA2D3 is downregulated in gliomas and functions as a tumor suppressor', *Molecular Carcinogenesis*, 56(3), pp. 945–959. Available at: <https://doi.org/10.1002/mc.22548>.
- Kameneva, P. *et al.* (2021) 'Single-cell transcriptomics of human embryos identifies multiple sympathoblast lineages with potential implications for neuroblastoma origin', *Nature genetics*, 53(5), pp. 694–706. Available at: <https://doi.org/10.1038/s41588-021-00818-x>.
- Kearney, C.J., Randall, K.L. and Oliaro, J. (2017) 'DOCK8 regulates signal transduction events to control immunity', *Cellular & Molecular Immunology*, 14(5), pp. 406–411. Available at: <https://doi.org/10.1038/cmi.2017.9>.
- Kuleshov, M.V. *et al.* (2016) 'Enrichr: a comprehensive gene set enrichment analysis web server 2016 update', *Nucleic Acids Research*, 44(Web Server issue), pp. W90–W97. Available at: <https://doi.org/10.1093/nar/gkw377>.
- Lan, C. *et al.* (2021) 'Association between ABHD1 and DOK6 polymorphisms and susceptibility to Hirschsprung disease in Southern Chinese children', *Journal of Cellular and Molecular Medicine*, 25(20), pp. 9609–9616. Available at: <https://doi.org/10.1111/jcmm.16905>.
- LANDIN-MALT, A. *et al.* (2016) 'An evolutionary, structural and functional overview of the mammalian TEAD1 and TEAD2 transcription factors', *Gene*, 591(1), pp. 292–303. Available at: <https://doi.org/10.1016/j.gene.2016.07.028>.
- Lee, J.W. *et al.* (2020) 'Deciphering UV-induced DNA Damage Responses to Prevent and Treat Skin Cancer', *Photochemistry and photobiology*, 96(3), pp. 478–499. Available at: <https://doi.org/10.1111/php.13245>.

References

- Li, Bin *et al.* (2022) ‘Benchmarking spatial and single-cell transcriptomics integration methods for transcript distribution prediction and cell type deconvolution’, *Nature Methods*, 19(6), pp. 662–670. Available at: <https://doi.org/10.1038/s41592-022-01480-9>.
- Li, B. *et al.* (2022) ‘[Expression and clinical significance of COL1A1 and COL1A2 genes in malignant pleural mesothelioma tissues].’, *Zhonghua lao dong wei sheng zhi ye bing za zhi = Zhonghua laodong weisheng zhiyebing zazhi = Chinese journal of industrial hygiene and occupational diseases*, 40 7, pp. 487–494. Available at: <https://doi.org/10.3760/cma.j.cn121094-20210621-00298>.
- Li, H. *et al.* (2023) ‘A comprehensive benchmarking with practical guidelines for cellular deconvolution of spatial transcriptomics’, *Nature Communications*, 14(1), p. 1548. Available at: <https://doi.org/10.1038/s41467-023-37168-7>.
- Lin, C.-M. *et al.* (2013) ‘Clinical and Prognostic Implications of Transcription Factor SOX4 in Patients with Colon Cancer’, *PLoS ONE*, 8. Available at: <https://doi.org/10.1371/journal.pone.0067128>.
- Liu, B. *et al.* (2020) ‘An entropy-based metric for assessing the purity of single cell populations’, *Nature Communications*, 11, p. 3155. Available at: <https://doi.org/10.1038/s41467-020-16904-3>.
- Liu, B., Li, Y. and Zhang, L. (2021) ‘Analysis and Visualization of Spatial Transcriptomic Data’, *Frontiers in Genetics*, 12, p. 785290. Available at: <https://doi.org/10.3389/fgene.2021.785290>.
- Liu, M. *et al.* (2024) ‘Spatial transcriptomics reveals segregation of tumor cell states in glioblastoma and marked immunosuppression within the perinecrotic niche’, *Acta Neuropathologica Communications*, 12(1), p. 64. Available at: <https://doi.org/10.1186/s40478-024-01769-0>.
- van Loo, G. and Bertrand, M.J.M. (2023) ‘Death by TNF: a road to inflammation’, *Nature Reviews Immunology*, 23(5), pp. 289–303. Available at: <https://doi.org/10.1038/s41577-022-00792-3>.
- Ma, C.Z. and Brent, M.R. (2020) ‘Inferring TF activities and activity regulators from gene expression data with constraints from TF perturbation data’, *Bioinformatics*, 37(9), pp. 1234–1245. Available at: <https://doi.org/10.1093/bioinformatics/btaa947>.

References

- Maris, J. *et al.* (2008) 'Chromosome 6p22 locus associated with clinically aggressive neuroblastoma.', *The New England journal of medicine*, 358 24, pp. 2585–93. Available at: <https://doi.org/10.1056/NEJMoa0708698>.
- Martínez-Pacheco, M.L., Hernández-Lemus, E. and Mejía, C. (2023) 'Analysis of High-Risk Neuroblastoma Transcriptome Reveals Gene Co-Expression Signatures and Functional Features', *Biology*, 12(9), p. 1230. Available at: <https://doi.org/10.3390/biology12091230>.
- Mayanagi, T. and Sobue, K. (2011) 'Diversification of caldesmon-linked actin cytoskeleton in cell motility', *Cell Adhesion & Migration*, 5(2), pp. 150–159. Available at: <https://doi.org/10.4161/cam.5.2.14398>.
- McInnes, L., Healy, J. and Melville, J. (2020) 'UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction'. arXiv. Available at: <https://doi.org/10.48550/arXiv.1802.03426>.
- Mei, R. *et al.* (2021) 'Evidence That ITPR2-Mediated Intracellular Calcium Release in Oligodendrocytes Regulates the Development of Carbonic Anhydrase II + Type I/II Oligodendrocytes and the Sizes of Myelin Fibers', *Frontiers in Cellular Neuroscience*, 15, p. 751439. Available at: <https://doi.org/10.3389/fncel.2021.751439>.
- Monclair, T. *et al.* (2009) 'The International Neuroblastoma Risk Group (INRG) Staging System: An INRG Task Force Report', *Journal of Clinical Oncology*, 27(2), pp. 298–303. Available at: <https://doi.org/10.1200/JCO.2008.16.6876>.
- Muckian, C. *et al.* (2002) 'Genetic variability in the extracellular matrix as a determinant of cardiovascular risk: association of type III collagen COL3A1 polymorphisms with coronary artery disease.', *Blood*, 100 4, pp. 1220–3. Available at: <https://doi.org/10.1182/BLOOD-2002-01-0283>.
- Norris, M. *et al.* (2012) 'Abstract 789: GSTP1 expression is regulated by MycN and is a marker of poor outcome in childhood neuroblastoma', *Cancer Research*, 72, pp. 789–789. Available at: <https://doi.org/10.1158/1538-7445.AM2012-789>.
- Ohkura, T. *et al.* (2004) 'Detection of the novel autoantibody (anti-UACA antibody) in patients with Graves' disease', *Biochemical and Biophysical Research Communications*, 321(2), pp. 432–440. Available at: <https://doi.org/10.1016/j.bbrc.2004.06.162>.

References

- Palla, G. *et al.* (2022) ‘Squidpy: a scalable framework for spatial omics analysis’, *Nature Methods*, 19(2), pp. 171–178. Available at: <https://doi.org/10.1038/s41592-021-01358-2>.
- Park, J. *et al.* (2019) ‘Switch-like enhancement of epithelial-mesenchymal transition by YAP through feedback regulation of WT1 and Rho-family GTPases’, *Nature Communications*, 10(1), p. 2797. Available at: <https://doi.org/10.1038/s41467-019-10729-5>.
- Patel, A.G. *et al.* (2024) *A spatial cell atlas of neuroblastoma reveals developmental, epigenetic and spatial axis of tumor heterogeneity*. preprint. Cancer Biology. Available at: <https://doi.org/10.1101/2024.01.07.574538>.
- Patel, A.P. *et al.* (2014) ‘Single-cell RNA-seq highlights intratumoral heterogeneity in primary glioblastoma’, *Science (New York, N.Y.)*, 344(6190), pp. 1396–1401. Available at: <https://doi.org/10.1126/science.1254257>.
- Peng, C. *et al.* (2016) ‘Cyclin-dependent kinase 2 (CDK2) is a key mediator for EGF-induced cell transformation mediated through the ELK4/c-Fos signaling pathway’, *Oncogene*, 35(9), pp. 1170–1179. Available at: <https://doi.org/10.1038/onc.2015.175>.
- Piccolo, S., Dupont, S. and Cordenonsi, M. (2014a) ‘The Biology of YAP/TAZ: Hippo Signaling and Beyond’, *Physiological Reviews*, 94(4), pp. 1287–1312. Available at: <https://doi.org/10.1152/physrev.00005.2014>.
- Piccolo, S., Dupont, S. and Cordenonsi, M. (2014b) ‘The Biology of YAP/TAZ: Hippo Signaling and Beyond’, *Physiological Reviews*, 94(4), pp. 1287–1312. Available at: <https://doi.org/10.1152/physrev.00005.2014>.
- Pliner, H.A. *et al.* (2018) ‘Cicero Predicts cis-Regulatory DNA Interactions from Single-Cell Chromatin Accessibility Data’, *Molecular Cell*, 71(5), pp. 858-871.e8. Available at: <https://doi.org/10.1016/j.molcel.2018.06.044>.
- Probability Inequalities for Sums of Bounded Random Variables: Journal of the American Statistical Association: Vol 58, No 301* (no date). Available at: <https://www.tandfonline.com/doi/abs/10.1080/01621459.1963.10500830> (Accessed: 27 June 2024).

References

- Qiu, H. *et al.* (2022) ‘LncRNA CASC9 promotes cell proliferation and invasion in osteosarcoma through targeting miR-874-3p/SOX12 axis’, *Journal of Orthopaedic Surgery and Research*, 17(1), p. 460. Available at: <https://doi.org/10.1186/s13018-022-03340-w>.
- R Core Team (2023). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>
- Rey, D. and Neuhausser, M. (2011) ‘Wilcoxon-Signed-Rank Test’, in M. Lovric (ed.) *International Encyclopedia of Statistical Science*. Berlin, Heidelberg: Springer, pp. 1658–1659. Available at: https://doi.org/10.1007/978-3-642-04898-2_616.
- Rodríguez Cruz, P.M. *et al.* (2021) ‘Presynaptic congenital myasthenic syndrome due to three novel mutations in SLC5A7 encoding the sodium-dependant high-affinity choline transporter’, *Neuromuscular disorders: NMD*, 31(1), pp. 21–28. Available at: <https://doi.org/10.1016/j.nmd.2020.10.006>.
- Russell, M.R. *et al.* (2015) ‘CASC15-S Is a Tumor Suppressor lncRNA at the 6p22 Neuroblastoma Susceptibility Locus’, *Cancer Research*, 75(15), pp. 3155–3166. Available at: <https://doi.org/10.1158/0008-5472.CAN-14-3613>.
- Shah, S. *et al.* (2016) ‘In Situ Transcription Profiling of Single Cells Reveals Spatial Organization of Cells in the Mouse Hippocampus’, *Neuron*, 92(2), pp. 342–357. Available at: <https://doi.org/10.1016/j.neuron.2016.10.001>.
- Shi, W. *et al.* (2020) ‘Five-mRNA Signature for the Prognosis of Breast Cancer Based on the ceRNA Network’, *BioMed research international*, 2020, p. 9081852. Available at: <https://doi.org/10.1155/2020/9081852>.
- Shiah, J.V., Johnson, D.E. and Grandis, J.R. (2023) ‘Transcription Factors and Cancer: Approaches to Targeting’, *Cancer journal (Sudbury, Mass.)*, 29(1), pp. 38–46. Available at: <https://doi.org/10.1097/PPO.0000000000000639>.
- Shim, J. *et al.* (2023) ‘Abstract 3552: YAP-TEAD2 binding mediates therapy resistance in RAS-driven neuroblastoma’, *Cancer Research*, 83(7_Supplement), p. 3552. Available at: <https://doi.org/10.1158/1538-7445.AM2023-3552>.

References

- Shu, J. *et al.* (2015) ‘GATA family members as inducers for cellular reprogramming to pluripotency’, *Cell Research*, 25(2), pp. 169–180. Available at: <https://doi.org/10.1038/cr.2015.6>.
- Ståhl, P.L. *et al.* (2016) ‘Visualization and analysis of gene expression in tissue sections by spatial transcriptomics’, *Science (New York, N.Y.)*, 353(6294), pp. 78–82. Available at: <https://doi.org/10.1126/science.aaf2403>.
- Stuart, T. *et al.* (2021) ‘Single-cell chromatin state analysis with Signac’, *Nature Methods*, 18(11), pp. 1333–1341. Available at: <https://doi.org/10.1038/s41592-021-01282-5>.
- Sun, G. *et al.* (2021) ‘Transcription Factors BARX1 and DLX4 Contribute to Progression of Clear Cell Renal Cell Carcinoma via Promoting Proliferation and Epithelial–Mesenchymal Transition’, *Frontiers in Molecular Biosciences*, 8. Available at: <https://doi.org/10.3389/fmolb.2021.626328>.
- Takei, Y. *et al.* (2021) ‘Integrated spatial genomics reveals global architecture of single nuclei’, *Nature*, 590(7845), pp. 344–350. Available at: <https://doi.org/10.1038/s41586-020-03126-2>.
- Tan, Y. and Testa, J.R. (2021) ‘DLX Genes: Roles in Development and Cancer’, *Cancers*, 13(12). Available at: <https://doi.org/10.3390/cancers13123005>.
- Tang, F. *et al.* (2009) ‘mRNA-Seq whole-transcriptome analysis of a single cell’, *Nature Methods*, 6(5), pp. 377–382. Available at: <https://doi.org/10.1038/nmeth.1315>.
- Tavares-Ferreira, D. *et al.* (2022) ‘Spatial transcriptomics of dorsal root ganglia identifies molecular signatures of human nociceptors’, *Science Translational Medicine*, 14(632), p. eabj8186. Available at: <https://doi.org/10.1126/scitranslmed.abj8186>.
- TOSKA, E. and ROBERTS, S.G.E. (2014) ‘Mechanisms of transcriptional regulation by WT1 (Wilms’ tumour 1)’, *The Biochemical journal*, 461(1), pp. 15–32. Available at: <https://doi.org/10.1042/BJ20131587>.
- Traag, V.A., Waltman, L. and van Eck, N.J. (2019) ‘From Louvain to Leiden: guaranteeing well-connected communities’, *Scientific Reports*, 9(1), p. 5233. Available at: <https://doi.org/10.1038/s41598-019-41695-z>.

References

- Uen, Y.-H. *et al.* (2015) ‘VAV3 Oncogene Expression in Colorectal Cancer: Clinical Aspects and Functional Characterization’, *Scientific Reports*, 5(1), p. 9360. Available at: <https://doi.org/10.1038/srep09360>.
- Ullmann, T. *et al.* (2021) ‘High-Throughput Screening for CEBPD-Modulating Compounds in THP-1-Derived Reporter Macrophages Identifies Anti-Inflammatory HDAC and BET Inhibitors’, *International Journal of Molecular Sciences*, 22(6), p. 3022. Available at: <https://doi.org/10.3390/ijms22063022>.
- Ulz, P. *et al.* (2019) ‘Inference of transcription factor binding from cell-free DNA enables tumor subtype prediction and early detection’, *Nature Communications*, 10(1), p. 4666. Available at: <https://doi.org/10.1038/s41467-019-12714-4>.
- Vahid, M.R. *et al.* (2023) ‘High-resolution alignment of single-cell and spatial transcriptomes with CytoSPACE’, *Nature Biotechnology*, 41(11), pp. 1543–1548. Available at: <https://doi.org/10.1038/s41587-023-01697-9>.
- Virtanen, P. *et al.* (2020) ‘SciPy 1.0: fundamental algorithms for scientific computing in Python’, *Nature Methods*, 17(3), pp. 261–272. Available at: <https://doi.org/10.1038/s41592-019-0686-2>.
- Wang, R. *et al.* (2013) ‘CCCTC-binding factor controls its own nuclear transport via regulating the expression of importin 13’, *Molecules and Cells*, 35(5), pp. 388–395. Available at: <https://doi.org/10.1007/s10059-013-2283-z>.
- Wang, Z.-Q. *et al.* (2016) ‘CD74 and intratumoral immune response in breast cancer’, *Oncotarget*, 8, pp. 12664–12674. Available at: <https://doi.org/10.18632/oncotarget.8610>.
- Wei, L. *et al.* (2019) ‘Oncogenic ADAM28 induces gemcitabine resistance and predicts a poor prognosis in pancreatic cancer’, *World Journal of Gastroenterology*, 25(37), pp. 5590–5603. Available at: <https://doi.org/10.3748/wjg.v25.i37.5590>.
- Wendt, K.S. and Grosveld, F.G. (2014) ‘Transcription in the context of the 3D nucleus’, *Current Opinion in Genetics & Development*, 25, pp. 62–67. Available at: <https://doi.org/10.1016/j.gde.2013.11.020>.

References

- Williams, C.G. *et al.* (2022) ‘An introduction to spatial transcriptomics for biomedical research’, *Genome Medicine*, 14(1), p. 68. Available at: <https://doi.org/10.1186/s13073-022-01075-1>.
- Wolf, F.A., Angerer, P. and Theis, F.J. (2018) ‘SCANPY: large-scale single-cell gene expression data analysis’, *Genome Biology*, 19(1), p. 15. Available at: <https://doi.org/10.1186/s13059-017-1382-0>.
- Wong, Alissa Michelle Go *et al.* (2013) ‘Characterization of CACNA2D3 as a putative tumor suppressor gene in the development and progression of nasopharyngeal carcinoma’, *International Journal of Cancer*, 133(10), pp. 2284–2295. Available at: <https://doi.org/10.1002/ijc.28252>.
- Xavier da Silveira dos Santos, A. and Liberali, P. (2019) ‘From single cells to tissue self-organization’, *The FEBS Journal*, 286(8), pp. 1495–1513. Available at: <https://doi.org/10.1111/febs.14694>.
- 10x Genomics Xenium; 2024, [www.10xgenomics.com/platforms/xenium].
- Xie, D. *et al.* (2021) ‘Emerging Role of E2F Family in Cancer Stem Cells’, *Frontiers in Oncology*, 11, p. 723137. Available at: <https://doi.org/10.3389/fonc.2021.723137>.
- Yu, G. *et al.* (2022) ‘LRP1B mutation associates with increased tumor mutation burden and inferior prognosis in liver hepatocellular carcinoma’, *Medicine*, 101(26), p. e29763. Available at: <https://doi.org/10.1097/MD.00000000000029763>.
- Yu, W. *et al.* (2022) ‘Bioinformatic analysis of FOXN3 expression and prognostic value in pancreatic cancer’, *Frontiers in Oncology*, 12, p. 1008100. Available at: <https://doi.org/10.3389/fonc.2022.1008100>.
- Zeineldin, M., Patel, A.G. and Dyer, M.A. (2022) ‘Neuroblastoma: When differentiation goes awry’, *Neuron*, 110(18), pp. 2916–2928. Available at: <https://doi.org/10.1016/j.neuron.2022.07.012>.
- Zeisel, A. *et al.* (2015) ‘Brain structure. Cell types in the mouse cortex and hippocampus revealed by single-cell RNA-seq’, *Science (New York, N.Y.)*, 347(6226), pp. 1138–1142. Available at: <https://doi.org/10.1126/science.aaa1934>.

References

Zeng, Q. *et al.* (2024) 'Knockdown of BUB1B Inhibits the Proliferation, Migration, and Invasion of Colorectal Cancer by Regulating the JNK/c-Jun Signaling Pathway', *Cancer Biotherapy & Radiopharmaceuticals*, 39(3), pp. 236–246. Available at: <https://doi.org/10.1089/cbr.2023.0070>.

Zhang, S. *et al.* (2017) 'Early growth response 3 inhibits growth of hepatocellular carcinoma cells via upregulation of Fas ligand', *International Journal of Oncology*, 50(3), pp. 805–814. Available at: <https://doi.org/10.3892/ijo.2017.3855>.

Supplementary Materials and Methods

Package Versions

The analysis was performed using python 3.12 with the packages listed in **Tab.S1**. R 4.3.1 was employed to generate some of the plots in this thesis.

Table S1. Important package versions.

Package	Version	Reference
Analysis		
squidpy	1.2.2	(Palla <i>et al.</i> , 2022)
scanpy	1.9.8	(Wolf <i>et al.</i> , 2018)
scipy	1.12.0	(Virtanen <i>et al.</i> , 2020)
tangram	1.0,4	(Biancalani <i>et al.</i> , 2021)
MaxFuse	0,0,2	(S. Chen <i>et al.</i> , 2023b)
spaGE	no version available	(Abdelaal <i>et al.</i> , 2020)
cytospace	1.0,6	(Vahid <i>et al.</i> , 2023)
Plotting		
matplotlib	3.8.0	
seaborn	0.13.2	
ggplot2	3.4.4	

AI-based tool and their applications

Table S2. AI based tools employed in this thesis.

Tool, version	Supplier	Section	Purpose
ChatGPT-4o	OpenAI, inc.	all sections	Grammar and spelling check, assistance in rephrasing for better legibility

Supplementary Tables and Figures

Table S3. CellAssign marker genes for scRNA-seq data annotation.

Broad cell type	Cell Markers
Neuroblasts	ISL1, STMN2, ALK, NEFM, PRPH, GAP43, NTRK1, KRG1, SYN3, HAND2, GATA3, PHOX2B, TFAP2B, CCND1, MYCN, SOX11, ST8SIA1, ENO1, AHCY, FBL, SLC7A5, RUVBL1, NCL, NPY, SEMA3E, IL7, FOXO3, CREB5, TERT, EBF1, EPAS1, STAT3, ZNF467, SOX6, ZNF423, ICAM1, TNFRSF1B, IL6R, TOP2A, MKI67, PRRX1, VIM, FOSL2
Bridge	ASCL1, ERBB4, CTTNBP2, CDH9, CSMD3
Chromaffin cells	DBH, TH, CHGA, PNMT, SLC18A1
Endothelial cells	PTPRB, FLT1, EGFL7
Connecting progenitor cells	CDH13, DDC
Mesenchymal cells	COL1A1, PDGFRB, FN1
Adrenal cortex	CYP11A1, CYP11B1, STAR, NR5A1
Schwann cells	MBP, MPZ, S100B, PMP22, CDH19, SOX10, ERBB3
Myeloid mononuclear cells	CD14, FCGR3A, FCGR1A, ITGAM, CD68, CD163, CD83, THBD, CD1C, BTLA, CLEC9A, ITGAX, FUT4, FCGR3B
Lymphoid immune cells	CD3E, CD4, CD8A, CD7, CD2, FOXP3, CD19, NKG7, NCAM1, NCR1, KLRC1

Table S4. Xenium SRT clinical information.

Batch	Plate ID	patient ID	subtype	stage	Disease_period
1	0010562	576	LR	1	initial - primary tumor site
1	0010562	639	TERT	4	initial - primary tumor site
1	0010562	808	MYCN	4	initial - primary tumor site
1	0010685	217	LR	2,2	initial - primary tumor site
1	0010685	325	MYCN	4	inital - metastasis (KM) oder post chemo - primary tumor site
1	0010685	827	ALT	1	initial - primary tumor site
2	0018252	26	TERT	NA	NA
2	0018281	833	ALT	NA	NA
2	0018319	526	ALT	NA	NA
2	0018252	476	ALT	NA	NA
2	0018319	644	MYCN	NA	NA
2	0018319	596	TERT	NA	NA
2	0018281	398	LR	NA	NA
2	0018249	61	ALT	NA	NA
2	0018281	369	TERT	NA	NA
2	0018249	395	LR	NA	NA
2	0018249	720	TERT	NA	NA
2	0018252	614	MYCN	NA	NA
2	0018252	827	ALT	NA	NA
2	0018319	217	LR	NA	NA
2	0018249	325	MYCN	NA	NA

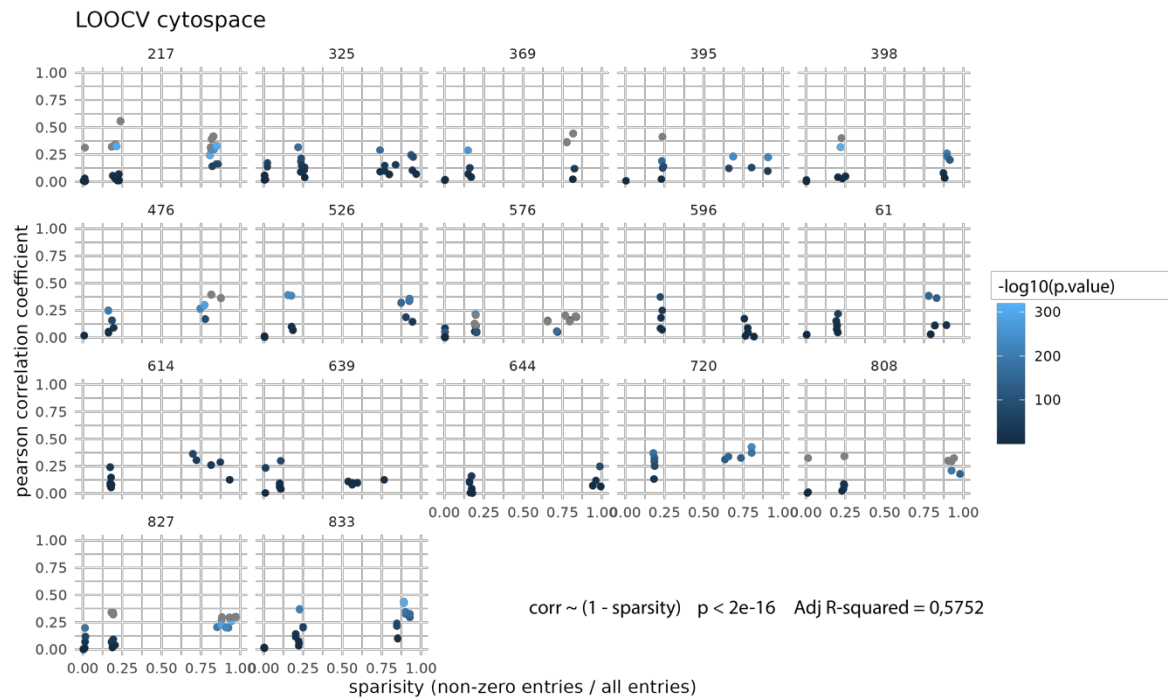


Figure S2.1. LOOCV results by sample for CytoSPACE. For each sample, the sparsity of the selected genes is plotted against the Pearson correlation coefficient between inferred and measures transcription counts. Grey dots have p values $< 10^{-16}$ and NAs were returned by `scipy.stats.pearsonr()`.

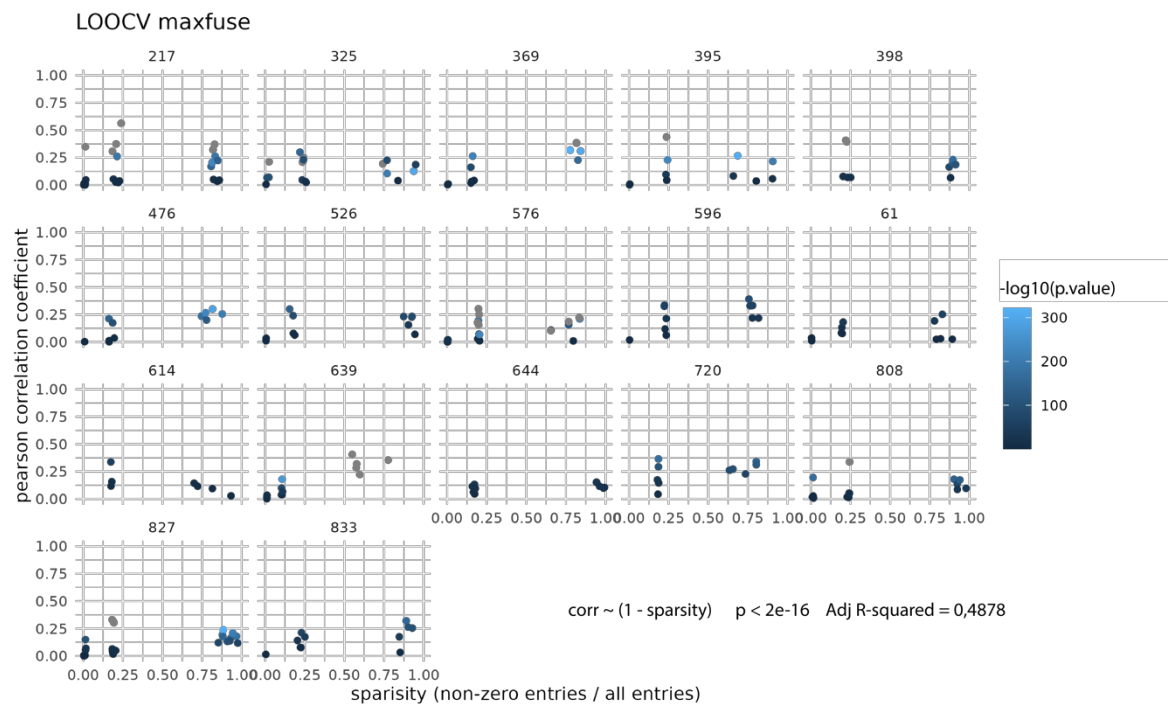


Figure S2.2 LOOCV results by sample for MaxFuse. For each sample, the sparsity of the selected genes is plotted against the Pearson correlation coefficient between inferred and measures transcription counts. Grey dots have p values $< 10^{-16}$ and NAs were returned by `scipy.stats.pearsonr()`.

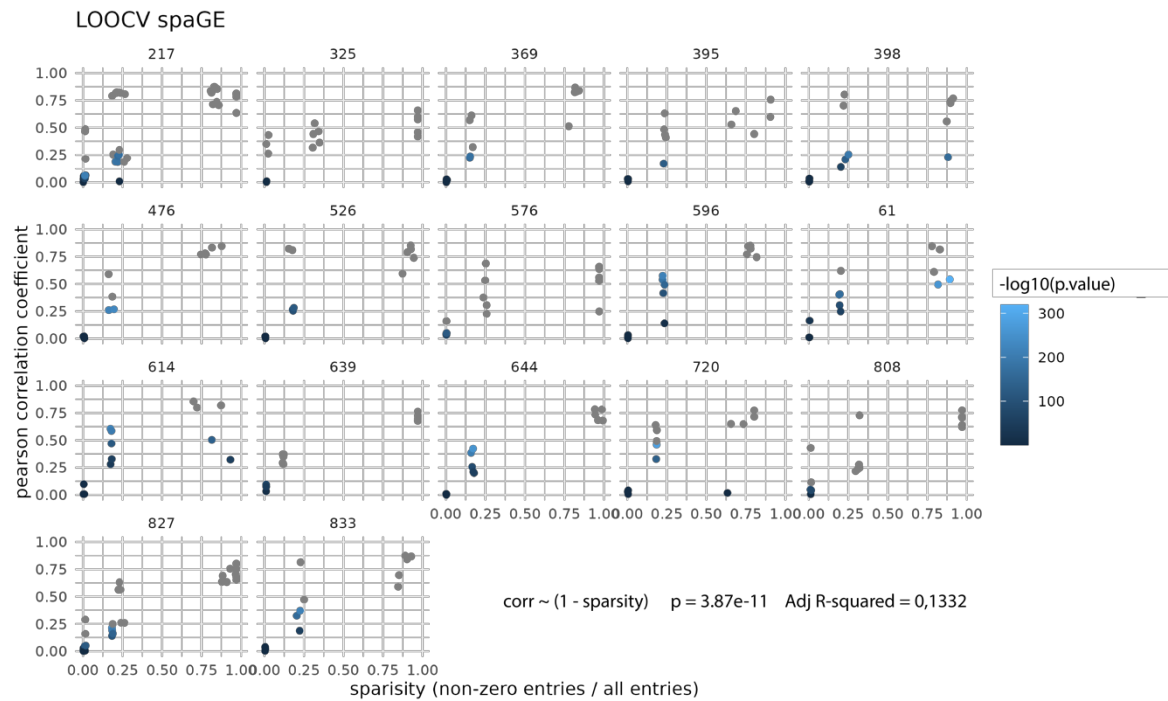


Figure S2.3 LOOCV results by sample for spaGE. For each sample, the sparsity of the selected genes is plotted against the Pearson correlation coefficient between inferred and measures transcription counts. Grey dots have p values $< 10^{-16}$ and NAs were returned by `scipy.stats.pearsonr()`.

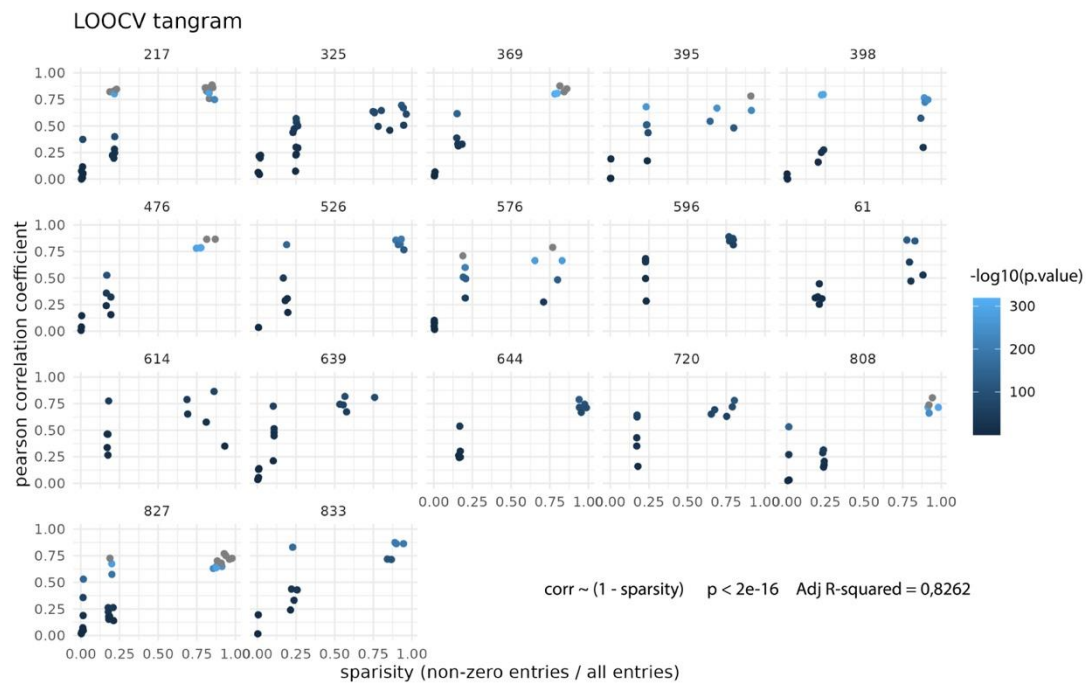


Figure S2.4 LOOCV results by sample for spaGE. For each sample, the sparsity of the selected genes is plotted against the Pearson correlation coefficient between inferred and measures transcription counts. Grey dots have p values $< 10^{-16}$ and NAs were returned by `scipy.stats.pearsonr()`.

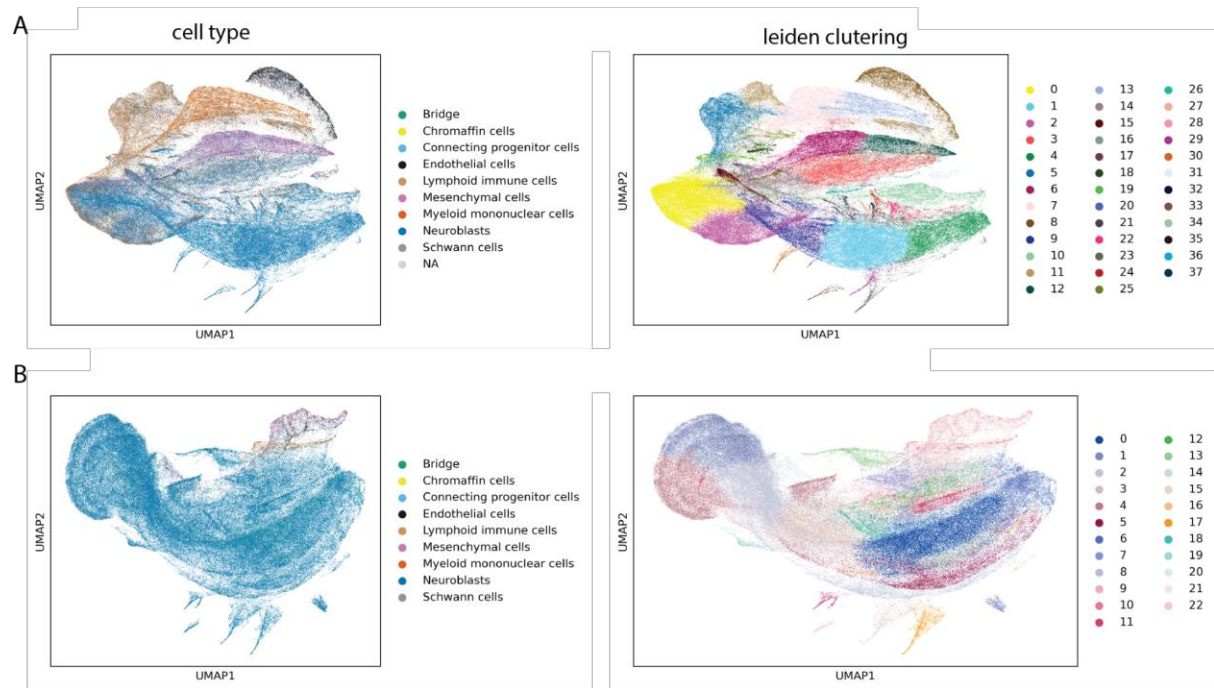


Figure S3. UMAP projection of log-normalised cell transcript counts imputed by spaGE by cell type (left) and Leiden cluster (right) for samples 576 (A) and 808 (B).

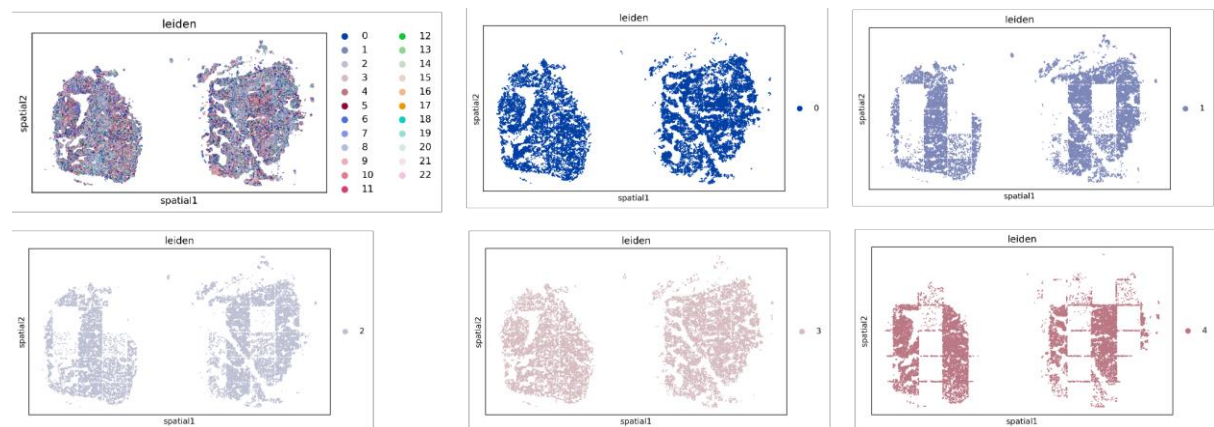


Figure S4. Leiden clustering based on SpaGE inferred gene expression.

Table S4. Entropy scores of gene expression, TF activity and gene activity mapped with Tangram and MaxFuse.

plate + sample Id	subtype	Neuroblast fraction	gene expression		transcription factor activity		gene activity	
			Tangram	MaxFuse	Tangram	MaxFuse	Tangram	MaxFuse
0010562_576	LR	0,4926	1,5652	1,3985	1,8104	1,3000	1,9068	1,3725
0010562_639	TERT	0,6908	1,1073	0,7119	0,8612	0,8416	1,2396	0,9158
0010562_808	MYCN	0,8298	0,8254	0,8358	0,8339	0,7092	0,8672	0,7373
0010685_217	LR	0,7710	0,7438	0,6671	0,6565	0,5850	0,918	0,6430
0010685_325	MYCN	0,6037	1,3099	0,9735	1,4232	1,2652	1,8428	1,29479
0010685_827	ALT	0,9496	0,4365	0,3892	0,276	0,2449	0,3721	0,2100
0018281_833	ALT	0,9325	0,8371	0,7489	0,4322	0,3746	0,524	0,2925
0018319_526	ALT	0,9240	0,5944	0,5705	0,4256	0,3888	0,5226	0,3121
0018252_476	ALT	0,9454	1,5331	1,5086	0,3584	0,3094	0,4272	0,2503
0018319_644	MYCN	0,81131	0,9769	0,943	0,9113	0,8149	1,0236	0,8234
0018319_596	TERT	0,6366	0,8762	0,8822	0,8815	0,8512	1,3582	1,0337
0018281_398	LR	0,8587	1,3739	0,5896	0,5908	0,5062	0,7147	0,5052
0018249_61	ALT	0,8532	0,9138	0,7674	0,5436	0,6037	0,8623	0,4542
0018281_369	TERT	0,8257	1,636	1,5911	0,5107	0,5707	0,7902	0,6127
0018249_395	LR	0,5651	2,0779	1,8971	0,9400	1,1518	1,7164	1,2097
0018249_720	TERT	0,5134	1,5053	1,3793	1,1015	1,2563	1,8355	1,3136
0018252_614	MYCN	0,6954	1,3392	1,2852	0,6958	0,9311	1,3518	1,0945
0018252_827	ALT	0,8994	0,5762	0,4993	0,4472	0,4300	0,6468	0,3928
0018319_217	LR	0,7015	0,9589	0,8336	0,8339	0,7384	1,0843	0,7990
0018249_325	MYCN	0,0187	0,9385	0,9176	0,9122	0,7926	1,1277	0,7829

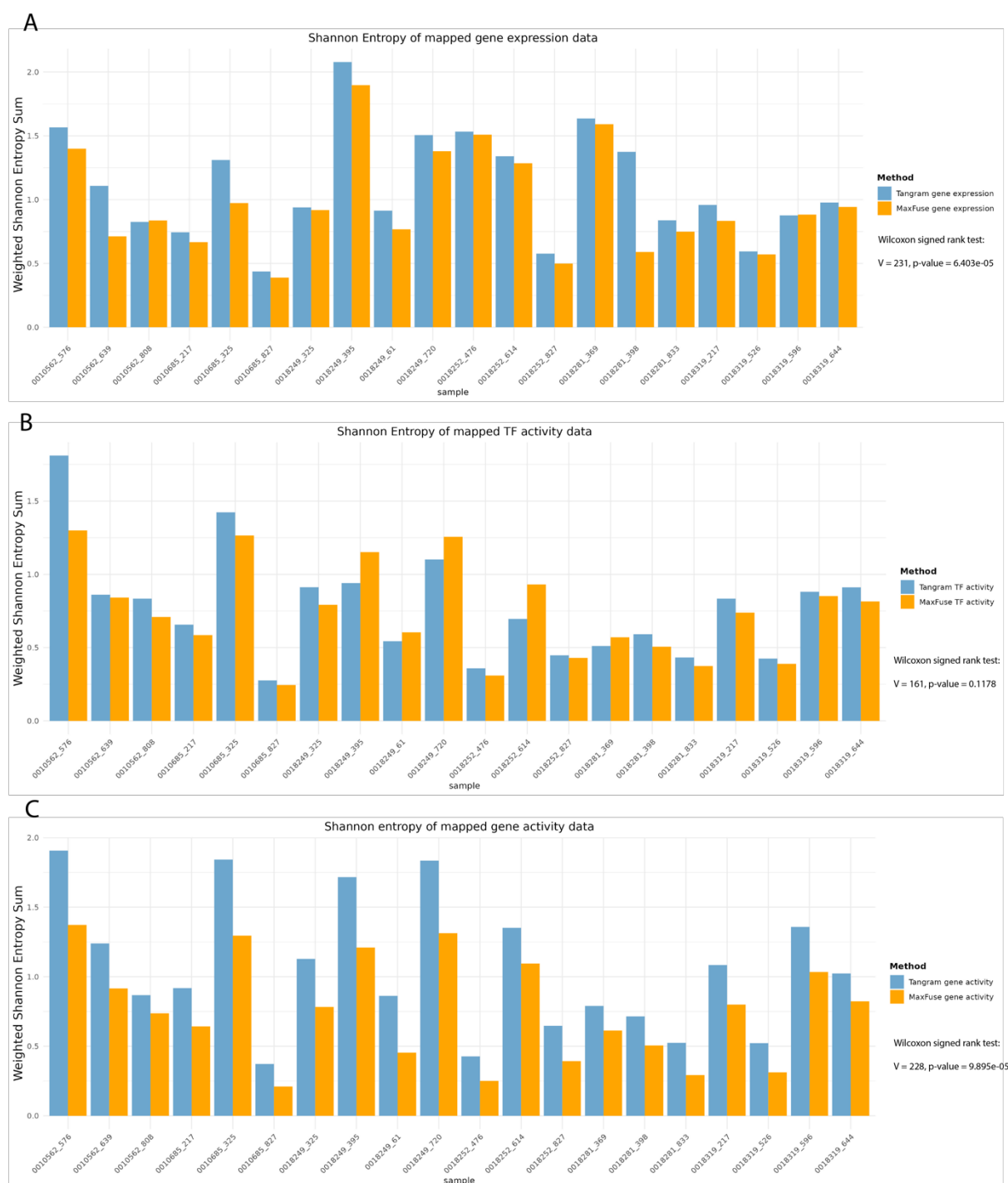


Figure S5. Weighted Shannon Entropies for each mapping across all samples. Gene expression (A), TF activity (B) and gene activity (C). A paired signed Wilcoxon rank test was performed for each modality. MaxFuse outperforms Tangram significantly in mapping of gene expression and gene activity but not TF activity.

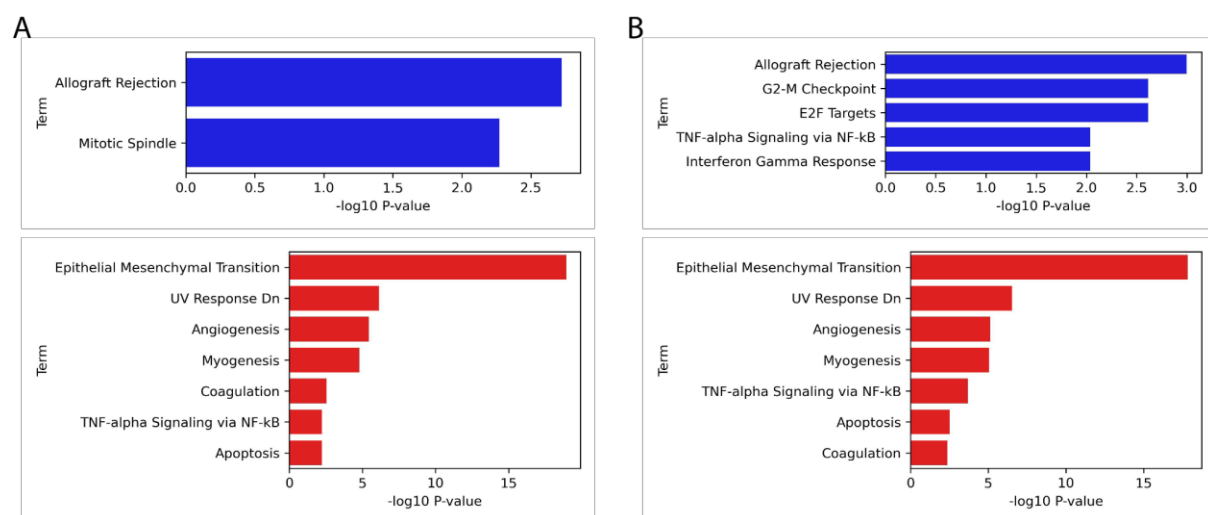


Figure S6. Enriched enrichR terms results of proliferative neuroblasts in sample 576 (LR). EnrichR was run using only the differentially over- (up) and underexpressed genes (down) (A) as well as using differentially expressed genes and differentially active TFs and their targets (B). The threshold for the adjusted p values was set at 0.01.